

Econ 310 Monetary Theory

Numbered Transcript

ECON 310 • Dr. Lawrence White

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Lecture 1: Money: Private or Public?

At a glance — White introduces monetary theory and whether money is private or public.

- 1.1 Good morning. We're starting eight lectures on monetary theory, which is a little bit esoteric, but I'll try to bring it down to earth and try to relate it to the issues that are important in our economy. I want to start with a big issue, which is what is the role of markets versus the role of governments in providing money, and what do we know from history, and what do we know from theory about the relative merits of government provision versus private provision.
- 1.2 But I guess we should start by defining what we're talking about. People have different ideas about what is meant by money. The standard definition among economists is that money is anything that serves as a commonly accepted medium of exchange. Now, that needs to be unpacked a little. A medium of exchange is something you acquire in one transaction, not in order to consume it, not in order to plant it in the ground and grow something from it or put it into a factory, but in order to trade it away later. It's something you acquire for the sake of carrying forward the exchange process. Why that's a useful strategy, I'll talk about in a minute.
- 1.3 The commonly accepted part: it's a fuzzy term. Not clear how commonly it has to be accepted to be commonly accepted. So there are going to be borderline cases. But what that rules out is money in a one-person economy. Robinson Crusoe has no use for money. And really, in a two-person economy, there's no use for money because you give money to the other person, they're just going to give it back to you, so you might as well just trade the goods right up front. So you need at least three people and three goods to make sense of a monetary economy.
- 1.4 Something can be a medium of exchange for a particular person without being a commonly accepted medium of exchange. Before money emerged, there were exchanges in which one person might be trading what they produce for peppercorns and using those to buy what they want to go home with. Somebody else may be using salt, somebody else may be using barley. So there are two things we need to explain: one, why people would move from barter to indirect exchange, make two trades instead of one, and then why a society would converge on a common medium of exchange. So the question of how money arose is the question of how some good acquired this property of being a commonly accepted medium of exchange.
- 1.5 Now, we know that since ancient times, governments have been involved in producing money, in coining money in the earliest days. But that's not the only thing we find in the historical record. We also find private provision of the earliest monies and private provision of coins. Coins actually come later. There is money before there are coins. There are shells used as money. There are pieces of silver that are not coined and used

as money. So, when we see governments getting involved, we need to ask: has that improved things or not improved things?

- 1.6 For that, we need a benchmark. We need to understand what a monetary system looks like without any government involvement. So, I'm going to talk about the origin of money and then the technological development, which is coinage. We'll see that private markets have done these things, and so it wasn't necessary for governments to get involved. Then we can ask whether it's an improvement for governments to get involved or have they improved things.
- 1.7 There are a group of economists who think that the state had to be involved, that you can't imagine money emerging out of barter. And they call their own theory the state theory of money. So it's pretty straightforward. To caricature it a little bit, it's the idea that there must have been some wise king who came up with the idea of switching from barter to monetary exchange.
- 1.8 The alternative that I'm going to spell out is what you might call a market theory of money, that we can talk about the incentives of individuals leading them, without any central plan, without any top-down direction, into practices that give rise to the convention, you might call it, of accepting some commodity as a medium of exchange, some common commodity. That's, to use Adam Smith's phrase, "an invisible hand" theory of the origin of money.
- 1.9 The state theory of money advocates want to deny that there's a coherent invisible hand theory of money, but I think it's clear there is a coherent invisible hand theory of money. Then it becomes a historical question: which came first, market-based money or state-based money? It strikes me as somewhat implausible to think that this really useful contrivance, this convention, this practice, would have come out of the palace as opposed to out of the marketplace, where people are looking for ways to make life more convenient for themselves.
- 1.10 The first really satisfactory spelling out of this theory that money emerges from the marketplace is from the Austrian economist Carl Menger. Menger was a professor at the University of Vienna, and he's the founder of what's now called the Austrian School of Economics. Menger said to put yourself in the position of a barterer, somebody who's producing goods and then coming to market to trade them for the goods that he wants to consume.
- 1.11 An example I like to use—it's a little bit far-fetched—you're producing asparagus, you're an asparagus farmer, and you come to market, and you want to go home with a plaid shirt. Now, you've got a problem. The person who is producing plaid shirts and selling them may not like asparagus, may not want to be paid in asparagus. What do you do then? Well, you ask him what he will accept payment in. He says, "I'll accept payment in salt." You think, "Okay, then if I can sell my asparagus for salt, I can use the

salt to buy the plaid shirt. So salt then will be my medium of exchange." So you go to the salt sellers and see if they'll accept asparagus.

- 1.12** But when you've been to the market a few times as an asparagus seller, you learn that it pays to trade your asparagus for anything that raises your odds of having it accepted for the thing you want to go home with. So you start looking for more commonly accepted goods. There may be several that you use on different occasions to serve as media of exchange. But that's the basic idea as to why people move from trying direct exchange to indirect exchange, trading for a second good and then using that second good to buy what they want to consume.
- 1.13** And then, the second part has to be explaining why people in a marketplace would converge on a common medium of exchange, because one person might be using salt and another pepper, as I said. Once you get the idea that trading your asparagus for anything that's more commonly accepted will raise your odds of achieving the trades you want to achieve, you start looking for more commonly accepted media of exchange. That means you're on the lookout for what other people are accepting. People become alert or aware of what is more commonly consumed, what more people want, will accept.
- 1.14** In that situation, if something is more popular than what you've got, you'll accept it, and you'll be, as I said, on the lookout for what's most commonly accepted. In that way, you'll accept something when you see other people accepting it, even if you don't want to consume it yourself, you'll be alert to what is commonly accepted. Under those circumstances, if you start accepting salt because you see other people accepting it, you enlarge the group of people who accept salt in payment by one person, and now that makes salt more attractive to other people who want to trade with you.
- 1.15** An example of this phenomenon, where people are attuned to what other people are doing, is when you go to the airport and the bookstore has a shelf which has the top 10 selling books listed for you, 1 through 10. Now, you might ask yourself, "Why do I care what the top-selling book is? I know what I'm interested in reading. Why do I care what other people are reading?" I think the answer has to be, well, you might go to a cocktail party where you want to talk about this book with other people. So, you want to know what other people are reading so that you're in the loop. But you also care about the intrinsic merits of the book. It's not the only thing you care about; what other people are reading.
- 1.16** But in the case of a medium of exchange, all you care about is whether other people accept it because you're not going to consume it. So popularity becomes self-reinforcing. That's how you get spontaneously, without any top-down direction, convergence onto a common medium of exchange. In light of that, we can say money was not invented. It's not like the product of a laboratory. It didn't take some genius or some wise king to come up with the idea.

- 1.17** If you look technologically at, say, cowrie shells used as money, they're not any different from the cowrie shells used as jewelry. They're the same shells. So it's not a technological breakthrough. What's changed are the practices of people in accepting them as a medium of exchange. So it's a social convention. It's not a technological breakthrough.
- 1.18** Menger summed this up by saying, "Money is not an invention of the state. It's not the product of a legislative act. Even the sanction of political authority is not necessary for its existence. Certain commodities came to be money quite naturally, as a result of economic relationships that were independent of the power of the state"—that is, the incentives of people trying to achieve their trades.
- 1.19** I'm going to argue that we can say the same thing about later steps in the development of money. So the development of coinage is not an invention of the state. Transferable deposits, the thing we use commonly every day now, is not an invention of the state. Banknotes are not an invention of the state. The development of money is an important development in the history of world economies because it makes possible a much finer division of labor.
- 1.20** If it's difficult to trade what you're producing for what you want to consume, you won't specialize so much. My example of the asparagus farmer going to market not knowing whether anybody wants his asparagus is a little anachronistic. I'm putting the cart before the horse to make it dramatic. In a world where people don't want asparagus—sorry, very few people want asparagus, or it's not commonly accepted as a medium of exchange, and it's difficult to find somebody who will trade you what you want for it, you won't just produce asparagus.
- 1.21** And in early economies, people were much more self-sufficient. They produced everything they wanted to consume because it was too difficult to trade for it. So in the United States, people used to produce their own candles if they lived on the frontier because it was too difficult to get candles otherwise. The transaction costs were too high.
- 1.22** Now, the state theory of money people are inspired by a book called "The State Theory of Money," written by a German economist named Knapp. But if you look at Knapp's argument, it's not fundamentally empirical or historical. It's wordplay. It's a linguistic argument. Knapp says, I'm going to distinguish between public money, which is money accepted by the state, in tax collections and in legal judgments, and private money. He says that, sure, there is private money. He says, a bank and its customers, quote, "form, so to speak, a private pay community."
- 1.23** But then he goes on to neglect private pay communities and focus on the state. And he's right that state acceptance delimits the state monetary system, but he's not right in saying that's what determines the monetary system as a whole. It just determines the public monetary system. The private pay community, what he calls the private pay

community, can be much larger and more important than the state community. It depends on how big the state is. So states have more influence, the bigger the state is as a share of the total output of the economy. But there's nothing that picks out the state as uniquely important in determining what is a commonly accepted medium of exchange.

- 1.24 It's true that states can compel people to accept— they can compel people to pay their taxes in a certain medium, but that may or may not be enough to get people to use it elsewhere. I mean, Walmart compels people to pay in a certain medium. That's pretty important in determining what money people want to carry. So the fact that it's not official money until the state says so does nothing to overturn Menger's story or his account of how a commonly accepted medium of exchange comes about without state intervention.
- 1.25 Let's move to coinage. The state theory of money people, also known as Chartalists, want to argue that the acceptance of currency rests on the insignia of the sovereign being stamped on the coins. Charles Goodhart, a well-known British economist, wrote that "currency becomes money primarily because the coins are struck with the insignia of sovereignty."
- 1.26 But let's back up a step. Before coinage, there are monies. They don't bear insignias. Shells don't bear insignias. People paid in loose bits of silver, known to monetary historians as hack silver, just hacked up pieces of silver. They didn't have any insignias on them. So it can't be that an insignia is necessary for something to become money.
- 1.27 And the types of money that do have insignias: coins, banknotes—it doesn't have to be a state insignia. Banknotes, paper currency, was widely issued by private commercial banks in the 19th century, and that was the most popular money. So it would say the name of the bank that issued it, but it's not a liability of the state. It's not issued by government. That was the most popular money. So it can't be that the insignia of sovereignty is what's essential to it.
- 1.28 Even coins have been issued with private rather than public insignias. So you want something stamped on the coin to tell you who issued it because you want to be able to trust it. So you want a coin issued by somebody with a reputation for issuing good coins, but it doesn't have to be the state. It could be a private mint master. There are periods in history where privately issued coins predominated.
- 1.29 In early India, in the earliest places where we find coins, the other being the civilization called Lydia, which is in what's now Turkey, there were lots of coins, some issued by the state, but others don't seem to have been issued by the state. It's a little unclear because we don't know whose insignias they are, but probably not the state.
- 1.30 Now, coinage depends on metal being the money that people want. What a coin is, is a certified piece of metal. The stamp tells you how pure it is—if not directly, then indirectly—the purity and the weight of the coin. But how did we get to metallic

coinage? There were other commodities that had served as money: salt, I mentioned, cowrie shells, beads. There's even an episode where oxen are used as a commonly accepted medium of exchange in ancient Greece.

- 1.31** There are problems with non-metallic monies that you don't have with metallic money. So you want something that's easy to store. You want something that's relatively precious. So when salt becomes abundant, it becomes very bulky to carry around. Salt as your medium of exchange, people put that aside and look for something that's easier to carry around, so something more precious.
- 1.32** You want something that is easy to make a change with. That's a problem with oxen. Once you make a change with an ox, it's not so durable, not so easy to carry around. When it's alive, it walks itself from market to market, but not so good to make change with.
- 1.33** And you want something that is available in either a uniform quality or where there's not a lot of hassle between you and your trading partner over what quality you're offering. So that's sometimes a problem with foodstuffs, whereas with advances in metallurgy, you can get uniform pieces of metal. So, there was a social winnowing process or survival of the fittest among commodity monies that gave rise to silver and gold.
- 1.34** So what coinage does is to take a raw piece of metal or a bunch of gold dust and turn it into a piece of uniform, ideally, uniform purity and weight. And that adds value to the raw material because if you're asking somebody to accept gold dust, gold dust comes in various purities. It could be 5% silver and copper mixed in with the gold. It could be 12% other metals when it comes out of the earth. You don't know. By having it coined into uniform pieces, you take away that doubt and make it more valuable. So it adds value to the metal to take it to a mint.
- 1.35** The earliest coins in ancient Lydia were blobby. They were made crudely by hammering designs into the metal. But then the resulting piece of metal wasn't exactly round. And most importantly, they didn't have any devices at the very beginning to indicate where somebody had shaved off some of the metal, sharp operators. If other people say, "Oh, it's got a lion's head on it. I know what that's worth," if the people accepting the coins don't weigh them or don't have some way of determining whether they've been filched, whether they've been shaved, sharp operators are going to shave off some of the metal.
- 1.36** So advanced coins have these devices that make any tampering evident. Like the caps on a bottle of aspirin, so you know whether it's been tampered with. Advanced coins are tamper-evident. But in the Lydian civilization, one numismatic historian says, "We find more than 20 distinct markings on these coins, and we know there weren't 20 different kings." So whose insignias are these? They are probably independent gold merchants. I said it's not clear, but it's certainly not clear that they were all royal coins.

- 1.37 And as I said, it seems to me plausible on the face of it that merchants would come up with this as a device for making their transactions easier. If somebody pays you in metal, you have to weigh it and try to guess how pure it is, but then you would mark that piece of metal. You might stamp something on it for your own sake to indicate how much it's worth. If somebody else trusts your stamp, then you've made your life easier. If they see that other people trust your stamp, they may bring you raw metal in order to have you stamp it so that it's more useful for them, less hassle in trading it away.
- 1.38 So people who are experts at assaying metal, that is testing the purity, and working with the metal to form it into coins, could make a business of producing coins. We know there were hundreds of coin makers in ancient India, and there seemed to have been dozens in Lydia. Later, after the fall of the Roman Empire—the Roman Empire, of course, monopolized the production of coins—but after the Empire fell, there were hundreds of coiners, private coiners in Europe, because there was no one to enforce the monopoly.
- 1.39 Now, one problem for the state theory of money is to explain why coins are made of gold and silver. Because their view is that these coins are valued because of the insignia, not because of the content. And some of them even refer to coins as tax anticipation tokens, that is, kings issued coins in order to pay soldiers. So the soldiers could spend the coins. They made the public pay their taxes in coins. So that's their theory as to what gave these coins circulation.
- 1.40 But if that's the case, why use expensive metal to produce these tokens? You could produce them on iron or parchment or copper. Why use gold and silver? On Menger's account, you want gold and silver because it's more convenient to have a precious metal. It's less bulky. But if they're just tax anticipation tokens, no explanation for why it would be gold and silver.
- 1.41 Another problem with this tax anticipation idea is that people use gold coins issued by, let's say, the Spanish mint, which had a good reputation, not for its silver coins, those were terribly debased, but for its gold coins. People around Europe used Spanish gold coins who had no tax obligations to the Spanish government. They were using the coins because they were valued for their content. And it's an undeniable, notorious fact that coins are valued in the market according to their gold content or silver content in the case of silver coins, not the insignia.
- 1.42 So as early as 1605, we find a Spanish priest named Juan de Mariana, who was writing about the problem of debasement. He says, "People don't value a currency any more than the quality, and the amount of metal allows." Even if there are laws against this—some kings didn't want people to discount the coins that they had debased; they made it illegal to value coins according to the metallic content—but of course, people did it anyway.

- 1.43 All right. I asked the question early on whether state involvement improved the quality of the money. In the case of coinage, it's pretty clear that it didn't. So ancient kings, medieval kings, are notorious for debasing the coinage, that is, reducing the metallic content and lying about it, trying to get people to accept coins at full value, which are under the standard. So it can't be that they intervened in order to improve the quality of the coinage. Or if that was the original intention, it didn't stay the intention.
- 1.44 Why are they debasing the coins? It's a source of revenue. Once you have a mint monopoly, you can charge high fees for turning raw metal into coins and earn a profit that way. But you can earn an even bigger profit by claiming that this is a one-ounce silver coin and only putting half an ounce of silver in it, if you can either fool people into thinking that it's still an ounce, or you can compel them to accept it at the value of an ounce through legal compulsion. So debasement usually goes hand in hand with these restrictions. You can't discount—you have to treat the coin based on its face value, not based on its silver content.
- 1.45 But it's predictable what happens when these laws are passed. You'd be a fool to pay in a full-weight coin when you're not allowed to get any more for it than a half-weight coin. So the full-weight coins disappear. People melt them down, or they export them to where they get the full value, and only the debased coins remain. This phenomenon is known as Gresham's law, the idea that if you have two coins that have a legal equivalence enforced by the law, the bad money is going to drive out the good. That is, the undervalued money is the one people are going to spend because they'd be foolish to spend the higher-valued money. The higher-valued money is going to disappear.
- 1.46 It's an extreme case of a price control that causes a shortage. You put a price control on the high-valued coin, saying it can't sell for any more than the low-valued coin; you're not going to find it. So it's a revenue motive that explains why the state got involved in coinage, at least judging by their practices.
- 1.47 If you look at the metallic content of medieval coins, it's on a steady downward path. Silver is being filched out of silver coins. Silver, especially, is debased because that's the local currency. That's small-value transactions. The Spanish mint issued silver coins, which it debased because it could compel people to use those coins locally. It didn't debase the gold coins because it had a market throughout Europe for those coins, so it wouldn't have paid to debase them. But if you can force people to accept the low-valued coins, it pays to debase.
- 1.48 That's the common path for a dozen different European mints, royal mints, monarch mints, until really the first mints that stopped debasing. Around 1700, the British mint decided to stop debasing, after centuries of citizens pleading with it, "Please stop debasing the coins."
- 1.49 I want to talk about an episode in private coinage that's not very well known. I want to talk about it because I published an article about it a few years ago. The article was

entitled "The Private Mint in Economics," and the title I borrowed from the title of a famous, among economists, paper by the Nobel Laureate Ronald Coase. His paper was entitled "The Lighthouse in Economics." He said that the lighthouse is an example of something which economists take for granted has to be provided by the government because they can't imagine how you could get users to pay for a lighthouse. The ship is just sailing by the lighthouse. How can you charge them? He showed that, in fact, most lighthouses were by harbors. And if you docked at the harbor, they charged you a fee, which went to cover the cost of the lighthouse.

- 1.50** The mint turns out to be a similar thing. People have not been able to imagine how private incentives would provide a reliable gold coinage. But in fact, they did. So some famous economists have made some pretty silly statements about how it's not feasible to have private coinage, but it's entirely an armchair argument. They didn't look at the history of private mints. It's actually scandalous that they would make these statements without actually looking at the history of it, including Carl Menger, whom I mentioned earlier. Surprisingly, he says that it's really not in the interest of anybody in the private economy to produce coins.
- 1.51** But clearly it is. We have examples of private mints, and particularly the examples I want to emphasize late in the game in the United States in the 19th century. There were defenders of private mints. There was the economist and sociologist Herbert Spencer, who was very much a fan of free markets, who said there's no reason why free markets can't provide coins. But his was an armchair argument, too. He never looked at the history of private mints. So I think it's informative to look at the history.
- 1.52** In the United States, there are three gold rushes. The one in California, of course, is well known. But before that, there was a gold rush in the Southern Appalachians. In Dahlonega, Georgia, the federal government eventually established a mint, and now it's a tourist trap. You can still go there and pan for gold. There was a later gold rush in Colorado, around Pikes Peak. In each of these three cases, gold is coming out of the ground, and it's a long way from the government mint in Philadelphia. So what do you do with the gold that's coming out? How do you get it into a form where it's easy to spend?
- 1.53** Well, you could ship it to the mint in Philadelphia and then wait weeks until your coins came back, that is, ship the gold dust, and they will turn it into coins for you and ship it back to you. Remarkably, the U.S. mint did not make a profit at this; they didn't charge anything. It was actually subsidized by taxpayers. But to the miner, there's the expense of shipping, there's the expense of insurance, and there's having to wait to get the coins back. So that's not very attractive.
- 1.54** Alternatively, there are people locally who will buy the gold dust for you, assayers. They will evaluate how pure it is, and they will pay you accordingly, and then they will ship it to the mint in Philadelphia. But they're going to pay you less than the intrinsic value of

the gold, that is, the value that the gold would have if you were at the mint in Philadelphia, because they've got to cover the costs of shipping and insurance.

1.55 There's a third alternative, which is when a private mint master says, "I'll bring the mint to you. I'll set up a mint near the gold mines. You bring the gold to me, and I'll turn it into coins. I'll turn it into my own coins, coins with my stamp on it." So, say you bring me gold dust, which would be worth, my numerical example is worth \$102.50 in coins. I will pay you \$100 in coins for that gold dust, and I'll keep the \$2.50.

1.56 Now, why is that attractive to you as a gold miner? Because you can't get \$100 worth of goods for it when it's in the form of gold dust. People don't trust it. They don't know how pure it is. So they might pay you—it's based on newspaper accounts—they might pay you \$96. They're going to discount it to the lowest quality that's around because otherwise, they're giving you an incentive to give them the worst quality. So, it's worth \$96 if you try to trade it locally, but the private gold mint will give you \$100 in their own coins.

1.57 Do you take that offer? You do if the \$100 in locally-produced coins actually spends at the same value as \$100. That is, \$100 official mint dollars. In the case of successful private mints, it did. So, here's a business that can make a profit without debasing the coinage. The profit comes from charging for the service of converting gold dust into coins because coins are more valuable because their quality is certified and their weight is certified. So coins are easier to spend because nobody has to weigh them, nobody has to test them.

1.58 It's like the operation in the old days of a grist mill. So farmers would bring their wheat to the mill. The mill would turn it into flour and keep some of the flour to pay for the price of the operation. So you bring in your wheat and come back a week later, and you get bags of flour. So these private gold mints operated that way.

1.59 The most important of them was a mint operated by the Bechtler family, Christopher Bechtler and later his son, Augustus Bechtler. They were in Western North Carolina, so they're mining gold that's coming from the Southern Appalachian mines. And they stamped them with the name Bechtler, so you knew which mint it came from, with the weight in grains of gold—that was the unit—and in fineness. The fineness of gold is measured in carats: 24 carats is 100% gold, 22 carats is 91.7% pure. So they were assuring people that this piece of coined gold has this much gold in it, this much pure gold in it.

1.60 The reason they stamped it with the different carats was that some of the gold that came to them from the miners was lower-carat gold—more silver was mixed into it naturally— and the other was higher quality gold. So rather than try to blend everything together, they sorted it out according to its purity, made uniform coins, and with lower quality gold, they made the coin a little heavier. With higher-quality gold, they didn't have to make it as heavy to have the same pure gold in all the coins. So

these coins have been tested. They were full gold weight, and they circulated at their face value. So by the official definition, they had \$5 of gold in them, so they circulated at the same value as an official \$5 gold coin.

- 1.61** We have a remarkable account of a visitor who came to the Bechtler Mint. In the 19th century, they didn't have the Internet. People didn't know what it was like in other countries except by reading books about it. So, travel logs were very popular, and this travel writer named Featherstonhaugh came to the Bechtler mint. He heard about this remarkable thing. He was British. He never heard of a private mint before. He went to see it.
- 1.62** When he sees Bechtler—he recalls this conversation in his book, he says, "I asked him, you're an expert in metals. You could make a coin that was the same weight as your gold coin but had less gold in it and palm it off on people as a pure gold coin or as a 23-carat when it was actually much less. Why don't you do that?" And Bechtler indignantly says, "Well, that would be dishonest." And then he gives a more practical reason. He says, "I would lose my business if I did that because people are always testing my coins. They're taking them to the Philadelphia Mint and asking them how pure this is. And if I get a bad test, that's the end of my business. People won't trust me anymore. If they don't trust me anymore, they're not going to bring me their gold. I'm out of business." So Bechtler discovered that honesty was the best policy if he wanted to stay in business. And the Bechtler Mint stayed in business for more than 20 years, basically until the gold ran out in that area.
- 1.63** By contrast, there was an earlier mint in Georgia run by an assayer named Templeton Reid. Reid produced some gold coins, but he was apparently incompetent at assaying because he valued the coins as though they were 99% gold when they were, in fact, only about 95% gold. Somebody else tested the coin. Somebody else was a competent assayer, tested his coins and reported it to the newspaper: "These coins are underweight." His business completely dried up. As far as we know, he only stayed in business for a few months. He didn't issue all that many coins, maybe \$7,000 worth. Whereas the Bechtlers stayed in business for 20 years and issued millions of dollars' worth of coins.
- 1.64** In California, there's a similar contrast. There are some mints which are set up in 1849 to coin the gold that's coming out of the hills. Big gold rush. They've brought their mining—once they heard there was a gold rush, they brought their mining, their minting equipment, sorry, from the East Coast out to California and set up shop. Some of them were incompetent or dishonest, but they were quickly found out, and they quickly went out of business. So the examples we have of mints that issued underweight coins, they only lasted one year. They're stamped 1849 or 1850, but there's never a second year.
- 1.65** The newspapers would report the results, and people are interested in this information, obviously. Newspapers would report on tests of the coins, and the ones

that were underweight, people would either discount them heavily or refuse to accept them. But the mints that were honest survived to mint another year, and the dishonest mints were replaced by honest mints that followed them.

- 1.66** Among the mints that were established in 1850 and afterward, I only found one that was dishonest. The others stayed in business for several years. What finally brought an end to this industry was that the federal government set up a mint in San Francisco to mint the gold coming out of the California hills, and they didn't charge any fees. The private mints had to charge fees. So they basically took all the business away. But not because the private market failed to provide honest coins. The coins were quite popular: something like \$36 million worth was willingly circulated.
- 1.67** Then, I mentioned there was a third gold rush in Colorado. The biggest mint there was called Clark, Gruber & Company. They produced about \$3 million in coins. They actually made their coins slightly overweight when they were first minted so that they could wear a little bit, because people actually carried gold coins in their pockets in those days, and still be full weight. And so they circulated at par in the area where they were issued, and people who took them back east were able to sell them at a premium because they were slightly overweight if they weren't worn.
- 1.68** What happened to the industry? To end it, finally, the federal government passed a law against it. Private coinage was outlawed during the Civil War. If you ask why you would outlaw private coinage during the Civil War, the answer is: the Union government was issuing its own currency called greenbacks, and they didn't want any competition. They didn't want people to refuse greenbacks because they preferred gold coins. The Bechtler coins, by the way, continued to circulate during the Civil War because they were more trustworthy than greenbacks or Confederate currency. So they continued to be popular.
- 1.69** Now, one of the economists who argued that this was not conceivable was Henry Sidgwick. He was an economics professor at Cambridge. He was the author of "The Principles of [Political Economy]," which was a standard textbook. And he said that there's a divergence between private interest and public interest in coining. It's in the interest of the community, he says, that the coins should be hard to imitate, should be hard to tamper with, tamper-evident, and don't wear easily. But somebody who's bringing their gold to a mint doesn't have any interest in making sure of those things. They just want to pass it on. They don't want to keep it. Therefore, they don't care about the quality. Therefore, we can't rely on self-interest to get us good coins. It's false. He seems to think that nobody will ever refuse a coin. And if that were true, then yes, the quality would deteriorate.
- 1.70** But people did in California refuse bad coins or discount them heavily. So if you're going to a mint with your gold dust, you don't want to go to a mint that doesn't have a good reputation. You have to go to a mint with a good reputation if you want to be able to spend the coins easily. It is in your private interest to secure excellence in hard to

imitate, hard to tamper, doesn't wear out easily. So these incentives are brought to bear on the individual who's ordering the coins. There's not a problem of divergence between private and social interest.

- 1.71** I mentioned Charles Goodhart earlier as a Chartalist. On private coinage, Goodhart wrote that the mint operator is always going to be tempted to debase the coinage because then he gets a higher return. Well, look, it's a higher return until you get found out. And if you get found out quickly, it's not worth it. It's better to stay in business. And if you want to stay in business, you produce honest coins.
- 1.72** And that's what happened. I mean, mint masters found out that honesty was the best policy. You've got to resist the temptation if you want to stay in business. So, private mints did produce high-quality and full-weight gold coins. By doing so, they stayed in business. Mints, which either through dishonesty or incompetence produced low-quality coins or low-weight coins, underweight coins, went out of business.
- 1.73** So this is not an example of a market that fails on account of what the economist George Akerlof called the "lemons problem." The lemons problem is where buyers aren't sure of the quality, so they stay away, and then the market collapses. It's called "lemons" because a low-quality used car, one that's going to give you trouble and break down, is a lemon. And he raised the question of whether there's a problem where people won't buy used cars because they're not sure of the quality, and so the market disappears. Even for people who have high-quality used cars, how can they convince people?
- 1.74** It turns out, of course, that's not a lemons market either. You can get the car inspected, or you can buy it from a reputable second-hand dealer like CarMax, where they stand behind the quality of the car. So, anyway, coins were not subject to this problem.
- 1.75** There's a similar story told by George Selgin in his book "Good Money," which talks about private copper mints. Copper coins were the small change of the day, but the Royal Mint in Great Britain was not producing enough of them. Private button makers had the technology, so they started producing copper coins. And businesses who paid their workers in copper coins would buy these coins from the button makers because the Royal Mint wasn't producing them. And they kept up the quality because it was in their interest to do so. So, it's not a lack of quality that explains the end of private mints; it's legal restrictions, legal restrictions by governments that wanted a monopoly and wanted a monopoly because it's a source of revenue.
- 1.76** What's the contemporary relevance of all this? Well, there's talk about having a central bank digital currency. I don't know if you've been following the debate over that. But the advocates of a central bank digital currency take it for granted that private financial institutions can't produce a trustworthy digital currency. But that's just an assumption. If you look at it, you find out people do trust private digital currency. They trust stablecoins, they trust Venmo, they trust WeChat Pay, Alipay, Cash App, and so on.

Really, you need to look for evidence of a market failing before you have a good case for the government intervening.

1.77 Let's go back to Europe, where they were struggling with government-issued debased coins. What did they do about it? Private enterprise came to the rescue here in the form of bankers who said that coins are not only bulky, but they're not trustworthy. You have to keep weighing them and testing them for purity. "Bring your coins to us. We're experts at weighing"—that's not too difficult, but testing them for purity—"bring us your debased silver coins and we'll give you an account balance in pure silver units based on how much pure silver was in the coins you brought us. So they created money of account, deposit, bank deposits, basically. But with the bank deposits, you didn't have to worry about debasement because the bank deposits were in pure silver units.

1.78 Historians have given a funny name to these units because they weren't embodied in any existing coins. The existing coins were all debased. These pure units were called ghost monies. They were unembodied in any existing coin. But it's not that mysterious. It's just a pure silver unit as opposed to the variable silver content of actually existing coins.

1.79 When merchants went to big trade fairs—that was a feature of trade in the Middle Ages—they would make sure that their banker was coming along so that they could make transactions without having to bring bags of coins and take home bags of coins. They would make a lot of transactions by transferring, basically writing checks, transferring account balances on the books of the banker. The use of bank-issued money, I'll say more about in the third lecture.

1.80 So, that's the case for believing that private enterprise has, in fact, produced trustworthy money. On the face of it, there's no reason not to trust them to continue to produce trustworthy money. Of course, we now live in a world where the basic money has been monopolized. It's produced by central banks. It's not redeemable for gold and silver. But we still have private money in the form of bank deposits.

1.81 The debate now is over whether these bank deposits, in order to maintain their quality or maintain their value in terms of the basic money, have to be restricted by the state, or whether we can allow private parties to issue dollar-denominated claims without following the restrictions imposed by the state. So that's the debate over dollar stablecoins. They are a privately issued dollar-denominated money.

1.82 The view that they are untrustworthy, well, there have been some hiccups, but the ones that survive are the ones that are well-managed to maintain their dollar value. So recapitulation of the private coinage story. On the face of it, I don't see a reason not to think that the survival of the fittest will serve us well here.

Lecture 2: The Gold Standard

At a glance — The gold standard and its properties.

- 2.1** Welcome back. In this lecture, I want to talk about the properties of a gold standard. In the first lecture, we talked about the evolution of gold over other commodity monies. The standard questions to ask about a gold standard: What is it that determines the quantity of money in the system? That's probably more of interest to economists and analysts. But what's of interest to members of the public, if they're considering whether a gold standard is a good idea, is, what determines the purchasing power of the money and its behavior over time? Is there going to be inflation? Is there going to be deflation? Those questions.
- 2.2** Now, I've got a whiteboard here because I may have to draw a supply and demand diagram. That's going to get a little technical, but nothing too obscure, I hope. So again, I want to start with the definition of what we're talking about. When we're talking about a gold standard, well, if you look up in The [Concise] Encyclopedia of Economics what a gold standard is, you will sometimes find a definition that says "the gold standard was an arrangement under which central banks did the following." And that's not a general definition. That's a description of what the so-called classical gold standard looked like in the late 19th century.
- 2.3** But a gold standard, more generically, doesn't require a central bank. It's simply a system in which gold defines the monetary unit. So, the monetary unit, economists sometimes call the unit of account, the unit that appears on price tags and in account books, things are in dollars, the dollar is a unit of account. In a gold standard, it might be—the defining unit is just an ounce of gold. Then that's very simple. The purchasing power of an ounce of gold is how many goods you can buy with an ounce of gold. If you have a piece of money that says 10 ounces of gold, it has the purchasing power of 10 ounces of gold.
- 2.4** Often, though, under the gold standard, the monetary unit had a particular national name. In the U.S., it was the dollar. But a dollar meant so many ounces of gold or so many grams of gold. Actually, the units gold was usually measured in were grains. But today we're familiar with the troy ounce. That's how we measure the price of gold.
- 2.5** So, under the gold standard, the dollar was defined as such-and-such a fraction of an ounce of gold. If you flip it to ask how many dollars per ounce of gold, after the Coinage Act of 1834, it was \$20.67 per ounce of gold. But that meant that \$1 was defined as basically one-twentieth, or one over 20.67 ounces of gold. So the dollar was just a name for a certain mass of gold. And likewise, other units like the pound or the franc or the mark, those were weights of gold.
- 2.6** Gold plays another role because it's the defining money. Gold serves as the medium of redemption or the thing that's in bank vaults to redeem, to pin down, the value of bank

deposits and banknotes and other forms of money that are claims to gold. Gold defines the unit of account, and it defines the medium of redemption.

- 2.7** Now, there are different ways to define the gold unit. The most common, and the one that emerges because people want coins rather than bullion, is a gold coin standard. A gold coin standard means that, if you have a \$10 bill, you get a \$10 gold coin when you go to redeem it at the bank, as opposed to a bullion standard. A bullion standard was a proposal by some economists, in particular David Ricardo, in the early 19th century, who wanted to make it cheaper for the government to be on a gold standard, by saying, "Well, you don't have to issue goldcoins. You can pin down the value of the money just by making it redeemable for big bars of gold in international trade. But then we can take away gold coins from the public and substitute something cheaper."
- 2.8** Well, but if the public would rather have gold coins and something redeemable for gold coins in large quantities only, that's not an innocent or innocuous substitution. A gold bullion standard is one step away from a gold coin standard, where people can actually test the commitment to redeem for gold.
- 2.9** One step further away is what's called a gold exchange standard. And during some historical periods, governments would issue money under the Bretton Woods system, 1946 to 1971, where the pound sterling was redeemable for dollars, and dollars were redeemable for gold bullion. So you couldn't go to the Bank of England and get gold for your Bank of England notes. You could only get—I'm not sure what you could get, but the Bank of England could get gold. So that was more indirect and less securely founded.
- 2.10** You can also distinguish among gold standard systems based on who's issuing the money that's redeemable for gold. So, in the system I'm going to describe in the third lecture, commercial banks issue the ordinary money, bank accounts and currency. But historically, you've had systems where the government monopolized the issue of currency. So that was the case in the United States after the 1930s; commercial banks continued to issue banknotes up until the 1930s when the right was taken away. And that changes the rules of the game. At the same time, the public in the 1930s in the U.S. was denied the right to redeem. They had to turn in all their gold coins in exchange for Federal Reserve notes. So that's moving it one more step away from being securely pinned and that commitment being enforceable by anybody who goes into a bank.
- 2.11** Now, the institutions that issue claims to gold, banknotes, and checkable deposits may be free to determine how much gold they need in the vault to meet their redemption obligations. So you can have a gold standard without any reserve requirements imposed by law. Or you can impose a reserve requirement. Governments have imposed reserve requirements, sometimes on themselves. So the Federal Reserve had a gold cover requirement for some years.

- 2.12** But there are some advocates of a gold standard who insist it's not a real gold standard unless bank-issued money is backed 100% by gold in the vault. Banks are not really banks; they're just warehouses for gold. And they sometimes call that a real gold standard or a genuine gold standard. I want to say that's not true. It's not unreal, if banks meet all the obligations to redeem without having 100% reserves; that is, if they have less than 100% reserves, but hold enough to meet all the redemption demands they actually face. And we'll come back to that in the next lecture. But there's nothing ungentle about a gold standard where there's money which is only fractionally backed by gold, which was, of course, the most common arrangement historically.
- 2.13** Now, to talk about the supply and demand for money, we have a units problem. There's a subtlety here, which is, when we talk about the demand to hold money, we're talking about a stock—that is, so many units at a point in time. I want to have so many dollars in my bank account on average. That's not per year. That's at a point in time on average.
- 2.14** But in a gold standard, gold is being supplied by gold mines, and gold mines produce a flow of gold, so many ounces per year. So we have to take account of both things: the stock supply and demand, and the flow of supply and demand. Just for reference, currently, we're not on a gold standard anywhere in the world, but the World Gold Council keeps track of these things. Their estimate of the total stock of gold above ground was about seven billion ounces. The value of that stock— well, the price of gold, I didn't check this morning, but let's say it's \$4,500— that stock is worth \$31 trillion. So the stock of gold above ground is worth about \$31 trillion, at a point in time.
- 2.15** The flow is how many ounces of gold the mines are producing each year. Last year, their estimate was 3,661 tons. If you divide that by the end of the year stock measured in tons, you get that the stock of gold grew about 1.7% last year.
- 2.16** This is a characteristic feature of gold, that the stock is large in relation to the flow. Or to put it the other way around, the flow is a small percentage of the stock, and it's generally in the neighborhood of 1.5% to, during years of a gold rush, maybe 5% per year, but that's unusual. It's usually in the neighborhood of 2% a year. But as I'm going to argue, that 2% is not constant. It's going to vary depending on supply and demand; in particular, if the value of gold is going up, that's going to increase the flow. So, the flow supply will respond and thereby bring back down the value of gold if it's unusually high. I'm getting ahead of myself.
- 2.17** All right. People want to hold a certain amount of monetary gold in order to accomplish their transactions. We can draw a demand curve where we plot the number of ounces people want— on the horizontal axis here, I have ounces of gold— against the purchasing power of gold. By purchasing power of gold, I mean, how many bundles of goods do you get for an ounce of gold? In the U.S., we measure the consumer price index, which is the same question inverted: how many dollars does it take to buy a fixed basket of goods? Well, under a gold standard, you ask, how many ounces of gold

does it take to buy a fixed basket of goods? And if you invert that, you get, how many baskets of goods do you get per ounce of gold? So that's what I'm calling the purchasing power of gold.

- 2.18** Now, the higher the purchasing power of gold, the fewer ounces you need to accomplish any given planned volume of transactions. If gold buys twice as much, you need half as many ounces. If we plot out the implications of that, we get this kind of curve, and I'm going to call it the demand curve for monetary gold. At a high purchasing power of gold, fewer ounces are demanded. If you cut the purchasing power in half, you need twice as many. If I drew it right, it should look that way. That's the demand curve plotted against the purchasing power of gold. In some money and banking textbooks, they plot the demand for money against the interest rate, but that's just a confusion. That's a mistake. That's like plotting the demand for orange juice against the price of grapefruit juice.
- 2.19** Now, we need a supply curve to determine the equilibrium quantity of gold held by transactors and the purchasing power of gold. The simplest case is a case we're talking about just a small economy, like, say, a city. For any city, they're not determining the purchasing power of gold. Their demand is a trivial share. It's a negligible share of the world demand. To them, they pay whatever the world price is to get gold.
- 2.20** In this limiting case, we can draw the supply curve as flat. That's the supply curve for a small open economy where, by selling more goods, they can bring in more gold. But if they're offering even a slightly higher value for gold, they're selling goods a little bit cheaper than anywhere else, all the gold they want will flow in. So as I said, that's a limiting case. If we want to talk about the supply of gold for the world as a whole, we want an upward-sloping supply curve; that is, one where, at a higher purchasing power of gold, you get more ounces of gold.
- 2.21** Now, where are you going to get the gold from? Remember, this is at a point in time. So the answer is not from the mines; the mines don't instantly produce gold. But there is non-monetary gold sitting around, ready to be converted to monetary gold. If we assume that the mints are open, as the purchasing power, we consider a higher purchasing power of gold, there will be people who say, "Well, at a high purchasing power of gold, I don't really want to have gold picture frames anymore, gold candlesticks, gold jewelry. It's more expensive now. I'm going to monetize that." In the trade, this is called recycling gold, melting down gold objects to get the gold out of it.
- 2.22** I should make this curve—it gets steeper because you run into diminishing returns of what people are willing to melt down. And eventually, you get to the total stock of gold above ground, which is part monetary and part non-monetary. So eventually, you're going to run out of non-monetary gold to melt down. But there is then some response in this stock supply of gold to a higher purchasing power, which means that there is a point at which the quantity demanded of monetary gold equals the quantity supplied

by people who have the choice of having monetary gold or jewelry, non-monetary gold. And that's the equilibrium in the market for monetary gold.

- 2.23** Now, this is a static picture. This is a point in time. As the gold mines produce more, the stock supply curve is going to shift to the right. If there were no growth in the demand for gold, then you'd have a falling purchasing power of gold. But in a growing economy, you also have an increase in the output of goods and services. People's real incomes are rising, population is growing. That's going to shift the demand curve out. So it's a question of which is moving faster, as to whether the purchasing power of gold is rising or falling over time.
- 2.24** I'm going to argue that there's a tendency for them to move at the same rate, so that you get a relatively stable purchasing power of gold over time. Not perfectly stable. We can look at the historical record, and what we see is that the purchasing power of gold is pretty stable. You don't have inflation or deflation of more than 2% a year, very often. But more than that, it's not just a lucky accident that they move at the same speed. There are economic forces that will make the supply adjust. If demand is growing faster, it'll make supply grow faster.
- 2.25** I don't want to get ahead of myself too far. That's the market for monetary gold stocks. We also need a flow market where we have gold mining. Flow market is ounces of gold per year. How much are the mines producing? Let's continue over this purchasing power of gold. Here you've got an upward-sloping supply curve. It doesn't have any particular shape. I'll put an F there for the flow supply of gold.
- 2.26** What is that telling us? At a lower purchasing power of gold, mines are not going to produce so much per year. They're only going to get the gold that's easiest to come by. At a higher purchasing power of gold, they're going to dig a little deeper, and you're going to get a higher supply per year, a flow out of the mines per year.
- 2.27** What's the flow demand for gold? The flow demand for gold is the industrial demand for gold. Demand for the flow of gold. That's gold that goes into circuit boards, gold that goes into teeth, gold that goes into uses that don't become part of the stock of non-monetary gold, but which are, in a sense, used up. Those kind of uses are smaller than the uses that are like jewelry. But when they're big enough to take up all the gold that's coming out of the mines—and that's what this equilibrium indicates, every ounce being mined is being used by industrial users— then you have a stationary economy where there's no accumulation of gold. The gold stock is not growing.
- 2.28** The fact that, typically, the gold stock is growing, tells you that usually you're not at this kind of equilibrium. You're usually at one where the industrial demands are smaller, and so the quantity being taken up is less than the quantity being produced.
- 2.29** All right, so if we put the stock and assume a stationary equilibrium, just for reference, and put these two together— let me eliminate the clutter here— we've got the same purchasing power of gold in both markets because you can convert gold coming out of

the ground into coins, and you can go the other way. You can melt down coins and turn them into objects, non-monetary objects, if that's what it pays to do. So, there's an arbitrage that keeps the purchasing power of gold the same in the two dimensions.

2.30 Now, the purpose of drawing supply and demand curves is always not just to admire the equilibrium point, but to ask what happens if there's a change in supply or a change in demand. Here, the dynamics are a little more complicated than an ordinary supply and demand story because we've got both a flow dimension and a demand dimension. If you're talking about the market for apples, and you ask what happens if apple juice becomes more popular, shift out in the demand curve, price goes up, more apples are grown in equilibrium, bigger quantities supplied, and that's the end of the story. An equilibrium at a higher price.

2.31 Here we've got a couple of things going on. So, we've got a stock dimension we have to worry about. So here's the thought experiment. There's an increase in the demand to hold monetary gold. What would be an example of that? Suppose a country that wasn't on the gold standard was on a silver standard, say Germany in 1870, decides to join the gold standard. That's a pretty big increase in the demand for monetary gold because Germany is going to sell its silver coins, buy gold, mint gold coins.

2.32 So the initial impact of this increase in the demand for monetary gold, with this given stock supply curve for monetary gold, is going to be to push up the purchasing power of gold. We get more monetary gold by converting some of the stock, but only at a higher purchasing power of gold. The value of gold against goods and services has gone up. Or to put it another way, the prices of goods and services priced in gold is going to go down.

2.33 But that's not the end of the story because we haven't taken account of what happens in gold mining. So, at this higher purchasing power of gold, we now have a difference between the quantity supplied and the quantity demanded for industrial purposes. And this difference—how am I labeling it? —is a flow of gold into the mints. You're producing more gold than industrial users are taking up. You can always take the extra to the mint and get it coined. That's going to be an increase in the flow supply—sorry, an increase from the flow market that's going to increase the stock of gold gradually. It's going to accumulate as additional coins.

2.34 To show that, we have to go back to the stock market, and the supply curve is going to start moving to the right through the accumulation of additional coins. As the supply curve moves to the right, this gap starts to shrink, but it's still positive, so it's still growing. When is this going to come to an end? It's going to come to an end when we get back to the initial purchasing power of gold. Because only then is there no excess flow supply going into the mints.

2.35 Here's our final equilibrium, call this case 0 and this case 1. We're back at the original purchasing power of gold. We went up in response to the increase in demand, but the

increase in demand was eventually satisfied entirely from the gold mines, from an additional supply coming onto the market. We went up, we came back down, and there's no change in the purchasing power of gold when we reach the new equilibrium.

- 2.36** Now, this may take 20 years, but still, it's something. It means that the purchasing power of gold is pretty predictable at 20-year intervals. And because people anticipate, it doesn't actually move all that much in the short run. If people can anticipate that the purchasing power of gold is moving, then they will price goods so as to move it along faster.
- 2.37** But this is the price-level stability that we've come to expect from a gold standard. This illustrates the main argument people make on behalf of a gold standard, that it pins down the value of the monetary unit, not because there's a lucky accident where the supply and the demand are shifting at the same speed, but because when demand gets ahead of supply, it triggers additional supply. The supply of gold responds in a way that dampens movements in the purchasing power of gold. So you get long-run purchasing power stability, or if you measure it in terms of the price index, price-level stability.
- 2.38** Now, it would complicate things to show you a growing economy. But basically, in a growing economy, you're continually up here where there's a flow of metal into the mints, and that's sufficient to meet the continual growth in demand. This is just a one-shot increase in demand. Again, you have stability in the purchasing power of gold in a growing economy.
- 2.39** If that's true, then when the world economy started to grow faster, or at least the economies of areas that were on a gold standard started to grow faster because of the Industrial Revolution, you should not expect a permanent depression in prices and a permanent increase in the purchasing power of gold, but you should expect more output from gold mines. You should expect faster growth in the stock of gold in response to the faster growth in the demand for gold. That is, in fact, what you find.
- 2.40** So, if you divide the long 19th century, and I happened upon this data that was in two different sources, they almost line up. One is figures on the flow supply of gold from an economist named Hugh Rockoff, and the other is data on the real growth of the world economy from Angus Maddison. Between the early 19th century, say 1810 and 1870, you had relatively slow growth. It was about 2% a year in real output. And the growth in the stock of monetary gold was also about 2%. It was slightly under.
- 2.41** If you switch to the second half of the long 19th century, and by that I mean from 1870 to the First World War, you get faster growth in the real output of the economy, and therefore, faster growth in money demand. Now it's running a little above 2.5%, so it's gone from 2% to about 2.6%. If there hadn't been any response from the gold mining industry, you'd have had falling prices. But gold mining output, in response to the higher purchasing power of gold, cranked up bigger supply, responded to the higher

purchasing power of gold by more flow supply. And the supply of gold, the monetary stock of gold, grew about 2.5% a year, whereas before it had been slightly below 2%.

- 2.42** So, it's not a coincidence that the real growth in the economy and the growth in the monetary gold stock coincided. It's ensured by this mechanism where, otherwise, the purchasing power of gold would keep rising and that would create incentives to dig deeper and find additional sources of gold, which is what happened historically.
- 2.43** We just did a shift in demand to the right. We could do a shift in demand to the left, and that would depress the purchasing power of gold. That would slow down mining activity until the drop in demand was entirely satisfied by a reduction in the stock of gold from the mines slowing down and industrial users using it up.
- 2.44** We could shift the stock supply curve one way or the other, but the result is going to be the same, that we get back to the same purchasing power of gold. This is the price-level stabilizing property of a gold standard. Changes in the purchasing power of gold, when they're a result of shifts in the stock supply or demand, are self-reversing. The purchasing power of gold returns to where it started.
- 2.45** Now, that's not the only kind of exercise we could do. You go back to the beginning. We can think about changes in the flow market. The most famous kind of change in the flow market is a gold rush. The biggest gold rush is the California gold rush. How do we picture that? Well, at any purchasing power of gold, there are now going to be more ounces of gold coming out of the ground. That's a shift to the right in the flow supply curve. Now the new flow equilibrium is at a lower purchasing power of gold.
- 2.46** How do we get there? Well, at the initial purchasing power of gold, we again have an excess supply compared to the industrial demand. That difference is going to go into the mints. That's going to push the stock supply curve to the right. When is that going to stop? Well, again, the answer is, when we eliminate this gap. This is the new long-run equilibrium. Let's see, it's here. Now that's the end of the story because we're in equilibrium in both markets. So now we haven't gone back to the original purchasing power of gold.
- 2.47** It's not that nothing can ever change the equilibrium purchasing power of gold. A permanent change in the flow supply can do it. Then the question of how stable the purchasing power of gold is becomes a question of how often there are these shocks to supply and how big they are. But not every new gold rush is a shock to the system.
- 2.48** Sometimes it's catching up. Some of the gold rushes, like the Yukon, took place when the purchasing power of gold was unusually high. Because it was high, you not only got existing mines digging a little deeper, you got more prospectors going out looking for gold. And so, the Yukon discovery actually helped push the purchasing power of gold back down to its long-run trend. Colorado in the 1890s, likewise.

- 2.49** But the California discovery was not stimulated by a high purchasing power of gold. It was just a lucky accident. How big an impact did it have? Economists trying to estimate that did some of the first work in index numbers. They're trying to come up with an average measure of the purchasing power of gold or the price index.
- 2.50** And an early prediction was, it's going to reduce the purchasing power of gold by half. Well, it turned out not to be that big. It turned out, it reduced it by about 21% over 18 years. So this was a gradual process of the new gold spreading around the world. And if you average out what inflation rate that means, it was an inflation rate of 1.3 %. So by today's standards, that's price stability. The Fed considers that price stability. It was minor inflation. And then the trend reversed where there wasn't more gold; the California discovery was pretty much depleted, and the world economy was growing faster than the output of gold, and the price level went back down.
- 2.51** But I wanted to give you a concrete picture of how the gold standard responds to changes and how, normally, it's self-stabilizing. The events in which the purchasing power of gold is changed were historically small. If you compare historical gold standards to what we have today, fiat money, which we'll talk about in Lecture 4, what do you find by comparison? You find lower inflation, and you find lower expected inflation, and you find more certainty about where the price level is going to be in the future, because it has this mean-reverting tendency, when the change in the purchasing power of gold is due to changes in stock demand or stock supply.
- 2.52** And that has an important impact, which is that people are willing to issue long-term bonds under a gold standard that they aren't willing to issue today. Very few firms, if any, will issue a 50-year bond today because there won't be any buyers. And there won't be any buyers because who knows what a dollar is going to be worth in 50 years? There's so much uncertainty, it's hard to value that bond. So, 20-year bonds are about the limit to what private firms issue. The federal government insists on issuing 30-year bonds.
- 2.53** But under the gold standard, you had 50-year bonds, you had railroads issuing 100-year bonds. People were, in retrospect, a little too sanguine about the existence of the gold standard for the next 100 years, but they were confident in what the purchasing power of gold was going to be in 100 years. Because you had many countries on the gold standard and more joining as it did well, you got the benefit of having a single global currency network. That reduces transaction costs across borders. It reduces devaluation risk across borders.
- 2.54** And one last feature that I hope to get to is, having money that is outside the control of the tax and spending authorities, outside the control of the national government, means that you have more fiscal discipline under a gold standard than you have when government expenditures can be financed by printing money. Governments can't print gold. That's a virtue.

- 2.55** The evidence on inflation is pretty stark. Two economists who were writing for the Federal Reserve Bank of Minneapolis, Rolnick and Weber, did a study in which they compared inflation under commodity standards to inflation under fiat standards. And the way they did it was by finding as many countries as they could that had had both for a long enough period of time. Then they compared the average inflation rate in those countries under gold and silver standards to the average inflation rate in those same countries, country by country, under fiat money.
- 2.56** What they found was not only a difference in the averages, but that every country in the sample had higher inflation under fiat money than under the gold standard. There were different versions of their paper as it went through working papers. The sample of countries changed, and therefore, the average inflation rates changed. But in the version that was published, the average inflation rate under commodity standards was 1.75%, under fiat standards, 9.17%. So there's more than seven percentage points of difference.
- 2.57** In the first version of the paper, under commodity standards, it was -0.5%, and under fiat standards, it was 6.5%. So again, a seven percentage point difference. And that's even leaving out the German hyperinflation, which was in their sample, because you throw in a million percent inflation rate, it kind of skews the average. So toss that one out, and you still get this dramatic difference in inflation rates. It's true in every country. The inflation rate under commodity standards is close to 0%. Under fiat money, it differs dramatically from country to country. In some countries, you have single-digit average inflation rates, but in others, you have double-digit average inflation rates.
- 2.58** Now, that's pretty much understood that, in practice, inflation has been higher under fiat standards than under gold standards. But defenders of fiat money will say, well, central banks had to learn what they were doing, but now they've got it under control. Now everybody's targeting 2% inflation, so that's not a problem anymore. Well, they may be targeting 2% inflation, but a couple of years ago, they did not deliver 2% inflation. We had 9% inflation in 2022. Europe had 10% inflation. Great Britain had 11% inflation. Lots of countries had double-digit inflation. And of course, in the worst cases, you get triple-digit inflation. I'll come back to that when I talk about experience under fiat standards.
- 2.59** What's less commonly appreciated is the greater predictability of the price level under a commodity standard, under a gold standard. I did a paper with George Selgin and Bill Lastrapes, which included an attempt to measure price level uncertainty by asking, if you projected the price level, how big would your misses be using a six-year rolling deviation of actual prices from the trend? And it was much lower under the gold standard than under fiat standards.
- 2.60** It has come back down since the so-called Great Moderation. So since we left behind double-digit inflation in the '70s and early '80s, average inflation rates in fiat standards

have come down. So I don't want to deny that. There has been some improvement. But we'll know that we're back to the predictability of the gold standard when we start seeing 50- and 100-year bonds again, and we haven't.

- 2.61** I also mentioned fiscal discipline. So, a country that wants to borrow under a gold standard, runs a fiscal deficit and therefore has to borrow, if they want to borrow on the world market, then they have to issue bonds denominated in gold. So they have to promise to repay in gold, something they cannot print. Then they have to be committed to getting the revenue to repay the bonds. They have to be running a surplus on all their other expenditures, so that they have enough gold to repay the bond holders. That's going to make it a country reluctant to run deficits and have to borrow on those terms.
- 2.62** In a fiat standard, even if a country, let's say Greece, doesn't print its own money, it's on the Euro standard, there is some prospect that if they get in over their heads, if they become overly indebted, they'll be able to call on the European central bank or the EU to help bail them out, because the EU can print money to do that. That creates a kind of moral hazard—that is, a temptation to issue more debt than you would issue if you knew you were going to have to bear it yourself.
- 2.63** And so, you've got Greece and Spain and Ireland and Portugal becoming heavily indebted, and there was a sovereign debt crisis in Europe around 2010. But a gold standard doesn't create that kind of moral hazard because there's no authority who can print gold to bail out an overly indebted country. The economist Thomas Sargent, in an interview he gave before he got the Nobel Prize in economics, made this point, that if a country wanted to be on a gold standard, that is, wanted to issue gold-denominated bonds, the way he put it was, they have to balance their budget in a present-value sense, meaning they have to be committed to surpluses to offset the deficits with interest.
- 2.64** So that's the case for a gold standard: better price-level stability, better price-level predictability, benefits of having an international money and fiscal discipline. What's the main objection to a gold standard? It's long been that it's costly to dig up gold and then put it in vaults. We can measure this cost either by the amount of resources devoted to gold mining, or we can measure it by looking at the stock of gold bars underground and figuring the other things they could have been used for: making jewelry, filling teeth, making circuits.
- 2.65** Now, you can't count both of those because that would be double-counting. One's a flow measure, and one's a stock measure. But either way, that's the objection. It's costly. Compared to what? Well, compared to, on the blackboard, a fiat standard that has the same inflation performance as a gold standard, but without having to dig up metal and put it in vaults.

- 2.66** A famous estimate of the resource cost of running a gold standard was made by Milton Friedman. Milton Friedman, although he was a free-market economist, was not a fan of the gold standard. Well, I could qualify that. He said he was a fan of a genuine gold standard without any government involvement. What he didn't want was a central bank on a gold standard. But what was his alternative? His alternative was a well-managed fiat money. And toward the end of his life, he became increasingly disabused of the idea that he was going to get a well-managed fiat money or experience a well-managed fiat money in his lifetime.
- 2.67** But in the '50s, he wrote a paper on the costs of a commodity standard, and his estimate was that it would take 2.5% of annual output just to mine the gold necessary to stay on a gold standard. I think this is a big overestimate. It exaggerates the cost of a gold standard by about 50-fold. Why do I say that? If you look under the hood at how Friedman arrived at his estimate, he said, "Well, we know a gold standard provides a stable price level. Let's ask how much the money supply has to grow to give you a stable price level in a growing economy, and then how much gold it takes to get the money supply to grow that much." That's a reasonable way of breaking it down.
- 2.68** First, how big is the stock of money compared to GDP? If you take a broad measure of the money stock, M2, when he estimated it, it was about 62.5% of national income, was the size of the money stock. That's important because we're going to talk about how much it has to grow percentage-wise, and we need to know what to multiply that percentage by.
- 2.69** So, how much does the money supply have to grow to give you price stability? He said, "If the real demand for money is growing at 4%," that was his estimate, because real output is growing at 3% a year. And that was the U.S. experience, although since 2000 the average growth rate has been 2%, not 3%, so we've had a slowdown. But 3% was a reasonable estimate. And then on top of that, he estimated that people were holding more money per dollar of income at a rate that was growing at 1% a year.
- 2.70** So, in terms I'll come back to, when we talk about fiat money, the velocity of money was falling at 1% a year. That implied that you needed 4% money growth, 3% to offset the growth in real output, and 1% to supply the increasing demand for money per unit of output, to give you price stability, so 4% money growth. How much gold growth do you need to get 4% money growth? He assumed— he needed to assume something about the ratio of the gold stock to the broader money stock; he assumed it was 100%.
- 2.71** Now, this is the crazy part of his estimate because he's talking about M2, the broadest measure of money, which is mostly savings accounts. There's no reason on Earth for a savings account to be backed 100% by gold. But historically, even checking accounts were not backed 100% by gold.
- 2.72** So, if we plug in a reasonable estimate of a reserve ratio, one that banks actually used historically, it's 2%. That would be the number I would offer. That's one-fiftieth of

100%. So the amount of gold you have to dig up is one-fiftieth of Friedman's estimate. It's five-hundredths of 1% of GDP, not 2.5% of GDP. And because it's that low, it's reasonable to ask, is the cost of, the resource cost of digging up the gold greater or less than the cost caused by the inflation we do experience under a fiat standard?

- 2.73** Now, Friedman, to his credit, wanted the U.S. to go off the gold standard and to sell off all its gold, to eliminate the opportunity cost of having the gold in the vaults. In practice, when we switched in 1971 to a pure fiat standard, Friedman thought the price of gold would fall on world markets because central banks would sell off their gold, and private individuals who had been holding gold as an inflation hedge would sell off their gold that wouldn't be needed anymore because we had a well-managed fiat standard. Yeah, that didn't happen.
- 2.74** In fact, central banks only sold off a little bit, and the public accumulated more than enough to offset what the central banks were selling. So the amount of gold in monetary form has actually gone up since 1971—that is, central banks' gold in their vaults, plus the public's holding of gold coins and gold bullion. And that means the demand for monetary gold is higher than ever. The purchasing power of gold is higher than ever.
- 2.75** When we left the gold standard in 1971, it was \$35 an ounce, or the London price was \$42 an ounce. It's now 100 times that. We're talking \$4,500. We're in that neighborhood. While the price level hasn't risen 100-fold, only risen about 20-fold, so the real price of gold is higher than ever, and that means gold mines are digging deeper than ever. We've increased the resource costs. So switching to a fiat standard isn't enough. Central banks have to sell off their gold, and the public has to not accumulate gold. But the public has been accumulating gold because fiat standards have not been reliable, because they want an inflation hedge.
- 2.76** And to top it off, central banks have started accumulating gold again. Quietly, since 2008, the amount of gold held by central banks has started climbing again. In particular, China has decided they don't want to hold so much in U.S. Treasury bonds. They're selling those and buying gold instead. The Reserve Bank of India is accumulating gold. And this is remarkable since we're not on a gold standard. Why are they doing that? Well, they think it's a better asset than holding so much dollars.
- 2.77** If we compare the five-hundredths of 1% of GDP, which was my estimate of the resource cost, to the cost of an inflation rate, what inflation rate, how low would inflation have to be under fiat money to make it net beneficial to be on fiat money? There are different estimates out there of the elasticity of demand for money with respect to inflation, or to put it another way, the burden imposed by inflation.
- 2.78** But a reasonable estimate is 2% inflation is the break-even point. If you can't keep inflation below 2%, you're better off on a gold standard that gives you 0% inflation. Yes, there are costs of digging up gold, but they're not as burdensome as the burden

imposed by 2% inflation. But of course, average inflation under fiat standards since 1971 has been well above 2%. In the U.S., it's been 4%; in other countries, it's been much higher.

- 2.79** Now, there's a practical calculation that people do very much like that in countries with high inflation. In countries with high inflation, people say to themselves, "My money is melting away in value. This is costing me. Maybe I should switch to a different money that doesn't melt away so fast." And the leading candidate around the world is the U.S. dollar, which is not perfect, but better than, let's say, the Venezuelan bolivar or the Argentine peso. People dollarize themselves when the cost of inflation is higher than the cost of going out and acquiring those dollars.
- 2.80** In defense of fiat standards, the economists who recognize that inflation is higher, but we've gotten more real stability, they say. Under fiat standards, central banks can run a countercyclical monetary policy and dampen business cycles, and we all benefit from that. Well, they are assuming a fact which is not evidence, namely that business cycles have been dampened.
- 2.81** In the paper I mentioned with Selgin and Lastrapes, we measured the deviation of annual real output. And since 1971, it's been about the same as it was before the First World War under the classical gold standard, which is surprising. I mean, the economy is much better diversified, so you would expect less aggregate fluctuation. But despite that, and despite having a central bank with the power to conduct a countercyclical policy, they haven't managed to do it. They haven't reduced the volatility in the real economy, so worse on inflation and no better on business cycle volatility, even though the economy is better diversified, even though the banking system is no longer as panic-prone as it used to be.
- 2.82** And yet, the gold standard is not so popular among my colleagues in the economics profession. There was a poll conducted by a group called the IGM Forum where they asked a panel of economists from leading universities, seven from Harvard, six from Yale, six from Stanford, etc., "If the U.S. replaced its discretionary monetary policy regime with a gold standard," I'm reading from the poll question, "with a gold standard, defining a dollar as a specific number of ounces of gold," that's a good way to define a gold standard, kudos for that, "the price stability and employment outcomes would be better for the average American. Agree or disagree?"
- 2.83** Everybody disagreed. Nobody thought it would be better for the average American. Now, if I had written the question, I would not have put in "employment outcomes would be better." There's no reason to think employment outcomes would be better, based on a different monetary regime. That depends on labor market conditions. The effect of monetary policy on the labor market is transitory. It's not permanent. So I wouldn't expect any permanent increase or decrease in employment.

- 2.84 But price stability is clearly better under the gold standard than it has been under fiat money, and yet nobody disagreed. And some of them gave ridiculous reasons for thinking the gold standard was a bad idea. In particular, the idea that if you price the consumer price index against an ounce of gold, you'll see that it's quite volatile. "The instability of the relative price of gold is too high," is the way one economist put it. But look, you're not looking at a gold standard when you make that comparison. You're looking at gold today, demonetized gold, gold that serves as a speculative hedge.
- 2.85 And so, yes, the price of gold today is volatile, but that's because it's been demonetized. If you want to know what the price of gold behaves like under a gold standard, for God's sake, look at the historical gold standard. Don't look at the price of gold today. I think there's a great lack of understanding of the properties of the gold standard when it was a gold standard, when it was a monetary standard.
- 2.86 I'm sad to say that some defenders of the gold standard today have also made some weak arguments in its defense, in particular, the idea that gold is a measuring rod that is stable in value. Well, value is not something to be measured. Value is subjective. What we have are prices. Prices don't measure value. They measure exchange rates, exchange ratios between money and goods. That's not measuring something. To the person who buys the good, it's worth more than the money. To the person who sells it, the money is worth more than the goods sold.
- 2.87 If by measuring rod you mean the pricing unit or the unit of account, okay, under a gold standard, gold is a measuring rod. Under a fiat standard, the dollar, the fiat dollar is a measuring rod. And the gold measuring rod historically has been more stable, sure, but not fixed. The purchasing power of gold was not constant historically, so let's not oversell it. Supply and demand determines the purchasing power of gold and works to keep it stable. But it's not like the number of inches in a foot, the exchange rate between gold and a basket of goods.
- 2.88 An even worse misunderstanding is to think that only gold is the true measure of inflation, that we should look at the price of gold if we want to measure inflation, not at the consumer price index. Well, what people care about is how much their income will buy when they go to the marketplace. Their income is in dollars. The prices they face are in dollars. So we would still need a measure of that, even if you don't want to call that the inflation rate. But the nominal price of gold is not a good proxy for the consumer price index.
- 2.89 So it's not helpful to say that gold is somehow an objective indicator of inflation. When it's demonetized, it's one good among many. It's not the only or even the primary indicator of inflation. All right, so whether we can re-institute a gold standard, I mean, technically we could, but politically, it's an uphill battle. We've got so many interests that are vested in the status quo that that's going to be difficult, but I will end there and take questions.

- 2.90** Currently, a one-ounce gold American Eagle is legal tender of US\$50. If government were to increase that, how would that impact the overall economics as it relates to the dollar?
- 2.91** When you say increase, you mean make it legal tender at the market price of gold? At \$4,000-some dollars?
- 2.92** Anywhere above \$50, up to the market.
- 2.93** So at \$50, you'd be a fool to spend your Gold Eagle, right? Because you can sell it to a coin dealer for a lot more than that. So it would have to be at the market price for people to want to spend it. But I don't think legal tender laws are as important as people sometimes think. There's an innocuous role that legal tender laws play, which is, right now we're on a dollar standard. People make debts denominated in dollars. And then the question arises when the debtor comes to pay, has he discharged the debt if he offers a certain kind of dollars to the creditor?
- 2.94** And a legal tender law innocuously sets a default. It says that if you're paid in Federal Reserve notes, you've been paid. You can't say, "I haven't been paid because I wanted to be paid in gold." Now, if you put that in the contract, you've overridden the default, and you can make contracts denominated in ounces of gold. It's not entirely clear whether the court will compel specific performance, but they will at the least compel the person who owes you, I don't know, 10 ounces of gold, to pay you the equivalent of 10 ounces of gold, whatever that's worth in dollars.
- 2.95** People accept payments that are not legal tender all the time. Checks are not legal tender. People are happy to be paid in checks all the time. Gas stations are free to put out a sign that says, "We don't take \$100 bills." \$100 bill is legal tender if you had a debt to the gas station, but you don't. You're making a spot transaction. And they can say in spot transactions, "We don't take \$100 bills," and that's perfectly legal.
- 2.96** Legal tender, where it has an impact is in debt contracts. And where it's pernicious is where something is declared legal tender, contrary to the intentions of the people who made the contract. So, under Roosevelt, people were worried about a devaluation. They made gold clause contracts that said, "You have to pay me back this many dollars or the number of ounces of gold currently equivalent to that many dollars." So if the dollar is devalued, you have to pay me more dollars. And the courts struck those down and said, "Dollar is legal tender. You can't index the contract to the dollar price of gold."
- 2.97** Not clear whether that is still the law or whether they would enforce specific performance of an index contract. But if you don't want to just go with the default, then you need to specify in your contract what the payment medium is going to be.

Lecture 3: Free Banking

At a glance — Free banking under a gold standard.

- 3.1 In the last lecture, we talked about the behavior of a gold standard. Gold in a gold standard is the underlying money, but it's not the money that people use every day. That is, payments were not typically made in gold coins by ordinary households and businesses. They use the services of banks to make payments.
- 3.2 What I want to talk about today is the role of banks in issuing money and what economic forces shape their behavior. Why do people use the services of banks instead of just using gold coins or warehouse claims to gold coins? And so, for a banking system, we need to ask what determines the amount of money that a banking system can issue, and how does that affect the purchasing power of money?
- 3.3 I want to deal with the case where banks are not restricted legally, other than [being] held to their contracts. So if a bank says, "We will redeem this deposit on demand, or we will redeem this banknote on demand," they are held to that contractual obligation. But otherwise, they're not restricted by reserve requirements about how many reserves they need to hold or capital requirements or branching restrictions or anything like that.
- 3.4 If there's no law that tells them how much gold they have to have in the vault, what is it that limits the amount of redeemable liabilities they practically can issue? Some critics of banking have thought that banks could unlimitedly issue banknotes, that this was a real problem, or that banks can create money out of thin air, costlessly, gratuitously.
- 3.5 We can turn the question over and ask: why would banks hold reserves if they're not legally required to? But this idea that there's no natural limit to the operation of banking, even under a system of redeemability for gold, has been the basis for the view that there need to be restrictions imposed on banks, either reserve requirements, the limit up to 100% reserve requirements, or a central bank that will counteract any tendency of the banking system to over-expand or over-contract.
- 3.6 And I want to say the banking system doesn't have that problem. An unrestricted banking system will behave in a way made regular by market forces. So, I don't like to actually talk about unregulated banks. I want to talk about banks regulated by market forces. So, unrestricted banks is maybe a better way of describing what I want to talk about.
- 3.7 Now, a bank is a financial institution, and the way to illustrate how a bank is different from other financial institutions is to look at its balance sheet. So, we draw a big T, and on one side we put assets, and on the other side, we put liabilities, plus the difference between assets and liabilities, which is net worth.

- 3.8** So, some early deposit takers were not actually banks in the sense that a bank is an institution that takes deposits and makes loans. But they were just deposit takers that stored the money. They were warehouses. So a bank took in \$200 worth of coins from customer Alice, who brought in \$10, and customer Bob, who brought in \$10, and others. Who brought in \$280— sorry, \$180. Under 100% reserves or warehousing, they kept all \$200 in coins. So this is a money warehouse.
- 3.9** Under fractional-reserve banking, the bank might only keep \$40 in coins and lend out the other \$160. So that's loans. Or they might buy commercial paper or securities, in other words, get some interest-earning assets on the balance sheet. Now, the coins in the vault, which are the reserves of the bank, available to pay people on demand, are only a fraction; they're less than 100% of the sight claims against the bank, that is, liabilities that can be redeemed on demand.
- 3.10** Now, how did Alice and Bob historically know whether the bank was keeping their coins in the vault, or lending some of them out? Well, pretty early in the Middle Ages, a convention arose that made it clear. If you wanted to bring your coins to the bank just for safekeeping, you'd put them in a sealed bag. They were big on wax seals in the Middle Ages. So you seal the bag; any tampering would be evident. You give it to the banker, say, "Store this for me. You're not to go into the bag." And it'd be evident if they had gone into the bag. And then you expect the very same bag back. So that's warehousing.
- 3.11** But bankers began to offer a different deal, which is, "If you bring us loose coins, not in a sealed bag, but loose coins, we promise to pay you back equivalent coins when you want them, but in the meantime, we won't charge you a storage fee, and in fact, we'll pay interest on your account because"—I don't know whether they said this part out loud, but—"we're going to be lending those funds out, earning some interest, and we don't want to share it with you, but under competition, the bank across the street is offering interest, we have to offer interest too. We're gonna pay you some of the interest we earn."
- 3.12** So, in a competitive system, banks could only keep as much for themselves as was required to cover their costs of operation, and they paid interest to their customers. So, that was the distinction: sealed bag warehousing, loose coins. It's a loan—you're making a loan to the bank, on those terms, and then the bank is lending the money out to some ultimate borrower. So the bank is an intermediary between the savers and the people who are putting the money to work, the investors who are borrowing from the bank.
- 3.13** So under what conditions is this gonna work? Well, Alice and Bob and the others have to be satisfied to get equivalent coins back. It can't be that they bring in a special, rare collector's coin and say to the bank, "Please store this," and then the bank lends that out, because the bank's not gonna be able to pay them back that particular coin. So one condition is that the coins that are being deposited are interchangeable or fungible,

and all you care about, the customers care about, is getting equivalent coins back. You don't care which \$100 bill you get from the ATM; you just want it to be a genuine \$100 bill.

- 3.14** So then the question is: if they've only got \$40 in coins, is that enough? Yes, any of the \$40 in coins can be paid to Alice and Bob, 'cause they're interchangeable with the coins they brought in, but is \$40 enough to meet all the requests that are actually gonna be made? How is it feasible?
- 3.15** The great monetary economist Ludwig von Mises had an analogy which I like. He said that if you're issuing 100 bread tokens, issuing 100 tokens, each one is good for one loaf of bread, you can't get by with a fractional reserve. You're going to need 100 loaves of bread because the tokens aren't good for anything except claiming the bread. You can't eat the tokens. The only use is to redeem for bread.
- 3.16** But the marvelous thing about claims on a bank is that they can do the job that the coins used to do, that is, you can make payments with a claim on a bank in the form of a deposit that you can write checks on, or in the form of a banknote. Suppose the \$10 is banknotes in circulation. Well, actually, the banknotes would have to be in the others because the bank doesn't know at any moment who's got the banknotes. It knows who's got the deposit balances, but it doesn't know where the banknotes are.
- 3.17** So banknotes and deposits can do the job of coins in making payments. And when that's true, when businesses will accept payment in banknotes and checks, the bank can be pretty sure that not every coin is going to be called for because Alice and Bob can make payments without coming to get coins.
- 3.18** Early on, banks had customers who just wanted safekeeping, and then they came and got the coins whenever they wanted to make a payment. But when they discovered that other people were willing to be paid in claims on the banker, that was a more convenient way of doing it, and now the bank sees that not every coin is being called for, so we can get through the day with a fractional reserve.
- 3.19** How big a fractional reserve do you need to meet all the claims? Well, that's a practical question. It depends on the actual payment practices, but that's the kind of practical statistical question that bankers are pretty good at answering. But it's gonna take experience to learn, and you can probably see some banks underestimate what reserves they need, other banks overestimate what reserves they need, but they'll figure it out.
- 3.20** Now, there is an occasion on which a bank isn't gonna get by with a fractional reserve, and that's when Alice and Bob and the others all decide to redeem on the same day. We call that a bank run. I'm not going to talk about bank runs in this lecture, I'm going to talk about them in a later lecture where I talk about how fragile banking is, really. But for now, take my word for it that historically, banks that were solvent, banks that were reputable, were not threatened by runs, did not have to worry about, or did not

worry about runs. They were unusual and limited to cases where customers had some good reason to suspect that the bank was already insolvent.

- 3.21** So why does the bank want to lend out \$160 [worth] of the coins? So they can earn interest, that's pretty obvious. Why do Alice and Bob wanna let the bank do that? That one's not quite as obvious, and some people have jumped to the conclusion that they must not have agreed to it, so the whole thing must be a fraud. But I've already suggested why they might agree to it, namely, the bank will say to them, "I'm not gonna charge you storage fees if you bring in loose coins that I can then turn around and lend. And in fact, I'm going to pay you some interest on your account." So it's entirely reasonable for some people, maybe not everybody, but for some people to say, "I like that deal because the reason I brought the money in was not to store it, but to get access to useful payment mechanisms, namely to be able to write checks, and to be able to use banknotes, because those are more convenient than carrying around coins."
- 3.22** So people who wanted payment services from banks were happy to accept this deal: no storage fees, interest on the account balance, with the knowledge that the bank isn't quite as secure, but as long as that risk is small enough relative to the benefits, they agreed to it. They didn't agree not to get paid back, but they agreed to take the risk.
- 3.23** As long as people are willing to agree to that, then there's a synergy between two things that banks do. One is to provide payment services—checkable accounts and banknotes—and the other is funneling money from depositors to the borrowers, that's intermediation. That's why banks do both things. It's cheaper to do both things together than to do them separately.
- 3.24** If you insisted on 100% reserves on all bank accounts, you could still have lending funded by bank accounts that weren't redeemable on demand, that had definite maturity dates, time deposits, certificates of deposit— that would be one kind of bank. And then you'd have money warehouses that provided checks on the other side—that would be a different institution or a different segregated set of accounts. But it's cheaper to do both things together in one institution: payment services and lending.
- 3.25** So is it a fraud? It's not a fraud if people would agree to it, and I've given a reason why people would agree to it. Well, did it say on the contract that we're holding a fractional reserve? On the face of a banknote, no, it didn't say that. What it did say is, "We'll pay the bearer on demand £10 or \$10," or whatever the gold unit was. It didn't say anything about how many coins we're keeping in the vault. If Alice and Bob are getting no storage fees and getting interest, that's a pretty clear signal that there's a fractional reserve, because how else would the bank be able to afford to pay them interest?
- 3.26** So, nothing in the banknote promised 100% reserves, and if you read a deposit contract, if you read one today, it doesn't promise 100% reserves. In fact, it says the relationship between you and the bank is, "You are lending us money. We're not a

warehouse. This is not a bailment,"—which is the legal term for a claim on a warehouse—"you are lending us money." So, informed depositors agreed to this deal, and historically, fractional reserve banking, fractional reserve deposits, became very popular when the banks in London started offering interest on accounts that you could redeem on demand. That's deposits.

- 3.27** What about banknotes? Banknotes are actually not feasible without a fractional reserve, because if there weren't a fractional reserve, you would have a warehouse ticket, and for the bank to store things on behalf of somebody and give them a ticket, which they were allowed to sign over to somebody else, the bank has to know who the somebody else is, because they have to charge storage fees to somebody.
- 3.28** One way grain warehouses do it is that whoever comes to collect the grain has to pay the storage fees. But a typical banknote doesn't charge any storage fees, and that's only feasible if the bank is earning enough from its lending operation to cover the costs of issuing and maintaining offices to redeem the banknotes. I can't think of any way to get storage fees out of people when you don't know who they are. When the holders of a banknote are anonymous to the bank because the banknote is just circulating from person to person, and the bank doesn't know who's got it now, how are they going to charge them?
- 3.29** This distinction between warehousing and fractional reserves has been recapitulated today when we talk about stablecoins. So gold stablecoins—Tether Gold, Pax Gold, and I understand there are a few others—have 100% reserves. They say, "We're providing a warehousing service. We have the gold in a vault in Switzerland or somewhere, and you can come and get it if you bear the cost of coming to get it."
- 3.30** But dollar stablecoins like Tether and USDC, those have fractional reserves. They are claims to dollars, but in the vault of Tether or USDC aren't just stacks of \$100 bills. They have interest-earning assets. They have securities, treasury bills. Their balance sheets aren't completely transparent in the case of Tether; they have some funky stuff on there, too: loans and gold and Bitcoin, but it's not 100% ready cash to pay back their customers.
- 3.31** People sometimes call stablecoins 100% reserve liabilities, but that's because they mistake assets for reserves. They call all the assets reserves. Traditionally, reserves meant the cash assets, ready to pay out to redeem this minute.
- 3.32** There is a social benefit, if you want to call it that, of fractional-reserve banking that Adam Smith emphasized. So, the question is: why should we tolerate this system? Well, first answer: it's good for Alice and Bob. They agreed to it; they like it. Why should we stop them? But Smith said, beyond that, that there's a social benefit, which is that when it was all coins in the vault, then society's scarce capital is tied up in coins, and the coins are not producing anything—this is an exaggeration on Smith's part because he had a kind of physical notion of production.

- 3.33** People who want storage are getting a valued service from storage, but people who want payment services, they would rather have the payment services without fees and with interest. So, the coins in the vault are producing less for them than lending out the money would produce. But Smith referred to the coins as dead stock, and I think that's an exaggeration. It's not that they produce nothing. It's that for people who agree to it, more is produced by fractional-reserve banking. Or, to put it another way, it's a less costly way of providing the payment services that people want.
- 3.34** A very similar point was made by Mises about the social savings. So, in Adam Smith's view—sorry, I didn't really finish that thought. He said the coins are dead stock, but if you lend them out, now you have active stock. Now you've financed enterprises that are producing stuff. And as a country, if you switch from 100% reserves, or only gold coins, to fractional-reserve banking, you can export the gold coins you're not using anymore and import raw materials, import machines. That makes your economy more productive.
- 3.35** Mises made the point that if the economy had been confined to 100% reserve banking, then as the money economy grew, the demand for money grew, that would have put downward pressure on the price level or raised the purchasing power of gold, thereby stimulating more gold mining in the way we discussed in the last lecture. But that gold mining is a more costly way of providing a payment medium than fractional reserve banking. So, this increase in the purchasing power of gold, he said, would have undesirable consequences, namely, society's resources would have to be devoted to gold mining, that don't need to be devoted to gold mining, that can be devoted to producing other stuff. "The monetary function could have been fulfilled by a smaller stock of gold," is the way he put it. So again, when the substitution is voluntary, from coins to fractional-reserve liabilities, it's a lower-cost payment system.
- 3.36** But what constrains the system? Let's assume what was historically the case, that banks made their banknotes and deposits redeemable, historically, for gold and silver coin. No individual bank determined the purchasing power of gold, so treat the purchasing power of gold as just given to the bank.
- 3.37** And let's assume a competitive system where no one bank dictates the interest rate on loans or the interest rate on deposits. They have to compete with other banks, and so there's a competitive interest rate on deposits, there's a competitive interest rate on loans. The loan rate has to be a little bit higher than the deposit rate so that banks can cover their costs out of their interest income, net of their interest expense, but competition will reduce that spread to as small as possible.
- 3.38** And let's assume that banknotes circulate throughout the system at par; banks accept one another's notes. Those are the conditions you see in the free banking system in Scotland in the 18th and 19th centuries.

- 3.39** So let's look at the balance sheet of a note-issuing bank. Let me be a little more explicit here about not the names of the depositors, but the type of liabilities and assets the bank has. It's got reserves, which are the coins in the vault, and it's got interest-earning assets, so loans and securities. And then, it's funding these loans by having notes in circulation, and the bank doesn't know who's got them at any moment; it just knows the number in circulation. And it's got deposits. And then, of course, it's got net worth.
- 3.40** So, from the point of view of the shareholders, starting a bank is a way to take their money and multiply its earning power by attracting lenders to the enterprise. Depositors are lending money to the bank, and in a way, note holders are too: they're providing funds if they bring in coin in exchange for banknotes to fund the interest-earning portfolio of the bank.
- 3.41** Now, if we take the net worth as given, let's not worry about decisions the bank makes about raising more capital. So if that doesn't change, there's a balance sheet constraint imposed by the fact that assets have to equal liabilities plus net worth. If net worth's not changing, that implies that any change in assets has to be matched by a change in liabilities. So, if the bank wants to make more interest income by making more loans, or buying more bonds, they have to either get more notes in circulation, or attract more depositors, or some combination of those, or reduce their reserves.
- 3.42** That means that making loans is not costless; it's not out of thin air; it's not gratuitous; it has to be funded. So when they look at the additional revenue they hope to earn from making more loans, they have to balance it against the cost of reducing their reserves—I'll make that more concrete in a minute—or the cost of having more notes in circulation or the cost of having more deposits on the books.
- 3.43** To understand those costs, we can divide them into two types: the kind of operating costs of running a bigger operation— if you have more depositors, then you have to process more payments, you need more tellers, that sort of thing—and what I call the expected illiquidity cost, by which I mean, if you have more notes in circulation, there's a bigger risk, a higher probability that so many notes are going to come back just on a random day that they threaten to wipe out your reserves, and you can't let that happen because there's some penalty for running out of reserves, either legal, like you're declared in default and you have to pay a legal penalty for that, or you're shut down, or you have to scramble to cover your shortfall, and that's a costly operation.
- 3.44** So that's what I mean by expected illiquidity costs: the probability of running out of reserves times the penalty you have to pay for running out of reserves, or the cost you have to incur to avoid running out of reserves at the last minute.
- 3.45** To have more notes in circulation, you need to print more, and there are some physical production costs. But that's not really the important thing. The important thing in a world where the note holders have the option to redeem the notes, the important

thing is to persuade enough people to hold those notes and not bring them in for redemption—I'll come back to that in a second.

- 3.46** For deposits, it's obvious that you've got to process more checks, you've got to pay interest on the deposits, but the operating costs go up, and the threat to your reserves goes up the more deposits you have, because the bigger the chance, just randomly, that a big bunch of checks will come in for payment, and those will be debited against your reserves. Likewise, if you reduce your reserve position, you raise the probability of running out of reserves, and that comes with a penalty if you actually do run out, or you have to scramble to avoid running out.
- 3.47** So we've got operating costs, and we've got the expected illiquidity cost. If a bank wants to maximize profits, then it has to balance the costs and benefits of expanding its balance sheet. If they want to increase their loans and securities, like I said, they've got to fund it. So, the marginal benefit, the benefit of one additional dollar of loans, is the interest they earn minus whatever the cost of collecting the loan is, having loan officers, that sort of thing. That has to be balanced against the cost of keeping more notes in circulation if that's the way they're going to finance it, or the cost of having more deposits in circulation, if that's the way they're going to finance it, or the cost of reducing their reserves, if that's the way they're going to finance it.
- 3.48** Let me come back to the notes in circulation, because that's the one I think other economists have had the biggest trouble understanding. I actually had an exchange of letters with another economist who said that the only thing, the only reason people hold currency that doesn't pay interest is that the banking system is legally restricted from issuing currency. And I said, "Historically, banks were not legally restricted from issuing currency. They did issue currency. And yet, the currency didn't pay interest."
- 3.49** His prediction was that currency should pay the same interest rate as loans and securities, minus maybe 1% for the cost of intermediation. I said, "Banknotes did not pay interest historically, yet people accepted them." And his response was, "I don't understand then. It looks like there are unlimited profits available because the more loans you make, the more interest you earn. And if you finance them by issuing notes, you don't have to pay any more interest. Profit!"
- 3.50** What he didn't understand is that there's a limit to the number of notes the public will accept. It's limited by the services they're getting from notes, services provided by the bank, but which are costly to provide. It's cheap to print up notes and to lend them to borrowers, but that's not the end of the story. You haven't financed the loan if that's all you've done, because the borrower is not gonna just hang on to the banknotes and admire them. He's gonna spend them. He borrowed money to invest in something. The spending is gonna put the notes in the hands of somebody else. What's that person gonna do?

- 3.51** More than likely, most businesses, at the end of the day, when they have extra currency, they put it in their own bank. If the customer that the borrower spent on takes the currency and puts it in his own bank, what's his own bank going to do with it? They're going to bring it back to the bank that issued those notes and say, "Here, we've got a claim on you, please pay. You can either pay in our notes that you've collected"—that's what the clearinghouse does—"or you can settle up the difference in coins." That's the settlement that takes place at the end of the clearing session.
- 3.52** So, it's gonna cost the bank reserves when the payment goes through. The notes don't stay in circulation without expanding the clientele of the bank. So, the bank has to do something to expand the clientele of people who are willing to hold its notes if it wants the notes to stay in circulation. And cultivating more customers is expensive. How do you do it? Pretty much by getting more depositors is the way banks did it, because the way they got notes in circulation was by paying them out to customers who were their depositors who wanted currency.
- 3.53** You can see the same thing today if you go to one of the places in the world that still has private currency, private banknotes, namely, Northern Ireland, Scotland, Hong Kong, and Macau. So, you go to Scotland, and you go to an ATM, what you get out depends on whose ATM it is. If it's the Bank of Scotland, you get Bank of Scotland notes. If it's the Royal Bank of Scotland, you get Royal Bank notes. If it's the Clydesdale Bank, you get Clydesdale Bank notes. So, the way those banks get more currency in circulation is by having more depositors, more people coming to their ATMs. So, it's costly to attract and retain depositors as a way of keeping notes in circulation.
- 3.54** What else can the bank do? It can open more branch offices, which will make notes more attractive to redeem and also attract more depositors, hire more tellers, stay open longer hours, advertise. Particular to banknotes, they can make them harder to counterfeit.
- 3.55** The first time I ever gave congressional testimony was when they were redesigning the \$20 bill to put in the anti-counterfeiting devices it now has. I had written in my first book, which talks about the free banking system in Scotland, that counterfeiting wasn't a problem for these banks because when they issued notes, they very quickly came back out of circulation, and so if somebody introduced a counterfeit, it wasn't long before it came back to the teller of the bank who could detect the counterfeit.
- 3.56** The Scottish banks had an interesting policy. They would honor a counterfeit note if the person who brought it in convinced them that they were innocent, they weren't the counterfeiter, and would tell them where they got it, because then they could trace it back, and the trail was still fresh, because notes didn't stay in circulation more than a week, a couple of weeks at a time. They kept being paid back into other banks. So that was their policy, and of course, they introduced anti-counterfeiting devices into the notes.

- 3.57** So I was called to testify as to whether counterfeiting was just a technological problem or whether there was more to it. And I said, "Yes, if you have a system of competitive note issue, then the banks have an incentive to put anti-counterfeiting devices in their currency and to track down any counterfeiters. One of the congressmen in the room said, "Let's assume we're not going back to free banking anytime soon. Don't you think the Federal Reserve and the Treasury have good reason to want to make Federal Reserve notes counterfeit-proof?" And I said, "Sure, because you want the profit from issuing more currency, just like commercial banks do." You can design the notes so that people find them more attractive. If you look at the banknotes in Scotland, they put local heroes on them. Same thing in Northern Ireland: Scottish inventors and poets and architects and so on.
- 3.58** So, those costs of attracting note holders, since it's not by paying interest on notes—there isn't any practical way to do that, you don't know who's got the notes, how can you pay them interest—economists describe that as non-price competition. Non-price competition has a bad reputation as being wasteful compared to price competition, and it is if price competition is outlawed and non-price competition is used as a second best.
- 3.59** So when banks weren't allowed to pay more than 3% interest on savings deposits, and 3% became very cheap when interest rates went up on everything else, they started attracting depositors by giving away toasters, if you're old enough to remember that. Then they modernized it, and it became toaster ovens, but that was wasteful in the sense that people would have rather had cash than a toaster oven. If they wanted a toaster oven, they could always buy it with the cash. But it was less bad than depositors getting nothing. It existed only because of the legal restriction. But when you see non-price competition in the absence of legal restrictions, it's not presumptively inefficient. It's cheaper to deliver what customers want when there's no cheap way to pay interest on banknotes.
- 3.60** The marginal revenue from making loans also has to be equal to the marginal cost of attracting more depositors and paying interest on the deposits, including the threat to reserves that more deposits in circulation impose. And finally, reducing reserves in order to make loans imposes the cost of a higher probability of running out of reserves, for which there's some penalty. So you raise the probability, you raise the expected penalty, you get to a point where it's not worth it.
- 3.61** So there's, what you might call, an equilibrium volume of a bank's liabilities. And in practice, it's going to depend on what expenses they're incurring to attract their clientele. What if a bank makes a mistake and goes beyond the quantity of its notes and deposits that the public will willingly hold? This was known historically as an overissue, and there's a lot of worry about banks overissuing notes. In particular, the Bank of England was justly accused of overissuing notes when it had a monopoly and when there wasn't a very strong disciplinary mechanism to stop them.

- 3.62** But an individual bank surrounded by other banks of issue is in a different position than a monopolist. If a bank issues more notes than its customers want to hold, so it, say, makes more loans by putting more notes in circulation, and then the people who receive the notes spend them to other people who don't want to hold those notes.
- 3.63** Either, they're gonna come back through the clearing system and drain the bank of reserves—so, loss of reserves is a signal to the bank that we had too many liabilities in circulation; we would be foolish to reissue them, because that would just repeat the story: they'd come back again and drain us of reserves. So, we need to stop and actually replenish our reserves. We need to sell off those extra loans or securities we have and go back to where we started, now that we've gotten the signal from reserve losses. And so, reserve losses or adverse clearings, bankers call them, that's a signal to the bank of when it's issued more than the public is willing to hold.
- 3.64** A second reason a bank could overissue is not that it expanded, but that, for some reason, its liabilities became less popular. So, I don't know, it's in a town where the population is shrinking, customers are moving away. So, it's gonna have to contract, not gonna be able to keep as many liabilities in circulation. If it tries to, they're gonna come back for redemption.
- 3.65** So the actual amount of notes a bank has in circulation is gonna converge on the quantity the public is willing to hold, given the expenditures the bank is making on attracting them. The public re-establishing its desired portfolio of assets is what's going to discipline the bank.
- 3.66** There were some theorists in the 19th century, and some of them are still with us, who have what I think is a fallacious theory of what limits the bank. Their theory is that if there are excess notes in circulation, those very notes will come back and repay the loans, and the bank won't be able to lend more because the loans are being repaid. That doesn't really make sense.
- 3.67** So, the reserve losses bring adverse clearings, the adverse clearings tell the bank to replenish its reserves by selling off the extra assets. And so, the individual bank is being disciplined by reserve losses to the neighboring banks. So, no bank can expand faster than the demand to hold its liabilities.
- 3.68** Now, that's sometimes been put, not as accurately, sometimes been put, "A bank can't expand any faster than its neighbors." That's true only if nobody's demand is growing any faster than anybody else's. But putting it that way has raised the question: what if all the banks expand together? Because if a bank expands together with every other bank in the neighborhood, every bank is going to be collecting more notes from— let's say, Bank A expands, Bank B expands, Bank C expands, they all expand by the same percentage. Bank B is gonna be getting more Bank A notes brought into it on deposit, so is C, D, and so on. But meanwhile, A is collecting more Bank B notes, C notes, D notes, and so on. So when they meet to exchange their claims on each other, they're

gonna cancel out. So, no bank is going to systematically lose reserves to the other banks because they all expanded in unison.

- 3.69** So, what's going to discipline them if the adverse clearings is not going to do it? The answer is, if we're talking one country or one banking system in a world gold standard, they're gonna lose gold to the rest of the world. Because if all the banks are expanding, that's gonna increase spending in that economy. If the money isn't driven out of circulation immediately, that's gonna drive up prices.
- 3.70** It's gonna have two effects. First, there's gonna be more spending on imports just because people have more money to spend. But then, because it's gonna drive up domestic prices, that's gonna stimulate more imports because imports are now cheaper. Domestic prices are higher than world prices.
- 3.71** This is the mechanism that the philosopher David Hume—he was also an economist on the side—described as the price-specie flow mechanism. If there's an overissue of bank money that drives up prices, it will be corrected by a balance of payments deficit, that is, gold will have to flow out to pay for imports, and that will bring the money supply back down. So, that'll be a self-correcting process.
- 3.72** Hume's discussion of the price-specie flow mechanism is one of the earliest economic theories of a self-regulating system. What I described as adverse clearings is kind of the Humean price-specie flow mechanism brought down to the level of an individual bank, except that the price level doesn't play a role, it's just the adverse clearings. In Hume's story, it's the rise in the price level that generates the adverse clearings for the economy as a whole against the rest of the world. So, every bank is going to be losing reserves to the rest of the world, and that's going to correct the excess supply of bank-issued money, and it's going to signal that we can't do this again.
- 3.73** Now, this scenario of all the banks expanding together is kind of imaginative. But there have been economists who thought that this was a plausible possibility, that if all the banks could collude, that this is what they would do. They would all expand together so that none of them would be disciplined by losses at the clearinghouse. In fact, if all the banks could collude, they would act like a cartel or a monopolist, and the way a monopolist makes a bigger profit is not by expanding his output, because if he expands his output, he has to sell it at a lower price to sell it all. What a monopolist or a cartel does is restrict output, and that way get a higher price.
- 3.74** The analogy in banking would be if all the banks got together, what would they say to each other? "Let's raise the interest rate we charge on loans. Let's lower the interest rate we pay on deposits. That way, we get a bigger spread. Yes, if we lower the interest rate we get on deposits, we're gonna get fewer deposits. And if we raise the interest rate we make on loans, we're not gonna make as many loans. So we're gonna contract our output, but that's okay because we're making so much more per unit. We got a

bigger spread." That's what banks would do if they would collude. The idea that if they colluded, they would all expand in unison doesn't make any sense.

- 3.75** Anyway, from this idea that adverse clearings provide an effective check on overissue, there was a group of economists in the 19th century who, faced with the Bank of England overexpanding, causing a boom, then an outflow of gold to the rest of the world, especially the continent of Europe, and then having to contract suddenly to correct that, to stop themselves from running out of reserves, confronted with that scenario, they said, "The Bank of England is not being stopped fast enough from over-issuing. When the only check is the outflow of gold to the rest of the world, that takes a while. We've gone through a boom and a bust before that happens. We need something more effective. We need a system that checks individual banks in a competitive system like Scotland's."
- 3.76** So, they proposed, "Let's have free banking in London." My first book is basically about the debate over that proposal, and needless to say, it didn't get very far because the Bank of England organized the Parliament to oppose it. They didn't want competition in issuing banknotes or making loans. But the free banking school, I call them, these thinkers who drew this conclusion from the problem, they said, "Let's not restrict note issue to a single institution, the Bank of England. Let's allow free entry."
- 3.77** Some of these arguments spilled over into the U.S. in the debate over the First Bank of the United States and the Second Bank of the United States. Defenders of the Bank of England's monopoly, or at least its creation, and the First and Second Banks have said, "They did useful banking services." And that's true. It's better to have one bank than no bank. But it's better still to have multiple banks. And in the U.S., there was only one bank that was allowed to branch across state lines. State-chartered banks were limited to a single state. It would have been better if there were dozens of nationwide branch banks.
- 3.78** The extreme case of imagination is what if all the banks in the world over-expanded together? They're not going to lose gold reserves to Mars. So, what's going to stop them? The answer is— this is a little subtle, but there's going to be more spending, and with more spending, there's a bigger threat to each bank's reserves. By assumption, reserves haven't increased, but the volume of liabilities for every bank has increased. So, there's a bigger threat to the reserves of every bank, just from random adverse clearings. Each bank is going to feel itself having inadequate reserves, and if they try to increase their ratio of reserves to liabilities, that means they have to contract their liabilities a little bit to get reserves from the other banks. But if every bank does that, then we're back to where we started. So, the in-concert overexpansion is not going to last. George Selgin makes this argument in his book, "The Theory of Free Banking," which I will put on the reading list.
- 3.79** So, there's an equilibrium volume of banking system liabilities. It's determined by the cost functions that I talked about, the volume of system reserves at present. For the

world, the volume of reserves is given at the moment. For any small open economy, it's not; it flows in and out through the price-specie flow mechanism.

- 3.80** The probability of running out of reserves just based on random clearings and the level of safety—I mean, how low does a bank want to reduce the probability of running out of reserves? That's going to depend on the penalty for running out of reserves. So, it depends on the size of that penalty and on the actual size of these probabilities. But with those fundamentals, there isn't multiple equilibria, where any volume of bank liabilities is okay, provided all the other banks are along for the ride. The same logic rules out all the banks agreeing to reduce their reserves in concert.
- 3.81** Now, when demand shifts from one bank to another, or when a local economy grows, the banking system will provide them with more money. So you can describe the supply of bank-issued money as demand elastic. The phraseology in the 19th century was that the issue of banks met the needs of trade, which, if we interpret as meeting an increase in the demand to hold bank liabilities—not the demand to borrow, that's the mistake to conflate those two things, but the demand to hold bank-issued money—banks would be happy to satisfy that. And so, when demand went up, the volume of bank liabilities would go up. If there's a fall in demand for bank liabilities, that would go down.
- 3.82** It's a nice property that banks will satisfy the demand in that way, rather than leave people frustrated, or if they don't expand but the demand for money increases, you would have people trying to get hold of more money, bidding up what they're willing to pay to borrow it. So, you would have effects on interest rates. By having a demand-elastic supply, you avoid those needless disruptive effects on interest rates.
- 3.83** Some economists, especially the 100% reserve economists, are skeptical of this. They say, "If a bank expands its liabilities, that's gonna drive down interest rates, that's gonna cause all the problems that are described by the Austrian theory of the business cycle, where there's an artificially low interest rate based on monetary expansion." My reply is: you're confusing two cases. You're assuming that there's no increase in the demand to hold this money, in which case, yes, issuing more than people want to hold will have this disruptive impact.
- 3.84** But if, by assumption, there's been an increase in the demand to hold bank-issued money, it would be disruptive not to meet it by issuing more. By issuing more, you avoid a distortionary rise in the interest rate. So, it's a warranted increase in liabilities of the banking system. It's not like a central bank expanding without there being any additional demand for money.
- 3.85** In Mises's book, "The Theory of Money and Credit," he has a very clear statement. He says that some of the people defending the needs of trade doctrine had mistaken ideas about what the banks were doing when they met the needs of trade, but nonetheless, they were observing something real. If you think about it rightly, I'm quoting him now,

"A single bank will be able to circulate more fiduciary media"—that's banknotes and deposits over and above their reserves—"only if there is a demand for them," even when this bank is not charging a lower interest rate on loans than everybody else—sorry, I stopped quoting him at that point.

- 3.86** Quoting again: "Thus, the banks increase and decrease their circulation *pari passu*,"—that is, in equal steps—"with the variations in the demand for money." And then a little later in the paragraph: "In doing so, they help stabilize the objective value of money." So, by satisfying increased demand to hold money, the banks prevent an increase in the purchasing power of money, if this is like a system-wide phenomenon, and they are restrained by the public's demand because they can't issue more money without the public being willing to hold more money issued by banks.
- 3.87** So, this willingness of banks to meet the demand to hold money helps stabilize spending in the economy, which has some desirable features, because if they don't meet the demand, then people, in order to build up their money balances, are gonna have to spend less. Their spending has to be less than their income. So, banks provide this service that— the banks are willing to expand when the demand to hold their money increases, because holding means it's not entering the clearing system. Therefore, the chance of running out of reserves due to adverse clearings has not increased. In fact, it's gone down for a given volume of bank-issued money. They issue enough bank money to bring it back up to where they're at an optimum. And so stabilizing aggregate spending has nice properties in smoothing the behavior of the economy.
- 3.88** There used to be a lot of myths about the behavior of free banking systems. Even Milton Friedman, early in his career, subscribed to these myths. In his program for monetary stability, he said that a problem with banknotes is that they're likely to become overissued. This is what happened under free banking in the U.S., he says. But then, he read the historians who had actually studied free banking systems and how they behaved, how they performed, that overissue and fraud were not problems.
- 3.89** And later in his career, in a piece co-authored with Anna Schwartz, they wrote that "historically, producers of money have established confidence by promising convertibility. Many examples can be cited of long-continued and successful producers of private monies convertible into specie." So, this is very different from the view Friedman earlier had that the private issuers of money are gonna find it easy to commit fraud.
- 3.90** I don't have time to go into the historical track record of different free banking systems: Scotland, Canada, Sweden, Switzerland, Australia. There were more than 60 cases where there was a plural private note issuing, in most independent countries, most British colonies. But the track record in the cases that are least legally restricted is very strong. There's innovation, there's efficiency in payment practices. You don't see excess

expansion, inflation, the system crashing or collapsing. There are some cases of real crises, but the systems are not panic-prone or run-prone.

- 3.91** So, why did we get central banking if free banking had a pretty good track record? In some places, free banking never got a trial. The Bank of England managed to stop the movement to introduce it into England, and they managed to get legislation that consolidated the Scottish system into the English system instead of the other way around.
- 3.92** In the U.S., we got a central bank not because free banking failed, but because the legally restricted system put in place during the Civil War, the so-called national banking system, gave rise to financial panics, and the national banking system was created in order to help finance the Union effort in the Civil War. So, indirectly, it was to patch up a system that was weakened by rules imposed for fiscal reasons.
- 3.93** In other countries, it was more direct. The Banque de France was created by Napoleon to give cheap loans to the government of France. The Bank of Italy, similar story. So, it's not a market verdict that we need a central bank. It was a political verdict. I'm not optimistic about reducing that verdict anytime soon, but, in the U.S.
- 3.94** But there are countries where this is a very live issue. The current president of Argentina, Javier Milei, ran on a platform of abolishing the central bank. And replacing it with what? With letting people use U.S. dollars and letting the banks operate on U.S. dollars. That would have been a free banking system with the dollar as the anchor currency. It hasn't come about yet.
- 3.95** But other countries have taken that step. Ecuador is dollarized, Panama is dollarized. Their banks are not completely unrestricted, but the issue of whether to have a central bank—well, they don't have central banks. So that is still a live issue in countries that are considering adopting some system other than having their own fiat currency. That's banking. Questions?
- 3.96** How would a country like Argentina operate a free banking system if it's underpinned by the U.S. dollar, which is issued by our central bank?
- 3.97** The easiest way to answer it is to look at the precedents for that thing. Look at Panama. How do they do it in Panama? They don't have a central bank. How do they get more dollars if they need them? The banks in Panama have arrangements with banks in New York. The Federal Reserve Bank of New York ships pallets of \$100 bills to Panama and takes back worn bills from Panama. So the currency is pretty easy.
- 3.98** But the banks have arrangements with New York banks to borrow money on short notice, have lines of credit with the New York banks. The banks in Argentina would have to do something similar. The need to do that helps discipline the banking system, because to get a line of credit from Chase or Citibank, you need to be creditworthy. Whereas to get credit from the central bank, if you're pals with the central bank, you

might not need to be creditworthy. The banking system in Ecuador became much more stable after they dollarized and after they could no longer count on the central bank bailing them out.

Lecture 4: The Fiat Standard

At a glance — The fiat standard after commodity money.

- 4.1 In our previous lectures, we talked about commodity money, and we talked about bank-issued money, and the economic forces that govern those. In this lecture, I want to talk about the kind of money we have today at the basis of our monetary and banking system, and that's fiat money. Fiat comes from the Latin term meaning "let it be," when God spoke Latin. He said, "Fiat lux," let there be light. So fiat is money by decree, as opposed to money whose value is based on it being redeemable for some commodity.
- 4.2 I want to ask the same questions about a fiat standard that I asked about a gold standard. What determines the quantity of money in the system, and what determines the purchasing power of money, and what determines the evolution of that purchasing power, that is, the inflation rate over time? But maybe I should first address the question of how we got to where we are, because when we left off, we had a gold standard with bank-issued money.
- 4.3 Market forces gave us commodity money, gave us coinage, gave us bank-issued money. The history of banking is a history of innovations in ways to make it easier to spend and receive money, so checks, wire transfers, banknotes, electronic funds transfer today. But we can't tell a similar market-motivated evolution story when it comes to fiat money—well, back up. When it comes to governments taking over mints, that's not the result of market forces. That was legislation. The main monetary institution today, central banks, those are the result of legislation. They're not part of the natural evolution of monetary systems.
- 4.4 Once you have central banks on a gold standard, when you've got one issuer of money—that is, a characteristic of a central bank is having a monopoly on the issue of paper currency banknotes—then the stage is set for the redeemability of that money into gold to be eliminated, to be, usually it's called suspended as though it's temporary, but taken away.
- 4.5 It would take us pretty far afield to go through the whole history of central banks, and of course, it's different from country to country. But basically, central banks were institutions that were created by governments to lend to governments. That's the predominant reason. In the United States, the story is a little different, but the central banks in the world, having become central, having become dominant in the monetary system in each country, during the First World War, suspended the gold standard. That was the fatal blow.
- 4.6 Some countries tried to get back on gold after the First World War at the old parity. Some tried to get back at a new devalued parity. There was chaos in the interwar period. After the Second World War, well, actually during the war, the Allied countries

came together and tried to figure out a way to run something like an international gold standard, but without so much gold. And that was the Bretton Woods system. Under the Bretton Woods system, the only currency convertible into gold was the U.S. dollar, famously at \$35 an ounce. But it was only redeemable for central banks.

- 4.7 So, U.S. citizens were no longer allowed to hold gold coins. They couldn't go to the Fed and redeem their Federal Reserve notes. But foreign central banks, the Bank of England, the Bundesbank, the Bank of Japan, they had the right to redeem their dollars for gold at \$35 an ounce. But this was a very slender thread linking monetary systems to the gold standard, linking them to gold, because it depended entirely on the willingness of the United States government to keep redeeming. And when they decided it was no longer convenient to keep redeeming, that was the end of it.
- 4.8 In 1971, Richard Nixon closed the gold window because the U.S. was running out of gold. And the U.S. was running out of gold because it had issued too many dollars, too many claims to gold, and foreign central banks didn't want to accumulate these claims indefinitely. They started redeeming them, and the U.S. ran out. Nixon went on TV to announce that he was closing the gold window, but under advice from some of his economic advisors, to say something positive instead of just announcing that we are no longer redeeming, because that sounds like a sign of weakness. He announced wage and price controls. That was supposed to be a positive measure to show he was taking charge. In the same speech, he announced these two things.
- 4.9 Now, if you read standard textbooks about why we have fiat money, often they don't even address the question. They just take it for granted. Let's start with where we are. And so, this is our fourth lecture, but many textbooks begin where we're starting today, with the existence of fiat money.
- 4.10 But when they do try to explain how we got from commodity money to fiat money, they often make the argument or the suggestion that it's more efficient. We talked last time about the resource costs of a gold standard, and they suggest that we can save those resource costs, and that explains why we switched to fiat money, except that historically it doesn't fit. It's not the case that there was a big debate among economists about how to make the monetary system more efficient, and they decided, "Well, let's eliminate the gold part because then we won't have to mine the gold and store it in vaults."
- 4.11 Most economists, while the U.S. was on the gold standard, favored the gold standard. There were some dissenters, and the public also favored the gold standard. Famously, there were objections. There were dissenters like William Jennings Bryan. But William Jennings Bryan lost the presidency three times. He ran for President three times as a Democrat and lost every time. So the public was in favor of the gold standard, based on that.

- 4.12 The idea that it's a resource cost-saving to replace gold with paper is reminiscent of the argument that it's expensive when people lock up their bicycles, that they use these expensive metal locks, hardened steel and magnesium and so on. Wouldn't it be cheaper if we just replaced all the metal locks with paper locks? Well, it would be, if people respected the paper locks, but they don't. You can think of the resource cost of a gold standard as being an investment in protecting the money from those who would erode its value.
- 4.13 The best summary, I think, of why we switched to a fiat money was offered by the humorist Dave Barry in his book, "Dave Barry's Money Secrets," where he said, "All the governments in the world," I'm quoting him now, "having discovered that gold is, like, rare, decided that it would be more convenient to back their money with something that's easier to come by, namely, nothing." Gold standards were eliminated, and it's clearest during the First World War, because they were an obstacle. They were a constraint on issuing money to pay for things.
- 4.14 The governments that wanted to issue money to pay for their war effort couldn't issue unlimited amounts of money under a gold standard, so they said, "Let's take that away. Let's issue money that's not redeemable. We'll promise to redeem it later, so that we can get people to accept it, but we'll worry about redeeming it later, when later arrives." And some governments, as I said, never went back. Others tried to go back. In Great Britain, they did not manage to go back. Well, they did temporarily and then suspended immediately again, almost immediately.
- 4.15 The problem is that, once you've issued a lot of money to pay for the war effort, you've lowered the value of money, you've raised the price level. And if you try to redeem, start redeeming again at the old rate, you'll immediately run out of gold, because gold buys more than the paper representative of it. So you have to bring the price level down. You have to raise the purchasing power of the money unit, and that is a process that governments weren't willing to go through.
- 4.16 I talked last time about the comparison between fiat money and gold and silver standards on inflation. We saw that every country in the Rolnick and Weber sample had higher inflation under fiat money than under commodity money. Commodity money inflation rates are pretty close to 0%. Fiat money inflation rates, sometimes double digits on average, depending on the sample of countries, something like 7%, 13% in one sample. It's been better in the last 20 years, but still not down to near 0%, as we had over long periods under the gold standard.
- 4.17 And of course, this improved performance, we had a break from it in the pandemic. So 2022 was a bad year for fiat money. And if you average, so as not to just focus on one bad year, going into the 2024 election, voters looked and saw that, over the previous four years, the price level had risen 20%. The dollar had lost 20% of its value. That's an average of 5% inflation a year for four years. That was a worse performance than had been in the previous decade.

- 4.18 And around the world, the U.S. had lower inflation than a lot of other countries over this period, 2020 to 2024. In 2022, which was the worst year, you've got triple-digit inflation in some countries: Venezuela, Zimbabwe. You've got high double-digit inflation in Sri Lanka, in Sudan, in Turkey. And these inflation rates just aren't possible with a commodity standard, but they're eminently possible under a fiat standard for reasons I'm about to explain.
- 4.19 With the gold standard, I provided supply and demand diagrams to determine the purchasing power of the monetary unit. We can do the same thing with a fiat standard. The demand curve is going to be the same; that is, the demand for fiat money is the same as the demand for gold-backed money, in that people want enough money to purchase the goods they want to purchase. Based on their transactions, they'll want a certain volume of purchasing power. The higher the purchasing power per unit, the fewer units they will need. The lower the purchasing power per unit, the more units they will need.
- 4.20 We can draw the demand curve, as we did before, in stock space, only instead of ounces of gold here, I'll just call the unit dollar and plot it again against the purchasing power of money. Here, it's not the purchasing power of gold; it's the purchasing power of \$1. At a high purchasing power, you might need this many units. At a low purchasing power per unit, you'd need proportionately more units. Let's see, let's cut it in half. We need twice as many units. Cut it in half again. We connect these dots. We get the demand curve, plotted against the purchasing power of the monetary unit. So it looks the same as the demand curve for gold-backed money.
- 4.21 What's different about a fiat money system is not the demand curve, but the supply curve. In a gold standard, we had an upward-sloping supply curve at a point in time because there's non-monetary gold, which at a higher purchasing power per ounce will be considered too expensive to keep in the form of candlesticks and picture frames and jewelry to some extent. It'll be melted down and coined, and you'll get a larger quantity of monetary gold at a higher purchasing power per unit from the conversion of existing non-monetary stocks into monetary use.
- 4.22 There isn't any process like that in the case of a fiat money standard. There is no commodity use for a dollar. It doesn't do anything for you except serve as a medium of exchange. A technical term used by some economists is that fiat money is intrinsically useless, meaning it has no other use, no commodity use, than serving as a medium of exchange or a payment medium.
- 4.23 So, how is the supply determined? It's determined by the central bank. And the central bank can make it whatever it likes. I won't say a lot about the technical process by which they change it, but in terms of drawing the supply curve, it is what the central bank decides to make it. There's the supply of dollars. It's vertical— well, my artwork is not perfect; it's supposed to be vertical, meaning it's arbitrary, it's decided by discretionary monetary policy; it's not determined by market forces. Now, the central

bank can change the supply. It can shift the curve from left to right. We'll look at that in a second.

- 4.24** But behind the scenes, the way the central bank—I'll just say the Fed, to refer to the U.S. case—the way the Fed can change the quantity of money is that it has control over its own balance sheet. The basic money in the system, the currency that you and I have in our wallets, is our Federal Reserve notes. The currency that banks hold or the money that banks hold to redeem their checking accounts is also Federal Reserve liabilities, some of it in the form of currency. Their ATMs are stocked with currency.
- 4.25** But when it comes to covering the checks you write, that is, transferring funds to the bank of the recipient of your check-writing or electronic funds transfer, they have an electronic form of a claim on the Fed, which is a deposit on the balance sheet of the Federal Reserve Banks. That's bank reserves in a book-entry form.
- 4.26** So when the Fed wants to increase the quantity of money, it can go out and buy assets and pay for them just by creating new money. So it wires money into a bond dealer's account to buy bonds. The bond dealer's bank gets reserves transferred to it to cover that payment. It has more reserves in the form of this claim on the Fed. No other bank has less reserves. And so there's an increase in total bank reserves, and that will support an increase in the broader supply of money held by the public.
- 4.27** So, where it comes from, where new money comes from, is from the Fed creating it in asset purchases. And this is a kind of money that is created out of thin air. I argued, when we talked about commercial banks, they cannot create money out of thin air. They have to get people to willingly hold it when it's redeemable for something else. And so that constrains them. There's no constraint on the number of Federal Reserve liabilities that the Fed can issue.
- 4.28** One economist I met years ago used to refer to a bank note, a claim on gold issued by a commercial bank, as an IOU. A Federal Reserve note is an IOU-nothing. It's not redeemable for anything else. Its value just is what it is, based on the supply and demand of those units. It's not backed by anything.
- 4.29** You sometimes will hear loose talk about what the U.S. dollar is backed by. There's no asset that backs it. What gives people confidence, to the extent they have confidence in its purchasing power, is their confidence that the Fed won't do something crazy. But it's not that there's any mechanism by which the Fed is constrained, such as they issue too much, it gets redeemed, they start to run out of gold, they have to reverse course. That's no longer operative. The only thing that limits the issue of money is their own prudence or their commitment not to let inflation get out of hand. We'll talk about the Fed's commitment to a 2% inflation rate a little bit later.
- 4.30** But a vertical supply curve, in economists' talk, represents an exogenous supply of money. It's not coming from within the market; it's coming from outside. In a gold

standard, the quantity of money is endogenous. It's determined within the theoretical system. Here, it's outside. So the Fed has the arbitrary power to make it what it wants.

- 4.31** All right, we've got the demand curve, we've got the supply curve. Where they meet is a quantity of money and a purchasing power of money that coincide, that match, the quantity demanded at this purchasing power, equals—the quantity demanded is shown horizontally—equals the quantity supplied, also shown horizontally. There is a purchasing power of the dollar that clears the market, that satisfies holders of money. They're not trying to acquire more. They're not trying to shed excess money.
- 4.32** If the purchasing power of money were something higher, say up here, the quantity supplied would exceed the quantity demanded, which means there would be a surplus of money. People would have more money than they wanted to hold, given the value of each unit. What do you do if you have money burning a hole in your pocket? You spend it. People spending it would drive down the purchasing power of money. We have an excess until we reach this equilibrium point.
- 4.33** And on the other side, if the purchasing power of money for some reason were lower than that, lower than the equilibrium, we'd have a quantity demanded that exceeded the quantity supplied. There's a shortage of money at that purchasing power. What do people do in response to feeling that their bank account is too low, relative to the spending they want to do? They need to accumulate more money.
- 4.34** How do you do that? Well, you try to sell more or you borrow. People trying to sell more and buy less are going to raise the purchasing power of the dollar, because they're going to lower the price of goods, and the purchasing power of the dollar is how many goods you get per dollar. They're going to raise the purchasing power, till you get back to equilibrium. That's a standard supply and demand analysis of why this equilibrium is stable.
- 4.35** All right, now let's consider a change in monetary policy. Like I said before, you always draw a supply and demand curve so that you can start shifting them. Suppose the Fed pursues a more expansionary monetary policy, and the supply curve shifts to the right. Now I need superscripts. The new equilibrium is down here, which is telling us what? If we increase the quantity of money, if I've drawn it correctly, it's about doubled, we're going to cut the purchasing power of money in half.
- 4.36** Because what people care about is not the number of dollars they have, but what they can buy with the dollars they have in their wallets and in their bank accounts. So if you can't—sorry, at the initial purchasing power of money and the new supply, we got a big excess supply. That's going to drive the purchasing power of money down, as we discussed a minute ago. People are going to spend off their excess, and in doing so, they're going to drive prices up. That's tantamount to a reduction in the purchasing power of the dollar.

- 4.37 If we've drawn the demand curve correctly, it's proportional. Double the quantity of money, reduce the purchasing power by half. Because at a halved purchasing power of money, you need twice as many units to accomplish the same transactions. If prices are doubled, you need twice as many dollars. And we'll come back to this proportionality, but this is a feature of what's called the quantity theory of money. The quantity theory of money is a theory that is designed to apply to fiat money—well, maybe I shouldn't say designed to, but which applies to fiat money, not in the same way to a commodity money.
- 4.38 Now, I've been talking about the quantity of money as just one concept, although a minute ago, I guess I did distinguish between money held by the public and the money stock on the balance sheet that represents direct liabilities of the Federal Reserve system. So this underlying quantity of money is sometimes called the monetary base, so that's Federal Reserve notes held by the public, plus bank reserves, either in the form of Federal Reserve notes or in the form of deposits on the books of the Federal Reserve.
- 4.39 That's a narrow concept of money. But when we talk about households, we've got a lot more money than just currency. We've got bank deposits, spendable bank deposits, and those are supported by the bank reserves that banks have. So we need a broader measure of money held by business firms and households. And the term that's used by the Federal Reserve and monetary historians is M2, for currency held by the public plus bank deposits.
- 4.40 There used to be a distinction between M1 and M2, where M1 just counted checking accounts and M2 counted that plus savings accounts. But that distinction has been so blurred because there's now practically unlimited transfers on what used to be considered savings accounts. The Fed, in fact, redefined M1 in a way that made it almost identical to M2. It's not really worthwhile to distinguish anymore. Let's just talk about all bank deposits, M2.
- 4.41 Now, the purchasing power of money, purchasing power of the dollar, is usually measured by the consumer price index. What is the consumer price index? It's the number of dollars it takes to buy a typical consumer basket of goods. Index, meaning some base year, the number of dollars it took to buy a basket, is set at 100, and later years are defined—the number is relative to that. So it's not an absolute number of dollars; it's how many dollars it takes today by comparison to what it took in some previous year. So, if the base year is 100, it might be 225 today, meaning prices have gone up 125%.
- 4.42 So, the CPI, there are people with clipboards who go into supermarkets and write down prices and report them back to the Bureau of Labor Statistics, and they record the CPI. And the weight given to different goods, to housing, to fuel, to groceries, is based on what consumers report to surveys about what the proportions of their budget are spent on. So the change in the CPI is not an indicator of how much everything has gone

up, because everything doesn't go up by the same percentage, but it's supposed to be a weighted average of how much the cost of living has gone up.

- 4.43** There are alternative measures that, instead of using a consumer survey to set the weights, use the actual expenditures recorded. That's the GDP deflator. Or if you just count the household expenditures, the personal consumption expenditure deflator, so that leaves out business expenditures. But all of these are rough and ready measures of how much the purchasing power of your dollar has changed, because your basket of goods is not the same as the CPI basket. Not everybody consumes the same bundle of goods. So it's not like a scientific measurement of some physical constant. It's a rough and ready measure.
- 4.44** Now, there's one other important measure, and that's the measure of total income in the economy, or gross domestic product. It's kind of in the background here because I drew the demand curve here rather than somewhere else, based on how much income people in the economy have. The more income they have, the more transactions they plan to make, the greater number of dollars they're going to want. So, in the background is nominal income, that is income in dollars.
- 4.45** You can either measure income in dollars or you can adjust for changes in the purchasing power of the dollar. That measure we call real income. So to find real income, you take the total income in the economy, gross domestic product, GDP, which is total sales of final goods and services, and then adjust for the change in the CPI, so that if you compare two years, you can ask: how much did total spending go up just measured in dollars? And then you can ask: how much did total output or sales of goods go up in real terms, adjusting for rises in prices, so that we get at the quantity of goods? How much better off are people, really, in terms of how much consumption they're having in real terms?
- 4.46** I referred a minute ago to the quantity theory of money. The quantity theory of money is centered on an algebraic version of this diagram. Something called the equation of exchange summarizes it. The equation of exchange says total spending, which we can decompose into the product of the number of dollars spent and the times per year each dollar turns over, so this is dollars times per year. So if each dollar turns over five times a year, then total spending is five times the quantity of money. That's total spending.
- 4.47** Then total receipts of spending have to be equal to the total spending, but we can decompose it a different way into the product of the average prices paid. That's dollars per bundle. Then real income is total bundles per year, whereby a bundle, I mean an array of goods that the typical household buys. This is an equation. It's actually true by definition, because we define this variable V , sometimes called the velocity of money, although that's a little misleading. It's not about how fast your checks clear. It's not about speed, really. It's defined as— three lines for true by definition—total spending or income, divided by the money stock.

- 4.48 This equation of exchange is a truism. It's true by definition of V , but that equation by itself doesn't give you a theory, doesn't give you a causal explanation of anything. We have to make some assumptions about what causes what. I've already mentioned one, which is that the quantity of money is exogenous. It's not caused by the other things in the system. The quantity of money is not caused by velocity or the price level or real income.
- 4.49 Secondly, there's a desired ratio of money balances to income. If you divide M by PY , you get the inverse of velocity. But this desired ratio of money balances to real income is also something that is not determined by the quantity of money. It's a real variable. It's the percentage of their income that people want to hold in the form of money balances.
- 4.50 If you grant me those two assumptions, then—oh, sorry, I need a third assumption, which is that real output is independent of the quantity of money, at least in the long run. That is, the quantity of money just measured in dollars. I'll try to make that more plausible in a second.
- 4.51 That leaves one endogenous variable that's going to be determined inside the system, and that's the price level. So, the quantity theory of money, expressed in terms of this equation, is saying that when you increase the money supply, the long-run result is going to be a proportional rise in the price level. Because it's not going to raise real income, it's not going to change the desired ratio of money to income, which is expressed in velocity. Changes in the supply of money are neutral with respect to velocity and real income, at least in the long run. But I hope this isn't getting too technical.
- 4.52 It's an important distinction to make between nominal variables and real variables. Money supply is a nominal variable. Velocity and real income are real variables. Price level is a nominal variable. What does that mean? They've got dollars in their measurement. They're measured in dollars, either just dollars or dollars per bundle. But the other two don't have dollars in the measurement.
- 4.53 One way to express the quantity theory is to say that a change in a nominal variable, a change in the nominal money stock, can only affect other nominal variables, so the price level. If you multiply the price level times real income, you get nominal GDP, GDP measured in dollars. That's a nominal variable. That will also go up with an increase in the money supply. But how can I claim that it doesn't affect velocity or real income? Wouldn't it seem that if you introduce more money in the economy, that the economy becomes more lively? Well, it may be in the short run, but let's think about the following thought experiments.
- 4.54 Suppose you live in an economy and you're happy with your money balances. Everybody's happy with their money balances, given the price level. So we're in a monetary equilibrium. Quantity demanded equals quantity supplied. Now we decide

to, instead of measuring things in dollars, let's measure things in dimes. Well, the numbers on all price tags are going to be multiplied by 10. But so is your bank balance going to be multiplied by 10. And so is the money in your wallet going to be multiplied by 10. If we also take all existing debt contracts and multiply them by 10, nothing real has changed.

- 4.55** Nobody's richer. Nobody's cash balance buys any more than it used to. No relative prices have changed. The price of milk relative to the price of meat, they've both been multiplied by 10, so there's no change in the ratio of the two. If we were in equilibrium initially, we're in equilibrium still. This sounds fanciful— let's multiply all the units by 10. But some countries have actually done something like this. After they've been through a high inflation and price tags have many, many zeros on the ends of them, which is inconvenient, you sometimes get a monetary reform where they just lop off zeros.
- 4.56** Years ago, I was in Yugoslavia, when there still was a Yugoslavia, and they had just gone through a reform where they lopped four zeros off of the monetary unit. You had to be careful as a tourist, because there was old currency and new currency, and you had to not get fooled. In Brazil, they've done this a few times. And when you just lop zeros off of the monetary unit and create a new Brazilian real, you also lop them off of all contracts and all price tags, and nothing real changes in that moment. It just becomes a little easier to calculate without so many zeros.
- 4.57** So that's a pretty clear case in which real output is not going to change, just because you've increased the nominal quantity of money. Instead of multiplying by 10, suppose we clone all the existing money. For every dollar, you now have \$2 in your wallet. For every dollar in your bank account, you have \$2. At the same time, we multiply all price tags by two. Well, you've got enough money to pay these doubled prices because you have twice as much money, so pretty much the same thing. There's nothing real that's changed.
- 4.58** Now, to get a little more realistic, suppose the Fed doubles the money supply. Well, if the way they distribute the new money is to give everybody money equal to their existing money balances, then it's the same as the previous experiment. Everybody's money balances have been doubled. But wait, we haven't doubled price tags yet. Well, but spending off the money people have, which will be excess at the old price level, it's going to drive the purchasing power of money in half. It's going to double prices. That will bring about higher prices in proportion to the quantity of money.
- 4.59** In the long run, after all the price adjustments have taken place, you're back to equilibrium, and the price level has changed in proportion to the quantity of money changing. But there is this transition where, for a while, there's an excess quantity of money, and that, if people aren't immediately aware of it, can have an effect on real output. So unexpected or unanticipated or not completely taking account of changes in

the money supply, which is most of them in practice, can have temporary real effects until spending changes the price level in proportion to the change in the money supply.

- 4.60 So behind the scenes, real income is being determined by real factors. The output of the economy in terms of loaves of bread and tons of steel and so on is not determined by how many wrinkly pieces of paper are running around. It's determined by how much labor there is, the skills of labor, how much capital equipment there is, what raw materials there are, and so on, how many hours people work. Those are real factors.
- 4.61 But you can have this transitional effect if, when you double the money supply, people don't immediately go out and spend off all the excess. Then it will take a while to get back to equilibrium at a lower purchasing power of money. People do tend to treat their money balances as an inventory or a buffer stock, meaning you don't immediately go out and spend your income, or if you get a windfall, you don't immediately go out and spend it. People take a while to figure out what they want to do with the excess money they have.
- 4.62 People are okay with letting their money balances build up a little bit above normal. But if they keep building up and keep building up, then you've got money burning a hole in your pocket, you will respond and spend it. But because the responses are a little bit slow, a little bit lagged, you can have transitional changes in relative prices. People who spend off the excess money first can buy things at lower prices than people who take a while to adjust their spending. So you can have relative effects. So that's the money and the price level story.
- 4.63 There's a kind of prejudice, I think, in Keynesian economics, that more spending is good to stimulate output in the economy, and that if people sit on their money balances and don't spend them, they're somehow restraining output in the economy, because the assumption is the economy is, for some reason, below its potential. It's not producing as much as it would if people were spending more vigorously. So there's a kind of prejudice against hoarding money. Keynes, during the Great Depression, said this pretty explicitly.
- 4.64 But any time the economy is below potential, as people try to measure it, there is this idea that we need to stimulate more spending. And so people sitting on their money are restraining the economy, holding it back. Against that, we need to appreciate why it is that people hold money and don't spend it immediately. There is a benefit you get by having money on hand, namely that when something comes along that you do want to buy, you're able to buy it without having to wait until you earn more money or sell something else to get the money to buy it.
- 4.65 So, behind the demand to hold money, there is a convenience that's provided by money balances. The economist William H. Hutt talked about the yield from money held. And if you think about it, there has to be a yield from money held; otherwise, people wouldn't hold it when they could be earning interest. Instead of holding money,

you could go out and buy some share of stock or a bond or something that would pay you more interest than the cash in your wallet, which pays you nothing, or the money in your checking account, which pays you very little.

- 4.66** So there's some service yield or convenience yield provided by money. And so, it's a mistake to think that the money is somehow idle when it's sitting in someone's bank account, or it's not being used or that it's restraining the output of the economy.
- 4.67** So, we've determined the equilibrium purchasing power of money. Again, the purchasing power of money is how many bundles you get per dollar. It's the inverse of the price level, which is dollars per bundle. To say there's an equilibrium purchasing power of the dollar is the same as saying there's an equilibrium price level. If there were more money than that—sorry, if the quantity supplied exceeds the quantity demanded, that'll change the purchasing power of money.
- 4.68** This has been called the fundamental proposition of monetary theory, that the way equilibrium is established in the market for money balances is by adjustment of the price level. This is a fundamental proposition not of all monetary theory, but of fiat monetary theory. Because the quantity of money is exogenous, that's not going to adjust; whereas we saw under a gold standard, a country that has more gold than people want to hold will spend it on buying imports, and the gold will leave. That's the price-specie-flow mechanism.
- 4.69** On a gold standard, the quantity of money is endogenous. But here it's exogenous, and the thing that adjusts is the price level. So, when more money is introduced into the economy, what can you do when you have more money than you want to hold? You can spend it. But your spending doesn't make it disappear from the economy. You're just passing it on to somebody else. Now, they have more money than they counted on holding or wanted to hold, relative to their other assets. And so, money acts like a hot potato.
- 4.70** Maybe I need to explain the analogy, which means it's not a very good analogy if you need to explain it. But there used to be this very cruel children's game called Hot Potato. It's like musical chairs, except that you put a—so you did it at a birthday party. You put a potato in the oven until it's too hot to hold, take it out, and toss it to the first kid, and they scream, get second-degree burns, and toss it to the next kid. And the point of the analogy is that people pass the potato around, but the potato doesn't disappear. What happens is, the potato cools off until somebody is able to hold it, and then they win the prize, like the last kid in the musical chairs.
- 4.71** So spending the dollars doesn't make them disappear, but it does make them cool off in the sense that, by pushing up prices, it reduces the purchasing power until the total amount of money buys the same real output as the initial quantity of money, which means less purchasing power per unit. You get proportionality of the price level to the

quantity of money. If you increase the quantity of money to 10%, you're going to get a 10% higher price level.

- 4.72 Now, I haven't said anything about where the new money enters the economy. Well, I guess I did briefly. I said, typically, the Fed goes out and buys bonds, so the money enters the economy first in the bank balances of bond dealers. But there are other ways to distribute new money. In the Paycheck Protection Program during the pandemic, basically, the new money entered the recipients of these checks from the treasury, entered the bank accounts of the recipients from the treasury; and so it was very widely distributed to begin with.
- 4.73 Does that make a difference? Well, it might make a transitory difference to how relative prices change when the excess money is being spent off, because we have a different group of people spending it. They might buy different goods, they might raise relative prices. But if we ask: does it matter for determining the ultimate price level? Probably not much. It's probably pretty negligible. In the long run, we just get the price level rising, and any changes in relative price are probably negligible.
- 4.74 But in the short run, it can be important. In fact, whole theories of the business cycle are built on the idea that, if you inject money into bond markets first, you will drive interest rates down, and that will have spillover effects on the kinds of investments and the quantity of investment businesses make. You can overstimulate investment. That's the Austrian theory of the business cycle. You can have important transitional effects, but transitions can be very important. Transition here is the business cycle, before you get back to a long-run equilibrium.
- 4.75 So that's the price level. What about inflation? Inflation is simply the rate of change of the price level in percent per year. So, an inflation of 5% means the price level has risen 5% over a year. Inflation is an ongoing rise in the price level, the general level of prices. It drives me crazy when people talk about inflation in college tuition or inflation in hospital costs. Those are price changes which may or may not be representative of the broad array of goods you can buy with your money. So, let's reserve the term inflation to a change in the general purchasing power or the price level, where price level is being measured by a broad array of goods, not just by one good. If you want to express it in terms of the purchasing power of the monetary unit, inflation is an ongoing shrinkage in the purchasing power of the monetary unit.
- 4.76 It's helpful to think about it that way because it naturally leads you to ask, well, when the value of something goes down, it must be that supply is growing faster than the demand. And that's the case. If the purchasing power of money goes down, it indicates that the supply of money is growing faster than the demand to hold money; whereas if you just say the price level is rising, you don't immediately think of money as the cause. And there are administrations who have done all they can to distract people from money as a cause of inflation. They blame inflation on all kinds of things. But if the price

level is rising, it means the quantity of money is growing faster than the real demand to hold money.

- 4.77** Because the Fed controls the growth rate of the money supply, the Fed controls the inflation rate. So attempts to blame inflation on other things are distracting us or diverting us from placing responsibility where it belongs, which is on the central bank, which controls the quantity of money. So the Fed could make inflation 0% if they produced money growth, which they are capable of doing, consistent with keeping inflation to 0%. Of course, the Fed admits that by having a 2% inflation target. So if they can make inflation 2% by controlling the quantity of money appropriately, they could make it 0% by having money grow 2% less than was appropriate for 2% inflation.
- 4.78** What about some other popular theories of inflation? What about budget deficits? How are they connected to inflation? Well, if we don't have monetary policy changing the quantity of money, how can the price level go up through budget deficits? It would either have to be that real income goes down, but that's not what the advocates of running expansionary fiscal policy want to happen. Or there has to be a rise in velocity, that is, the turnover rate of money, which implies people don't want to hold as much money relative to their spending.
- 4.79** But the connection between a budget deficit and a rise in velocity is pretty weak. Budget deficit would have to raise interest rates by enough for people to say, "At these high interest rates, I don't want to hold as much money. I'm going to spend it faster." And both of those links are pretty weak empirically; that is, when the U.S. runs a big budget deficit, we don't see a big rise in world interest rates, because that is an increase in the demand for loanable funds, but it's small relative to the entire world market. Then the second part of the link: demand to hold money is not all that sensitive to interest rates.
- 4.80** But that was holding the money supply constant. There's another way a budget deficit can cause inflation, and that's by leading to an increase in the money supply. How does it do that? It does that if the central bank decides to do that, if they decide to respond to the deficit by printing more money. Why would they do that? Well, the treasury might say to them, "Look, if we try to sell all this debt, which we have to borrow to cover the deficit—deficit means we're spending more than we're taking in taxes; we have to borrow the difference—if we borrow a lot, that's going to put pressure on interest rates upward. We don't want to see that. Why don't you help us out by buying these bonds that we are trying to issue? Why don't you lend us the money, in effect?"
- 4.81** Many central banks around the world do that. They knuckle under to the demands of the treasury to print more money to help pay the government's bills. If you look at countries in hyperinflation, that's basically always the case. The reason inflation got out of control was that the central bank was printing money to pay the government's bills and just neglecting the effect it was going to have on the inflation rate.

- 4.82 In the U.S., we had a lot of money creation. We had the big expansion in the monetary base during the years of huge budget deficits surrounding the pandemic. It's hard to say it was causal, but it was certainly coincidental that we had a big monetary expansion at the same time we had a big budget deficit.
- 4.83 There was another thing going on, of course, which was, in the pandemic, people did want to build up their money balances relative to their spending. People cut back on their spending. And because of uncertainty, they wanted to have more money on hand. They had to cut back on their spending because restaurants were closed and the airlines weren't flying and there weren't as many things you could spend money on. So with an increase in the demand to hold money, it was not a bad policy for the Fed to create more money, to let people hold it, satisfy their demand, without requiring a big change in the price level to bring about a change in the real quantity of money. Because it would have been a depression. It would have been a sharp deflation.
- 4.84 But this increase in the demand to hold money during the pandemic was temporary. The Fed should have recognized that, when things return to normal, people are going to spend off these high money balances that they've accumulated. The Fed was very slow to respond to that, and that's how we got 9% inflation. Velocity returned to normal, spending returned to normal, and the Fed did not pull out the excess money.
- 4.85 What about greedflation? There was a coincidence in the recovery from the pandemic between rising prices and high corporate profits. But the profits are not causing the spending; the profits are a result of the increase in spending. People were unloading their excess money balances. Sales were brisk. That drove up business revenue. That drove up profits. So the high profits and the rising prices were the results of the surge in spending, which was the delayed result of the surge in the money supply during the pandemic. The profits weren't causing the inflation. They were both caused by the surge in spending.
- 4.86 This simple equation, and then the theory, the quantity theory of money, is very helpful in understanding the difference in inflation across countries, because the main difference is money growth—countries with high inflation or countries with high money growth. It could potentially be differences in velocity growth or real income growth, but empirically, those aren't that big, the differences across countries. Whereas a country with high inflation surely has high money growth. There's no other way to do it. Across decades, if you want to explain why the '70s was a decade of high inflation, it's because it was a decade of high money growth. Lots of empirical studies have been done across countries, and there's a pretty tight relationship between money growth and inflation.
- 4.87 One more thing I want to say about inflation, which is that, when people are lending money, they're taking inflation into account. If there's 10% inflation and you're lending money for a year, you know that the dollars you're going to be paid back a year from now are going to buy 10% less. How do you take that into account? You tack 10% onto the interest rate you're charging.

- 4.88 So, countries with high inflation also have high nominal interest rates—by nominal, I mean not correcting for inflation. Just look at the headline. Not far from here, in El Paso, Texas, you can get—I didn't look these numbers up this morning, sorry, but I'll make some numbers up—you can get 3% on a savings account, whereas if you cross the bridge into Juarez, you can get 8% on a savings account. So why doesn't everybody cross the bridge and get 8%? And the answer is: the 8% is not in dollars; it's in pesos.
- 4.89 And why is it higher? Because inflation is higher in Mexico. The purchasing power of those pesos is not going to be any higher, even after you add 8%, than the dollars after you add 3%. Countries with high inflation are going to have high interest rates, but only in nominal terms, not in real terms, not in terms of what you can buy with the return. So if you want your country to have low interest rates—and President Trump says he wants low interest rates—and if you want those low interest rates to be sustainable, they have to be based on low inflation, and low inflation has to be based on tight money, on low money growth. And yet, Trump and, say, Erdoğan in Turkey, have had the opposite view, that if we flood the market with money, we can drive interest rates down. Well, temporarily only, because once the inflation takes hold, interest rates are going to go up to incorporate that inflation.
- 4.90 All right, so that's the story with our fiat money. Supply of money is controlled by the central bank, therefore the inflation rate is controlled by the central bank. Therefore, nominal interest rates follow along. If we want low inflation, we have to convince the central bank to pursue a non-inflationary monetary policy.
- 4.91 The Fed says that it has a 2% inflation target, but it's more of a guideline than a rule. It's a very loose target, where inflation had to get up to 9% before they started reacting to the rise in inflation. They're not hitting 2% every period, and they're a little lackadaisical about bringing it back down. They claim they're committed to it, but inflation in the U.S. has been stuck at about 3% for a few years. They seem to be loosening monetary policy, rather than tightening it to bring inflation back down to 2%. It's a very weak commitment.
- 4.92 We'll come back in one of the later lectures to talk about ways to make that commitment stronger. Are there ways to get better performance out of central banks in a fiat money system?

Lecture 5: Government in Money

At a glance — Government's role in money.

- 5.1 All right. I want to return in this lecture to the question that was raised in the first lecture, which is the role of government vis-à-vis the market in the production of monetary services. Economists don't like to use the word "should." That's normative. That's for philosophers. But they do like to talk about what's in the public interest or what contributes to public welfare. So let me rephrase the question. Not "should" government play a role in money, but is there some welfare-improving role that government plays or can play in money or needs to play in money?
- 5.2 Some economists take it for granted that if we see government playing a role, if we see central banks, if we see government-run mints, there must be a good reason for that. There must be something in the public's interest to have it done that way. I don't make that assumption. I actually want to look at the case. I want to ask: when private markets produced coins, what did that look like? Are government mints an improvement over that? In that case, pretty clearly, they were not an improvement over that.
- 5.3 So the question of: is there a rationale for government to play a role based in improving public welfare is very different from asking: why did governments historically take on this role? Governments, especially ancient autocrats and medieval monarchs, probably didn't have the public welfare in mind when they decided to monopolize the mint; so they had other things in mind than public welfare.
- 5.4 The technical term for an improvement in public welfare that the economists who specialize in these questions use is, an improvement in welfare is a Pareto improvement. Pareto is the name of an Italian economist who talked about the idea that there are some things that can be done that make some people better off without making anybody worse off. To do that is now called a Pareto improvement. So, for example, an ordinary trade between two individuals, each of whom values the thing they're getting more than the thing they're giving up, makes both individuals better off, so win-win. Two people are better off. Nobody else is affected. That's a Pareto improvement.
- 5.5 We pretty much count on markets to bring about mutually beneficial trades, but there are well-known cases where that presumption is absent. Welfare economists talk about market failures, market failures where market forces don't bring about all the improvement in welfare we could get. So, a wise government can intervene in a way that makes some people better off and nobody worse off. That's the theory.
- 5.6 And what are these cases? What are the causes of markets to fail to achieve all the Pareto improvements that are available, or to bring about a Pareto optimum? There are two main categories that I'll talk about. One is called public goods, and connected

with that is the notion of external benefits and costs, meaning benefits to other people that are not brought to bear on the person providing them, so they provide too little, or burdens placed on other people that are not brought to bear, so too many of them are provided. So public goods and externalities, one category.

- 5.7** The other category is natural monopoly. If market forces bring about a single producer in an industry, that producer is in a position to impose monopoly prices; that is, restrict output and charge higher prices than a competition among producers would produce, it would bring about. There is potentially a role for limiting the monopoly pricing power of the monopolist. Having government as monopolist may or may not be an improvement, but in principle, the argument is that they can improve on the monopoly outcome.
- 5.8** Now, what I want to ask is: do either of these categories apply to the provision of money? Do they provide a sound rationale for the government to get involved in producing money or in restricting what's done by those who do produce money? Natural monopoly, I don't think there's much in that box. The production of coins is not a natural monopoly. The production of banknotes is not a natural monopoly. That is, the market will sustain many competing producers. There's not going to be a tendency for one firm to take everything over. We don't see that historically.
- 5.9** One place we do see a tendency toward concentration down to a single entity is in the innerbank clearing system. Banks who are accepting deposits in the form of checks written on other banks need to present those checks to the bank they're issued on and get paid for them. Where banks meet to do that is called the clearinghouse. Banks in a given city, it's most efficient for them to all meet in one clearinghouse, to meet in one place, so they can do multilateral clearing and settlement, rather than a series of pair-wise meetings.
- 5.10** So there is potentially a problem if the clearinghouse decides to impose monopoly prices, but clearinghouses don't do that because the clearinghouse is formed by the banks that are participating in it. It's almost always structured as a member-owned cooperative. They're not going to exploit their members. Their members would vote against it. Clearinghouses are a natural monopoly, but it's not a monopoly pricing problem. There's no monopoly pricing problem that arises because of that.
- 5.11** So the interesting arguments we're going to find in the public goods and externalities category. When I use the term public goods, I want to warn that I don't mean the public good in the sense that political philosophers sometimes use it, to mean what's really good for people, whether or not they want it. I don't mean that. I mean public good as a technical concept of a good that is underprovided by markets, relative to what people really want. We can get more satisfaction by arranging collective provision, is the idea.
- 5.12** Public good does not mean any good that is, in fact, provided by governments. Governments provide lots of private goods—that is, goods that could be provided

privately. So governments run bus systems. Lots of private companies run bus systems. Governments own tennis courts—local governments— tennis courts, golf courses. Those are not public goods just because they're provided by taxpayers. They're still private goods.

- 5.13** Public goods, in the sense I'm using it, is defined by two characteristics that I'm going to need to explain. The first is non-rivalry in consumption, which means that one person's enjoyment of the good doesn't diminish the amount available to anybody else. I'll give an example in a second. And the other is non-excludability. It's difficult when the good is provided to some people to prevent other people from also enjoying it. It's difficult then to charge users according to whether they're using it or not. There's some barrier to charging people.
- 5.14** So, an example of a private good, a good that is definitely rival in consumption, would be a chocolate donut. It's my favorite example. So I eat the chocolate donut; you cannot also eat the chocolate donut. My consumption of it is rival to your consumption of it. And so it's not that hard to have markets provide chocolate donuts. People who want to eat them have to pay for them, because you can't just enjoy the chocolate donuts that other people have purchased.
- 5.15** An example of a good exhibiting non-rivalry in consumption, and I choose this example deliberately because it's not provided by the government, is broadcast radio or broadcast television in the old days, when there was such a thing. So I can tune in a radio station and enjoy the programming. It doesn't diminish the amount that you can enjoy it. It doesn't exclude you from—sorry, it doesn't reduce the amount of signal that you get, that I tune it in.
- 5.16** And of course, radio is provided, over-the-air radio is provided privately. Well, except for— well, I guess NPR is now private. It used to be 90% privately funded. Now it's 100% privately funded. But mostly it's funded by advertising. Advertising is a device for getting somebody to pay for it by making you, the listener, sit through the commercials. The second characteristic I mentioned was non-excludability in supply. So when a city runs a bus service, they typically do charge people to ride the bus, but they may or may not cover their costs through the fares they charge. But it's easy to exclude people who don't pay from getting on the bus.
- 5.17** An example of something where it's difficult to exclude people, maybe you remember there was concern decades ago about a hole in the ozone layer. And dangerous radiation was coming through because we had depleted the ozone layer through the use of aerosol sprays and things like that. It would be difficult to go around and charge people, say, "We're going to remedy the hole in the ozone layer for you, but only if you pay. If you don't pay, we're going to leave the hole over your backyard." No, it doesn't work that way. You've got to fix the whole thing. So you can't provide the benefit only to the people who pay. If you fix the hole in the ozone layer, you fix it for everybody, whether or not they paid.

- 5.18** If a good has this characteristic, that it's hard to charge people, whether or not they're enjoying it, nobody's going to pay. As long as it's provided, I'm going to be able to enjoy it, because they're not going to be able to exclude me. And if nobody pays, well, maybe it doesn't get provided, but I would be a chump if I paid when I don't have to. But it's going to be difficult for for-profit businesses to provide something if they can't charge users.
- 5.19** So charities provide goods that it's difficult to charge users for, but arguably, they're not capturing all the potential gains. They're not providing enough to reach the optimum. They're not providing everything for which people are potentially willing to pay. The idea here is that nobody's worse off, meaning everybody's getting benefits in excess of what they're being asked to pay.
- 5.20** Now, I have a book that's not on money, called "The Clash of Economic Ideas," in which I talk about this public goods theory in more detail and provide some criticism of it. But that's not my purpose today. My purpose today is to just ask whether money has this problem of exhibiting non-rivalness in consumption or non-excludability or market failure.
- 5.21** So, non-rivalry? No, not on the face of it. The money in my wallet is not available to you; it's not providing any services to you. You're not enjoying the benefits that I'm getting. The money in my checking account, same story.
- 5.22** There's not a problem of excludability. You can't write checks on my bank account. You can't use the money in my wallet. And of course, we don't have any problem with private provision of checking accounts. And in the 19th century, before we outlawed it, we had no problem with private provision of currency, of banknotes. In the example of private coinage, we didn't have a problem getting private mint-masters to provide coins. So these are private goods. There's no evident market failure. There's pretty easy exclusion of non-payers.
- 5.23** Let me give an example of, a pretty weak example of something monetary being called a public good, and then we'll deal with the harder cases. Some economists have said that, under the Bretton Woods system, where the United States was providing a reserve currency for other nations, or even today, when a lot of countries either peg to the dollar or keep central bank reserves in the form of dollars or dollar bonds, some economists have said that the U.S. is providing an international public good when it does that.
- 5.24** Well, it's an international good, but it's not public. You have to pay for it. Dollars are rival, dollars are excludable. So any central bank that holds dollar assets had to pay to get them, had to give up something to get them. We're not just showering benefits down on people who don't pay, the way that happens with the provision of a public good where users are not excluded. So, the argument about money has to be a little more subtle than that. It's often couched in the form of an externality problem; that is,

there are significant benefits that are being provided that are not being taken into account and therefore are being underprovided.

- 5.25** There are two categories of these arguments that I want to address. One is, when people choose which money to use, they're not taking sufficient account of the benefits they're providing to other people. There are uncaptured external benefits in your choice of which money to use. That's the first set of arguments. And the second one is going to be that you don't take enough account of the benefits to other people when you choose how much money to hold. If you hold more money, the argument goes, this is providing benefits to other people which you're not taking into account; therefore, you're not holding enough money.
- 5.26** Now, the first one, that there are benefits to other people from your choice of which money to use, is usually centered on the idea that it's a benefit to other people if you choose to use the same money that they use. So commonness or convergence on a common money is the benefit you're providing to other people. But we talked in the first lecture about Menger's theory, pretty persuasively, that there's an incentive of market participants to converge on a common money because it's not only providing benefits to your trading partners, it's providing benefits to you to use the money that they would prefer you to use.
- 5.27** And how is it brought to bear on you, that you should choose money A rather than money B? Well, your trading partners, if they want you to use money A, are going to offer you a better price if you pay in money A than if you insist on trying to pay in money B. So in a sense, they're going to bribe you. They're going to give you a better deal. That internalizes the benefit to others.
- 5.28** And so, you want the money that's most common, has the most potential trading partners, so do the most other people want you to adopt that money. So there's no divergence there. I don't think that argument is going to work. So it has to be more subtle than that, that somehow we don't converge fully enough, or we don't converge on the right money, or we don't converge as quickly as we should. I'll come back to those.
- 5.29** So money does have this property that there tends to be convergence on a common monetary unit. It's most convenient for me if I don't have to change money between getting paid and going to the store. I like to be paid in the stuff that I can actually turn around and spend. Money has this property that's sometimes called—it's a network good, or at least the monetary unit is a network good. There are benefits to being a member of a large network rather than a small network. And so by joining the network and enlarging it, I'm benefiting other people in the network. I'm making it easier for them to trade with me, with a broader group of people.
- 5.30** I think that's true, but I don't think it's a problem because this benefit, as I just indicated a second ago, can be appropriated. It's not completely external to me

whether other people benefit from my using the money. The only way they benefit is if I trade with them. And when I trade with them, they can internalize it. They can offer me a better deal if I'm paying in the money they want to be paid in, than if I'm paying in something else or trying to pay in something else. So that internalizes the benefits. They aren't external benefits anymore. There's no divergence between the money it's in my interest to adopt and the social interest in my adopting a common money.

- 5.31** Now, there could be a problem of people choosing not to join a monetary network if there were some fixed cost of joining, and once they're in the network, they're benefiting a lot of people a little. If we could collect a little from this lot of people, that would cover the fixed cost of joining. But if that doesn't happen, then I'm not going to join. If there's a fixed cost of joining, conceivably, some people don't join the network who could be bribed to join the network, but it would take collective action to bribe them.
- 5.32** Well, I think that is also a non-problem, just evidenced by the fact that everybody does use money. We don't have any holdouts who are in barter, who are saying, "Well, I would use your money if you paid me to overcome the cost of switching to using money." There isn't any big fixed cost of switching from barter to using money. So I think that's an irrelevant hypothetical.
- 5.33** Now, it's a more subtle question whether we get convergence fast enough, because it's difficult to say what is fast enough. Convergence on a standard is a problem not just in money; it's a problem in all kinds of technologies. Some years ago, there was a battle between Blu-ray disc and high-definition DVD. Blu-ray won, HD DVD fell by the wayside. Of course, nobody cares about Blu-ray anymore because now everybody streams. But it's an example of a competition among standards. If you were buying a video recording device, a disc player, you had to worry, "Will I get orphaned if I buy the wrong system?" How do you avoid that? You read what other people are adopting, you read what analysts say is the technically better system.
- 5.34** To imagine that government would improve on this is to imagine that government would know better than people in the marketplace which is the better standard and which is the standard to which everyone else will converge. But there's no reason to think that the government knows that any better than the technical experts. There are examples, actually, where, in Japan, the government imposed a high definition TV standard, which quickly became obsolete because it was an analog standard, and then digital TV came along. So it didn't help along the process of providing the most technically advanced system.
- 5.35** There's a famous argument by two economists, Brunner and Meltzer. There was an act passed by the British legislature, going back into 19th-century history again, called Peel's Act of 1844, or the Bank Charter Act of 1844, which, basically, was responsible for giving the Bank of England a monopoly on the issue of currency. Smaller banks that were allowed to issue banknotes were now going to be no longer allowed to issue

banknotes. So we're going to go from multiple competing brands of banknotes to one issuer.

- 5.36** Brunner and Meltzer defend this by saying, "Well, this saved people from the hassle of trying to decide whether banknotes were trustworthy or not, and it just gave them the best banknotes." That seems to me to be a dangerous argument. I mean, you could make that about almost anything. There are too many different washing machines available at Home Depot. Somebody should standardize them and only provide the best system, and everybody should use the best system.
- 5.37** Competing banknotes are not that difficult to sort out according to quality. If you're offered a banknote that you're not familiar with, all you need to know is: does my banker accept it? Can I put it on deposit in my bank at 100 cents on the dollar? If not, I don't want it, or I want to discount it. But if all the banks are accepting one another's liabilities at par, which is what we saw in the Scottish free-banking system, because it's in the interest of banks to do that, there's no incompatibility of standards. They're all on the same standard, so not really a problem. So, I think it's very difficult to make the case that you make people better off by restricting their choices.
- 5.38** So, today, businesses have to deal with this problem of people presenting them with payments which are claims on different banks. Well, this was more of a problem when people wrote checks, but now people pay with debit cards. One debit card accesses your account at Citibank. Somebody else has a debit card that accesses their account at Bank of America or Chase or Wells Fargo. How do business firms, how do retailers know which ones they can accept? Well, they know which ones don't bounce, which payments actually arrive.
- 5.39** Back in the days when people wrote paper checks, there were various devices for them to check whether you had enough money in your account to cover the check. But today, the debit card, when you offer it, they immediately check whether you have enough money in your account to cover it, and then the payment goes through. And all these claims on different banks are interchangeable. They're all trading at a dollar for a dollar. Not a problem. So I don't think there's really any reason to think we don't get enough commonness or uniformity or convergence in our payment media.
- 5.40** Let me switch to a different argument, and this one, I think, holds a little more water. And that is, I can't enjoy low inflation in a dollar economy without you also enjoying low inflation. So I can't enjoy high-quality money without you also enjoying high-quality money. So there's something public about it.
- 5.41** Well, there's something public about it, or there's something shared about it, if we use the same money. If we have a system where we choose among monies, then we can have different experiences; and of course, people in different countries have different experiences with inflation because they use different monies. But there is a point to this example, which is that, yes, as long as we're both on the dollar standard, we're

both relying on the Fed to produce low inflation. And so, I can't have low-dollar inflation without you also having low-dollar inflation.

- 5.42 Well, how do we get low-dollar inflation? We have to pressure the central bank, we have to pressure the Fed, or we have to pressure Congress to pressure the Fed, or pressure the President to pressure the Fed, to produce low inflation. And these efforts at pressuring or lobbying, those are a public good. If I spend a lot of my time giving lectures, calling for low inflation and trying to make a persuasive case, the benefit, if I'm successful, redounds to everyone, not just to me.
- 5.43 Now, I'm a kind-hearted person, so I'm doing this as a public benefit, and lots of other economists do the same, but it may not be enough. I think, really, we don't get enough improvement in monetary policy because improvements are subject to the free-rider problem. And that's true of all public policy. Efforts made to improve public policy mostly redound to other people's benefits, and so their external benefits. You, the advocate of an improved policy, are not the only one to benefit. You don't capture all the benefit, and so you don't spend all your time doing it.
- 5.44 Something that is commonly acknowledged to be non-rival and non-excludable is information. I can know something without it excluding you from knowing it. Let's say a mathematical formula, the Pythagorean theorem, useful if you're tiling your bathroom. If I invent a new mathematical theorem that's as useful as the Pythagorean theorem, I'm not the only one who benefits from it. And this is the main argument made by academics when they call for federal subsidies of their research. The benefits of this research are going to redound to everybody.
- 5.45 And in the case of monetary policy, we might say, having better ideas or a better recipe for monetary policy is information that everybody can share. I use the term "recipe" advisedly, because I want to come back to my private good, the chocolate donut. There's a sense in which recipes are public goods because they're pure information. How do we get better recipes for chocolate donuts then?
- 5.46 Because you sell the donuts, which embody the better recipe, and that rewards the chocolate donut producer for having a better recipe, so even if, in some sense, anybody could use that recipe, only those who do use the better recipe get rewarded. In that way, we provide an incentive to use better recipes. And if you're first in the market with chocolate donuts made according to a new, better recipe, you do get a reward, and your sales go up relative to the people using old recipes.
- 5.47 We might say the same thing about improvements in monetary policy. If somebody comes up with a better way of running monetary policy to provide higher-quality money, the benefits are going to be not captured by them, the benefit is going to be captured by the issuer of money, but they have other incentives maybe we'll talk about. But anyway, we get too little improvement in monetary policy because it has this problem of non-rivalness and non-excludability.

- 5.48 If you wanted to come up with new money, would you have enough incentive? Would the market reward you for coming up with better money? So instead of trying to improve the existing money, introduce your own money. Can you do that? Well, even if there were no legal barriers to it, and I guess Bitcoin is an example of a project that aspires to be a better money, did the people who introduced Bitcoin get rewarded for it? Yeah, they did. People who bought Bitcoin early on were rewarded quite handsomely.
- 5.49 We have intellectual property law that rewards all kinds of intangible products. If I come up with a wonderful musical melody, I get it licensed, and people can't use that melody without paying me. We have agencies, ASCAP and BMI, that go around and monitor what music is being played by radio stations and movies and even bars, and they charge royalties that go back to the songwriter. So, we've provided an incentive to songwriters to come up with this intangible thing, which is really non-rival in consumption. I can hum it, isn't rival with your humming it, and so on.
- 5.50 So I don't think lack of intellectual property in monetary units is the reason we don't have more monetary units. I think the reason is the strong network property, that nobody wants to be the first to use money that nobody else is using. What's the point of that? The whole idea of having money is to be able to trade with people. So that's not the problem.
- 5.51 Historically, economies have evolved. I talked in the first lecture about salt and peppercorns and oxen being used as money. But economies moved away from that. They evolved. So it's certainly not an absolute barrier, the fact that there are network or convergence properties. It doesn't mean that people are permanently stuck on money if better money comes along. So Menger gives the example that, in the economies in ancient Greece, where they used oxen, that was fine as long as it was a rural economy. But in the city, oxen were not convenient to use. People didn't know the quality of an ox, so they were reluctant to be paid in an ox. And copper coinage became the most commonly accepted medium of exchange.
- 5.52 Economies don't necessarily get locked into an inferior unit of account, and we see in the world around us economies switching units of account, switching their monetary standards. In particular, in economies with high inflation, with really bad money, individuals in those economies put themselves on the US dollar. So it's true that it takes a little longer than if there weren't the network property giving the incumbent an advantage, but it's not an absolute barrier. So, at a high enough inflation differential, people in Argentina start using dollars, people in Lebanon start using dollars. And that's a switch in the monetary standard.
- 5.53 The most we can say is that the network property makes that happen too slowly, or it takes too big of an inflation differential for people to switch, so that if we were able to choose collectively, we would sooner switch to a better money. And I guess there's something to that. The problem is a practical problem. How do we know whether what

we're turning to is better money than the one we've got? Now, if you're in Argentina, it's pretty obvious that the U.S. dollar is going to hold its value better than the Argentine peso, or if you're in Lebanon, or if you're in Zimbabwe or Venezuela. But in other cases, it's not so obvious.

- 5.54** So, in Europe, they decided to ditch their—the initial 11 countries that formed the European Central Bank, which introduced the euro— they decided to ditch their 11 national currencies, the German mark, the French franc, the Dutch guilder, the Italian lira, and so on, and adopt a new currency, which is the euro. The practical problem is: how do you know the euro is going to be better, more trustworthy, lower inflation, than the currency you're giving up? It's not clear that it has been better.
- 5.55** For countries that had chronically high inflation, yes, it was better—Spain, Italy, Greece—but they always had the option to adopt the Deutsche Mark. In fact, many of those countries tried to peg their currencies to the Deutsche Mark. It's a mystery why Germany wanted to join the eurozone, or the Netherlands, because they already had a low-inflation currency. Why did they want to switch from a central bank run by German bankers to a central bank where the board was going to have one German and one Frenchman and one Spaniard and one Portuguese and one Italian and one Greek? What's their assurance that that's going to provide money as good as the one they've got?
- 5.56** They tried to write a constitution. They wrote a constitution that said the European Central Bank has one goal, and that goal is price stability. So that was meant to reassure the Germans. At the first meeting of the board of the European Central Bank, they passed a resolution that said, "When we say price stability, we mean inflation less than 2%." So that was a pretty firm commitment. Trouble is, it didn't really last. It's still there on paper, but the euro has been above 2% inflation for a long time without any urgency about returning to 2%.
- 5.57** So, to get Germany to participate, there had to be these assurances offered. But once it was ratified, Germany was no longer a veto player. They no longer could stop the central bank from pursuing more inflationary monetary policy. That's going to depend on the median voter, and the median voter is more like France than like Germany. So there are practical problems in switching to a new currency and identifying when it's better than what you've got.
- 5.58** An economist named Patrick Minford did a study of whether Great Britain should adopt the euro. Britain kept the British pound, but they were considering joining the euro; at least under Tony Blair, they were considering joining the eurozone. Blair always said he would have a referendum when the time was right, and he never had the referendum. I interpret "time was right" meaning "opinion polls show that it will pass," but the opinion polls never showed that it would pass, because the British public didn't like it. They trusted the pound more than they trusted the euro.

- 5.59 Some countries did have referenda. In Denmark, they had a referendum, and the voters said no. And so, Denmark still has the Danish krone. Swedish voters said no. Sweden still has its own krona. In Germany, opinion polls indicate they would have said no, so there wasn't any referendum. The Prime Minister wanted this project. So it's not clear that it was an improvement from the point of view of the people who were being asked to switch.
- 5.60 All right, so that's the problem of the choice of a currency. I mean, in principle, there is a little bit of, or in practice, I should say, there is a little bit of awkwardness in switching from one money to another. It doesn't stop the switch from happening when the difference is big. It may stop the switch from happening when the difference is small. And so, there may be some small benefit that's being left aside because of the network property of the incumbent currency. But it's not always clear, if you're trying to adopt a new currency, that it's going to be better than the one you're being asked to leave behind, if it's a new currency, especially.
- 5.61 Let me turn to the argument that there's some external benefit. But to summarize—sorry, to summarize the first part of the talk, I don't think the choice of which money to use is a big problem. I don't think people face incentives that diverge much, if at all, from what would be in the interest of everyone, from what would be socially beneficial.
- 5.62 Let me turn to the question of whether you don't hold enough money because you're not taking into account the benefits you're providing to your potential trading partners — not by using the same money, but by having enough money on hand. So how do you benefit them?
- 5.63 A well-known economist named David Laidler once wrote that, "Any one agent, holding cash balances of a given size, is less likely to incur the costs of running out of cash, the larger are the average balances of those with whom he trades." So if your trading partners have more money, that's good for you, because I guess you have access to it somehow. But how do you have access to it? I would argue that it's not an external benefit. You have to pay if you want to have access to somebody else's money balances.
- 5.64 How do you get more cash if you don't have it? You can borrow it. You can go to a bank. Banks will charge you for giving you money. It's not free. It's not an external benefit that the bank has cash ready to lend. You have to pay to get it. There are businesses that will buy your used goods and give you cash. They're called pawn shops. Again, they're not bestowing the money on you without charge. They are charging you. If you pawn your guitar and you want to get it back later, you have to pay more than they gave you for it. So these are not external benefits. These are internalized.
- 5.65 Now, there's a more subtle argument offered by the famous economist Paul Samuelson. I'm going to argue that he didn't state the argument properly, but Milton Friedman later did. In Samuelson, it was labeled "the non-optimality of money-holding

under laissez-faire." So Samuelson is worried about people holding the wrong size money balances, people in general, because their private incentives differ from what's best for society. And the basic argument is that people don't hold enough. There's, real money balances are too small. Another way to put it is, the price level is too high given the stock of money.

- 5.66** The way Samuelson made the argument was to say that we could have a social windfall if everybody agreed to cut prices in half. Why is that a windfall? Well, because now our money balances are worth twice as much, so we're all wealthier, so that's great. However, that's not going to last because we know that individuals who have larger money balances than they want to hold are going to spend them off. And this private action is going to cancel the social windfall by driving the price level back up.
- 5.67** So Samuelson is saying, isn't that a shame? That is an indictment of the private incentives that are leading people to spend off their money balances. They're not taking into account that they're canceling the windfall that other people were enjoying. This is a nonsensical argument.
- 5.68** One way to see it is, imagine we made it about wheat. We're a nation of wheat farmers. If we arbitrarily double the price of wheat, have we made all the wheat farmers better off? We've doubled the price of wheat, we've doubled the value of their inventories, doubled the value of their crops. Yeah, wait a minute. Not everybody can sell their wheat at double the price, because the price that was prevailing was the price that equated the quantity demanded to the quantity supplied, the price that cleared the market.
- 5.69** If you're going to double the market-clearing price, well, there are going to be fewer buyers at that doubled price, and more sellers. So you're going to have a big surplus of wheat at that price. Not everybody can sell at the higher price. So the idea that they're better off is an illusion. They can't actually sell the wheat at the higher price; it's not actually going to raise their real income.
- 5.70** The same thing is true of money. There's a market-clearing purchasing power of money. If you try to double it arbitrarily, it's not going to clear the market. You're going to have more people who want to spend, who feel themselves to have excess money balances. You're going to have an excess supply of money relative to the quantity people want to hold at that price level. And that's why they spend it off, and that's why the price level rises back up, or the purchasing power of money falls.
- 5.71** So it's no good to say there's a flaw in the market, that non-equilibrium prices can't stick. Non-equilibrium prices do not bring about a Pareto optimum, because they are prices at which not everybody can trade, make the mutually beneficial trades they would like to make.
- 5.72** So we need to amend the argument and say that it can't be just about arbitrary prices. It has to be about the equilibrium price. So it has to be that the equilibrium price level

is too high for a given nominal stock of money. And that is another way of saying, real money demand is too low. For some reason, people don't want to hold as much money as it would be socially beneficial for them to hold. And it can't be about wealth effects on other people, because wealth effects are generally not relevant when we think about whether we're exhausting all the mutual gains from trade.

- 5.73** One way to think about that is, if you're holding more money, you must be holding less of other assets. And so there's capital losses on the other assets that balance the capital gains to holders of money. So it has to be a different argument.
- 5.74** An economist named Martin Hellwig said, "We can think of it as a technological externality, not a capital gains externality or pecuniary externality." So, having higher real balances is not like we arbitrarily raise the price of refrigerators, and everybody now has a more valuable refrigerator. It's like a policy that improves the cooling capacity of refrigerators.
- 5.75** And how can we think of it that way? People hold money balances in order to spend, in order to accomplish transactions. And increasing the purchasing power of all the money in existence raises the purchasing power that people have. If we don't have enough real demand for money, we don't have enough purchasing power, then we're not capturing all the potential benefits we could be capturing. So he's arguing that the marginal benefits of increasing the real quantity of money, the purchasing power value of the stock of money, exceed the cost of doing so.
- 5.76** Let's think about what are the costs of doing so. It's useful to think about other kinds of money, other than fiat money. For gold coins, would there be a problem that the marginal benefit exceeds the marginal cost? No, because if there were, there would be greater production of gold coins. Competition in this private good is going to bring about marginal cost equals to price. How about bank-issued money? Same thing. If banks could make a profit by expanding, they will do so. But we saw in our discussion of unrestricted banking system that banks will expand to the point where marginal costs equal benefits, equal price.
- 5.77** So now we're left with fiat money. What's peculiar about fiat money? What's peculiar is that there's no such automatic market clearing, because we don't have decentralized markets. What we've got is the generalized economy in which the price level is determined. So it might be the case that the marginal benefit of increasing the real quantity of money exceeds the marginal cost, and there's nobody who's going to act on that. So it goes unremedied. Well, who can act on that? Central bank. How can they do it?
- 5.78** Suppose they just print more dollars. Will that increase the real quantity of money that people get to enjoy? No, because we already have seen that if you just lay more money on people in real terms than they want to hold, given the cost of holding it, they'll

spend it off, and that'll raise the price level in proportion to the quantity of money you introduced, and now nothing real has changed, so you haven't improved anything.

- 5.79** It's no good to just declare a lower price level. We already went through that. To make it sustainable that there's a larger real quantity of money, you have to shift the real demand curve. You have to shift it out. You have to get people to want to hold more real balances of money. How can you do that?
- 5.80** If you could raise the real output of the economy, then people would have more transactions, demand for money, and that would give you higher real balances in equilibrium. But monetary policy can't do that. Changing the quantity of money is not going to get you a permanent increase in the real output of the economy. Well, what's left? The opportunity cost of holding money. If you can lower the opportunity cost of holding money, you can get people to want to hold larger real balances.
- 5.81** This is the argument that Milton Friedman makes in a famous article called "The Optimum Quantity of Money." Why it's not the optimal quantity of money, I don't know. But here's Friedman's idea. You can raise real balances by increasing real money demand, and you should do that, or it's welfare-improving to do that, as long as the marginal benefit exceeds the marginal cost.
- 5.82** One thing that holds people back from demanding bigger real balances is the interest they're forgoing by holding money rather than other assets. So we can use monetary policy to lower the opportunity cost of holding money. One measure of that opportunity cost is the interest rate you're giving up. So we can use monetary policy to lower nominal interest rates. How do we do that?
- 5.83** I said at the end of the last lecture: nominal interest rate reflects expected inflation. If you want to lower nominal interest rates, you need to lower expected inflation. How can you do that? By lowering actual inflation, having a credible commitment to lowering actual inflation. If you do that—and how do you lower actual inflation? If you're the central bank, you lower money growth. You pursue a tighter monetary policy. So by lowering expected inflation, you lower nominal interest rates, you will get people to hold more real balances.
- 5.84** What's the limit to lowering the nominal interest rate? It's believed for a good reason that there's a zero lower bound. You can't get people to hold money if you pay them a negative return on their bank accounts, because they will just cash them out and hold cash. Cash pays you a zero return, and it doesn't cost that much to store. So the limit is maybe not zero, maybe it's a little bit negative until it's equal to the cost of storing cash. But that's the effective lower bound on nominal interest rates. Let's round it up to zero.
- 5.85** How do you get zero nominal interest rates? Well, there's a real interest rate which is positive. To offset that, to get a zero nominal rate, you need negative inflation. You

need inflation at the opposite of the real interest rate. So the optimum quantity of money prescription is for, not just low inflation, but negative inflation, deflation.

- 5.86** Estimates of the real interest rate—well, it's a price, so it varies from decade to decade. It varies with how many savers there are versus how many borrowers. Right now, we're in a period of low real interest rates around the world, because baby boomers have maxed out their IRAs and there's a lot of savings available. But estimates run down to maybe 1% as a reasonable estimate of the real rate of exchange between present and future. The premium and equilibrium people are willing to pay to have stuff now rather than later. So, if you're aiming at the negative of the real rate of interest, you're aiming at -1% inflation.
- 5.87** Now, there's another way to eliminate the opportunity cost of holding money, and that is to pay a higher interest rate on money itself. But there's no easy way to pay interest on currency. Rule that out. The way to get this equilibrium, this optimum, is to have deflation at some moderate rate, like -1%.
- 5.88** You can think of that as reducing the cost of holding money as low as it can go and lowering it to the cost to the central bank of producing that level of money-holding, by reducing inflation to -1%. So Friedman's assumption is, it doesn't cost anything to the system, anything to the central bank or to the government, to have -1% inflation rather than positive inflation, which means he's assuming a way that the revenue that governments get from printing money, which I will come back to. But this is the best thing for consumers, to have this really low inflation rate, even a negative inflation rate.
- 5.89** But notice what the argument is offering here. It's saying, set the marginal cost of holding money to the marginal cost of providing money, and both are zero. So it's a marginal cost-pricing argument. It's saying the central bank should not make it costly to hold money the way a monopolist makes it costly to buy their goods. The central bank should control the inflation rate to a point where it's like a perfectly competitive outcome. The price they're charging in the sense of the return you get from holding money is equal to the return you can get on other government assets.
- 5.90** So at -1% inflation, you get a 1% real return from holding money. If treasury bonds are paying 0%, because you've driven the nominal interest rate down to 0%, they're also yielding 1% in real terms as the purchasing power of the dollar is growing at 1% a year. So you're pricing Federal Reserve notes in the terms of the real return they pay equal to treasuries, or at least short-term treasuries. Those are the closest substitute. So it's a competitive pricing argument. If they don't do that, the government is price-discriminating. They're paying a better return to people who have treasury bonds than to people who have Federal Reserve notes. So it's an argument for competitive pricing.
- 5.91** There's another way to look at it, which is equivalent, and that is to think of paying anything less than a competitive rate on currency is tantamount to taxing currency.

And taxing currency distorts the money, the decisions people make about how much to hold. Higher tax, they will hold less. Zero tax, currency pays the same as T-bills.

- 5.92** So would this kind of deflationary policy really be costless? I mentioned that there are revenue implications. If it's shrinking the money supply or holding it constant, whatever is necessary, given the growth rate of the economy, to produce -1% inflation, if you have 2% real growth, you could have 1% money growth and still have -1 % inflation, assuming no change in velocity. For this to be costless to the fiscal authorities, you need to replace the revenue from printing money by some other tax. So the assumption has to be that there's another tax available that isn't any harder to collect or doesn't have any more dead weight burden than the tax on money.
- 5.93** Now, some economists or other observers have found this puzzling, that having a deflation is costless to the economy, because they think of a deflation—when they think of deflation, they think of the Great Depression. Prices were falling, and that really hampered the economy. And we don't want that. Even if it's only -1% a year, we don't want that. Ben Bernanke has, in many places, said there's something different about deflation that means we ought to fear it.
- 5.94** But I think they're confusing a case in which there was deflation because of unanticipated shrinkage in spending, and a case where it's perfectly anticipated. If it's your policy to have -1% inflation, everybody can take it into account. It can be priced into lending contracts. It can be priced into wage contracts. It's not going to be depressing the economy because it's not surprising anybody with a lack of spending on what they're selling. They're not going to price things too high to sell. They realize that they have to keep lowering the prices to keep up with a -1% inflation rate.
- 5.95** Second objection I've heard is that if currency pays the same rate of return as treasury bonds, 1% real was my numerical example, why would people save and invest in ways that build the productive capital in the economy? Wouldn't a 0% interest rate be bad for saving and bad for capital formation? Well, you've got to ask yourself, why do people ever invest in stocks and corporate bonds when they could buy risk-free treasury bonds? Risk-free treasury bonds don't pay all that much, but they are risk-free.
- 5.96** The answer, of course, is that people hope to earn more by taking a little risk. You're willing to put your money in the stock market, you get a return for bearing that risk, that over any short term, you could lose money. The same thing applies here. Even if the nominal interest rate is 0%, you can earn more than that if you're willing to take a little bit of a risk. So it's not going to discourage people from buying stocks or investing in companies in other ways, any more than the existence of treasury bonds does today.
- 5.97** So the conclusion that this mildly deflationary policy is the least costly feasible regime for a fiat money seems to me pretty plausible, seems to me pretty persuasive. Prescribing a -1% inflation to minimize the harm from inflation seems to me pretty reasonable.

- 5.98** If you could get a central bank to do that, then you would have money that was even better, or paid a better return to holders of money, than they experienced under a gold standard when the average inflation rate was zero. But that's a very big if.
- 5.99** So I don't think it's a strong argument that we should switch from a gold standard, or should have switched from a gold standard to a fiat standard, so that we could pay a higher return to holders of money. Real-world fiat standards have never produced long-run -1% inflation, predictable -1% inflation. They've all produced inflation rates much higher than those produced by gold and silver standards.
- 5.100** So I think it's a useful reminder to central banks issuing fiat money that they do better by the public the lower they can keep the inflation rate. But I don't think it's a very strong argument for having fiat money because central banks, in practice, have not taken that advice.

Lecture 6: Bank Fragility

At a glance — Bank fragility and policy rationales.

- 6.1** In Lecture 5, we talked about policy rationales. We talked about policies that have been offered or reasons that have been offered for a government to be involved in money. I shot most of them down, I think. I now want to turn to the similar consideration of arguments that have been offered as to why the government should be involved in the banking system. And again, I'm going to divide it between natural monopoly arguments and public good and externality arguments. So let's talk about natural monopoly first.
- 6.2** There's a famous finance economist who won the Nobel Prize in economics, named Eugene Fama. And Fama wrote an article once, and as an aside, it wasn't really the main thrust of the article, but said, "We let government issue currency, and it's efficient to do that," he says, "because when we observe privately issued currency, banknotes, they don't pay interest. Government-issued banknotes don't pay interest. So the public bears the same cost, or receives the same return, from government banknotes that they do from private banknotes. We let the government earn the revenue from that, and it's like a tax without a distortion because it doesn't change the prices the public pays." That's an efficient way to raise revenue.
- 6.3** A similar argument was offered by an economist named Scott Sumner in a piece where he was discussing whether the mint should be a government monopoly. And Sumner argued for having competition for the contract to be the mint, but he thought it was efficient to have just one mint—and he's talking about today, under a fiat standard—on the grounds that, if you had anybody allowed to issue \$1 coins, let's say, competition would drive the cost of those coins up to a dollar, because otherwise there would be an uncaptured profit opportunity. If you could produce a dollar coin that would circulate, and it cost you less than a dollar to produce it, you would do it.
- 6.4** So his concern is that competition among coin issuers, and this would be not competition on price, because the price is fixed at \$1 per coin, and not competition on interest yield, because by assumption or just practically, there's no easy way to pay interest on coins, but other things, what Sumner thought were useless costs—putting more valuable metal in the coin or more advertising campaigns or hiring celebrities to promote them, something like that, he thought would be wasteful. It would just be competition to get customers that didn't improve the product.
- 6.5** He offered us an analogy, a case we talked about earlier, which is, back in the 1970s, before the federal law was changed, banks were capped on how much interest they could pay to depositors. The interest rate was capped by the Federal Reserve. And so when it became profitable to get more depositors, because the interest rate they had to pay was way below what they could earn with the funds, banks began to offer prizes or incentives to open an account in the form of free toasters or toaster ovens.

- 6.6 And a bank would spend up to \$30 if that was the value of having the account. But the toaster oven that cost the bank \$30 might not be worth \$30, probably not worth \$30 to the customer. They already had a toaster oven. They would rather have the \$30. And Sumner says that likewise, if issuers of coins are competing on non-price dimensions, that is, not on the interest rate they're paying, but on some features they're putting in the coins, that would be wasteful.
- 6.7 It seems to me that these arguments rest on a non-sequitur. I worked on this topic jointly with a colleague of mine, Don Boudreaux. Boudreaux made the point that, more generally, that non-price competition has a bad reputation as being wasteful, and the example of the toaster ovens is often cited. But it does provide some benefits to consumers, and so it's better than no non-price competition. If you ask why firms are competing on non-price dimensions, well, in the case of currency, we argued, it's because there's no feasible way or no cost-effective way to compete on the price dimension. There's no easy way to pay interest on hand-to-hand currency.
- 6.8 Competing on other dimensions is a way of providing as much benefit to consumers as can be provided, net of the cost of delivering it. Non-price competition shouldn't be sneered at. It can be efficient if it's the least-cost way of delivering benefits to consumers because there's some obstacle to price competition, and in the case of currency, which is most convenient when it circulates hand-to-hand at a fixed face value—so, if you imagine paying interest on currency, how would you do that? Well, one way, you might have a schedule on the back of the banknote. On Monday, it's worth this value. On Tuesday, it's worth a few cents more. That would be a hassle. That would take away a lot of the benefit from using currency that you just look at it and you know what it's worth. You don't have to get out your calculator or look at some table on the back of it.
- 6.9 So, if the manner of competition isn't legally restricted, firms will only engage in non-price competition when that attracts more customers than price competition, given the cost of engaging in price competition. We use the example of shampoo. You can buy generic shampoo, and it cleans your hair, but it doesn't have lots of fancy features. It doesn't come in a fancy bottle, doesn't have a fancy fragrance. And yet, some people don't buy generic shampoo. They pay more to get these additional features, something that smells nicer or has a different texture or whatever. You can call that non-price competition. The firms are competing, not just on providing shampoo at the lowest price, but on putting qualities in it that people value.
- 6.10 Well, we shouldn't presume that that's inefficient. People want those qualities. They're willing to pay more for a fancier shampoo, so it's not wasteful to compete on those dimensions. Likewise, it's not inefficient to compete on other dimensions for currency because the zero interest rate on currency isn't due to legal restrictions. It's due to there not being any easy way to deliver currency to some—interest, sorry—to currency that's circulating hand-to-hand.

- 6.11** So, to attract clientele, banks that are competing in the issue of coins— sorry, firms that are issuing coins or banks that are issuing paper currency are going to look for the product features and services that come bundled with the product that will attract the most customers. And non-price competition is going to survive when it delivers benefits greater than the cost, but not otherwise. So, the argument for having a government monopoly for the purpose of eliminating non-price competition is it's going to reduce efficiency because it's going to eliminate benefits that people value. Only the valuable benefits, valuable features, are the ones that are going to survive in a competition among firms.
- 6.12** Let me turn to the—and I'll spend the rest of the lecture on—the number one argument for government involvement in banking. It's the argument that, if you leave banks unrestricted, unsupervised, you're going to have an unstable banking system. You're going to have too many bank runs. I'm going to break the argument down into three parts.
- 6.13** First, the claim is that an unrestricted banking system is inherently prone to runs. And because runs at one bank will inspire runs at another bank, it's inherently prone to contagion. If you just look at U.S. banking history—this is a plausible argument because the U.S. banking system was run-prone, and there were local contagions.
- 6.14** My explanation is not that this was an inherent problem, but this was a problem created by legal restrictions that weakened the banks. That argument is based on this not being a problem in other countries. It was a problem that arose from restrictions on banks that were almost unique to the U.S. But in Canada, you didn't have the runs and financial panics that the U.S. had in the late 19th century. So that's the first part of the argument: inherently prone to runs and panics.
- 6.15** Secondly, these are bad things. Probably can take that for granted, but runs and panics have negative effects, and not just on the depositors, but also on shareholders and also on bank borrowers. And that's going to be the least controversial part of the argument. That one, I'm pretty much going to accept, although I'm going to qualify when runs and panics have these harmful effects and when they don't.
- 6.16** Then third, there's a remedy. There's some government policy that can reduce runs and panics and deliver a benefit without a cost that exhausts the benefit, deliver a benefit that exceeds the cost of instituting this policy.
- 6.17** I'm going to mostly accept part B, that runs and panics are harmful. But I'm going to argue that the first part, that an unrestricted banking system is inherently run-prone, I'm going to argue that that's false. I'm going to argue that appropriate government policies can deliver a benefit less than cost. I think that's pretty doubtful, too. And of course, if unrestricted banking systems aren't inherently run-prone, then there isn't much benefit that government can provide in the first place to make them less run-prone because they're already not run-prone.

- 6.18** Arguments of this type, that banking is inherently fragile, is another way of putting it, have been used to justify, first and foremost, deposit insurance. But not only deposit insurance, also having a lender of last resort. Fragility and the financial panics of the late 19th century, in particular, the panic of 1907, that's what inspired the Federal Reserve Act. And the Fed was supposed to act as a lender of last resort; that is, provide more reserves to banks that were being stripped of their reserves by panicky depositors running on the bank.
- 6.19** It's also used, has been used, especially during the Great Depression, for other restrictions on banks, requiring banks to hold a certain minimum capital, restricting entry into banks. If you're afraid that competition will leave banks too weak, you want to ensure that they survive, then you will want to ensure they're profitable. To ensure they're profitable, you don't want too much competition. You want to limit entry, so that each bank has a little monopoly profit.
- 6.20** The interest rate ceilings that I've mentioned before that the Fed imposed, those arose during the Great Depression. The idea was, again, if banks have to compete too hard on paying higher interest to their depositors, they will skimp on safety in order to earn higher returns, and that will make them weaker and more failure-prone. That's always reminded me of the argument that, if you allow airlines to compete on price, you'll have too many planes dropping out of the sky, because airlines that are trying to cut their prices are going to skimp on safety. Turns out, the number of planes dropping out of the sky is no greater since we had price deregulation in airlines than it was before. But that would be an aside.
- 6.21** Reserve requirements have been justified by making sure banks have enough reserves to avoid being victimized by a run. Making sure banks don't invest in risky things or portfolio supervision. That's been rationalized by the idea that, if you don't do that, banks will take too many risks and be too failure-prone. And then the famous Glass-Steagall Act, which separated banking from other financial activities. So a bank cannot sell insurance, a bank cannot offer mutual funds, a bank cannot underwrite bonds. The rationale for that was, those are risky activities, and we want to keep them away from the bank. The theory is a little weak because the more activities an institution does, the safer it becomes through diversification, but put that aside. I'm going to focus on the argument for deposit insurance.
- 6.22** What do I mean by a bank run? A bank run is an occasion where lots of bank customers, many bank customers—it's a little fuzzy how many is many, but enough to cause a problem—try to cash out their deposits or redeem their banknotes out of concern that, if they wait, they won't get paid. So that's why it's a run and not a walk. You want to get there while you can still redeem. You don't want to be last in line. And a banking panic is when many banks, enough to cause a problem, are having runs at the same time. If we had the copyright permission, I would show you the bank run scene from Mary Poppins, probably the most famous cinematic bank run.

- 6.23** Let's start with the least controversial part about runs being harmful. It's pretty clear they're bad for bank shareholders. If there's a run on the bank, then the bank has to come up with reserves on short notice to satisfy the depositors who are redeeming. They've got to sell assets in a hurry. And selling assets in a hurry, they may be forced to take losses on them, sell them for less than they could get if they had time to sell them in an orderly way.
- 6.24** Secondly, there's some harm to bank borrowers who have a relationship with the bank and keep rolling over their loans. If the bank says, "We're being run upon. We have to contract our liabilities. We can't roll over your loan," that's probably the least of the harms. But clearly, it's harmful to depositors. It's certainly harmful to the ones who are at the end of the line, who get less than a full payout if the previous depositors have emptied out the bank. But more generally, it makes banks riskier to keep your deposits in if they are subject to runs. So you'd rather put your deposit in a less run-prone bank. Being run-prone, just the probability of a run, is a harm to you.
- 6.25** If the runs are generalized to many banks, then it's even more harmful. A panic is even more harmful because now the money that's withdrawn from one bank isn't going to be deposited in another bank. And so, the system as a whole is not going to retain the same amount of deposits; the system as a whole is going to shrink. And that means the borrowers who are turned away from the bank being run upon can't go to another bank to borrow, because that bank doesn't have any more depositors. There's going to be less intermediation, so a smaller supply of bank loans.
- 6.26** Runs are harmful. Then, of course, that is why banks want to avoid being run upon, to avoid these harms. But there are times when a bank is badly run, badly managed, is losing money, has become insolvent, and the question is: who puts an end to that? If you're a bank shareholder, what you want to do depends on your liability for losses beyond the point where the bank becomes insolvent. If you have liability for the unpaid debts of the bank over and above the value of your stock, then you would tell the bank, "When you reach zero value, you should stop. You should shut down. Let's liquidate the thing. We don't want to lose. We've already lost the value of our shares. We don't want to lose anymore."
- 6.27** But if you're in a limited liability system, where once the bank is insolvent, that is, its assets have become worth less than its liabilities— because it's made bad loans, is the main reason—if you have a limited liability, then you say, "Well, I've already lost the value of my stock because the bank has zero net worth, then each share of the bank's net worth has zero value. If nobody's going to close the bank, we might as well gamble for recovery."
- 6.28** Banks that are gambling for recovery are going to waste even more. So, it's good to close the bank before that happens. It's good to close an insolvent bank. The fact that an insolvent bank will get closed by a run when the depositors think that the bank is

already insolvent, that's a signal that there's not enough to go around if we all demand our money, so I don't want to be last in line. I want to get it now.

- 6.29 The threat of being subject to runs if a bank isn't clearly solvent forces banks to be more transparent, to be less risky, and to make it plain what their investment strategies are, so that the public will believe them, so that their depositors will be reassured.
- 6.30 Now, how does a depositor know whether the bank is solvent or not? Well, banks used to advertise how much net worth they had, and I suppose they were liable if they lied in those advertisements. But you can read, as a depositor or a potential depositor, you can read assessments of a bank's financial health the same way you can read assessments of a mutual fund before you invest in it these days.
- 6.31 If nobody ever runs on an insolvent bank, then the bank can continue squandering wealth until some legal authority closes it, and today, that's the FDIC. That's their job. When they notice that a bank is insolvent, they're supposed to close it. Actually, they're supposed to close it before it becomes insolvent, but they haven't done that. They haven't lived up to that legal mandate to intervene before the bank becomes completely insolvent.
- 6.32 So we have found out that regulators are sometimes asleep at the switch. We found that out most recently in the case of the Silicon Valley Bank failure. Silicon Valley Bank had a very risky portfolio. It became insolvent. When it became insolvent, the word spread, and depositors started pulling their money out. There was a run on the bank, and then the FDIC stepped in to close the bank after the run had begun. But they didn't anticipate the run, and they didn't notice that the bank was on the brink of insolvency.
- 6.33 We saw it earlier in the whole savings and loan industry fiasco in the 1980s. The savings and loans had risky portfolios because they were required by law to invest only in 30-year home mortgages, and that's a very risky asset. The value of a mortgage goes down when interest rates go up, and two-thirds of the savings and loans were insolvent before regulators started closing them. The ones who hadn't yet been closed, but who knew that they themselves were insolvent, were gambling for recovery, and that made the losses much bigger. So there was too much forbearance by the regulators. So compared to that, it's better to have a run and close it soon.
- 6.34 But the run is bad if the bank that's being run upon isn't insolvent. And there can be mistakes by depositors. They don't know perfectly whether the bank is solvent or insolvent. Even if they're responding to some news that suggests it might be insolvent, they might be mistaken. So if the bank is run upon that's not insolvent, that's still solvent, that's regrettable. There's no benefit to closing the bank to offset the harms that the run causes. So that's the runs-are-harmful part.
- 6.35 Let's go back to the first part of the argument, which says banks are inherently run-prone. Why would they be inherently run-prone? Well, one theory is, and this theory has become enormously popular among economists, and in fact, the economists

Diamond and Dybvig, Douglas Diamond and Philip Dybvig, got a Nobel Prize in economics for their article, giving an elaborate example of this problem.

- 6.36** Namely, the basic idea is that banks are inherently run-prone because the run itself can turn the bank insolvent. It turns the bank insolvent because it forces hasty asset sales. If a bank has to accept whatever it can get for its assets, then its net worth is shrinking, and it can become insolvent. So the run starts, there's a hasty sale of assets, they don't get much for the illiquid assets. The bank actually becomes insolvent. In that way, the run is self-justifying. Even if the bank was solvent when the run started, it's insolvent by the end of the run.
- 6.37** So the first people in line turned out not to be ridiculous. Turned out they were smart to do that because they got a full payout, and the people at the end of the line didn't. This is sometimes called the "me-first" problem. If there's not enough to go around, it's rational to run; and even if there's enough to go around right now, if there's not enough to go around, should everybody else join the run, then you want to get to the front of the line.
- 6.38** That means that any event, even if it doesn't have anything intrinsically to do with whether the bank is solvent, that makes people think the bank might be in trouble, will cause the bank to be in trouble. This is sometimes called a sunspot theory. Sunspots are these patches that appear on the face of the sun. There was an economist in the 19th century, Jevons, who thought that sunspots caused business depressions. Other economists thought that was so ridiculous, that sunspots were irrelevant, that the name they gave to any theory that invokes some irrelevant cause is a sunspot theory.
- 6.39** But what this self-justifying situation is telling us is that, if people think sunspots cause trouble at the bank, they will cause trouble at the bank because they'll cause a run. And by the time the run is over, the bank is in trouble. So whatever makes people anticipate a run will cause them to run, and that will validate the anticipation, even if what they're responding to has no inherent connection to the bank's solvency. So the basic theory here is that runs occur because depositors are worried that other depositors will run.
- 6.40** Now, what implications does that have? What should we see in the real world if that's true? Runs could happen anytime, anywhere. It's not clear what is going to trigger this fear. It could be set off by irrelevant events, so-called sunspots. What Diamond and Dybvig did was to formalize this idea. And as I said, they got a Nobel Prize for it because their model of a self-justifying run was so influential on later economic theorists. But the core idea is pretty old.
- 6.41** There's a letter in which Thomas Jefferson, after he left the White House, is talking to a friend of his about banks. And he says, "People think that the banknotes are safe because if they go to the bank, they can get silver coin for it." And he says that one or two people might be able to get silver coin for it. But if everybody tried to redeem their banknotes at the same time, there wouldn't be enough silver in the vault to pay them

all. And so, if people just knew, if they realized how little a silver reserve the bank has, we would have runs on all the banks. So if people thought that other people were going to run the bank, they would run the bank, too. So that's the sunspot theory.

- 6.42** Now, there's a problem with it, if we try to use it for historical explanation of when bank runs have occurred and when they haven't occurred, because it suggests that bank runs are random or maybe linked to irrelevant events, but that's not the historical pattern. If you look at a run-prone system like the United States in the 19th century, there's a definite pattern in bank runs. They typically occur at the beginning of a recession. So calling them random doesn't really explain that.
- 6.43** The second anomaly: not all banking systems are run-prone across the world. So you've got runs in countries where banks are weak and often fail. That's the United States. That's also England. But you don't get them in countries where banks seldom fail. So that's Canada, that's Scotland. This cross-country contrast, sunspot theory doesn't help you understand, because it suggests it could happen anytime, any place.
- 6.44** So is there a better theory for what sets off a bank run than whenever depositors get nervous? Yes. The better theory can be called the bad news theory. Depositors run when they get some information that suggests that the bank may already be insolvent. They get some bad news. And if you think the bank is insolvent, it's rational to run because there isn't enough to go around already. They can't pay all the depositors back.
- 6.45** That explains why bank runs would occur at the beginning of a recession, because at the beginning of a recession, what does the newspaper say? "This business just declared bankruptcy." Well, if they declared bankruptcy, then they're not going to pay back their loan to the bank. If you know what bank that is, that bank might be insolvent; it might be time to get your money out.
- 6.46** In the United States, we had a very much restricted banking system, a geographically restricted banking system. Banks were not allowed to branch across state lines, so there were no nationwide banks in the 19th century. And in many states, they weren't even allowed to branch within the state. You had so-called unit banking systems, where one bank, one office. Well, if you've only got one office in one town, then your loan portfolio is pretty much going to be businesses in that town. You're going to have a hard time lending money across the state, let alone across the country, to diversify your loan portfolio.
- 6.47** So you suppose you're a bank in Dahlonega, Georgia, which is the center of the carpet industry in the United States, and there's a downturn in the carpet industry, and carpet firms are going bankrupt. The banks in Dahlonega are going to be in trouble. Sorry, not Dahlonega, Dalton. Carpets of Dalton. Dalton, Georgia. Dahlonega is where the mint was. The banks have to be lent up to the carpet companies because that's the main business in town. If one bank fails, then, and it's not your bank, but your bank is across

the street, you'd better be worried about your bank because they've got pretty much the same set of loan customers, carpet makers.

- 6.48 So you had regional clusters of bank failures in the U.S., and that meant that a failure of one bank would set off runs on other banks because the other banks had pretty much the same portfolio. If you want to avoid that, you want nationwide branched banks that have well-diversified portfolios of assets.
- 6.49 The bad news theory also explains the difference across countries. If you've got a country with minimal restrictions on banks, such that they are able to strengthen themselves, they're not restricted in how much capital they raise or how diversified their branching is or their loan customers, thereby, you're going to have more solid banks. If you've got those restrictions, you're going to have weaker banks. They're going to be more prone to fail. You're going to have more nervous depositors for good reason.
- 6.50 So the difference between the U.S. and Canada was that the U.S. restricted its banks in ways that weakened them. Not intentionally, but the geographic restrictions were intended to provide sheltered markets to the local banks. And it was a side effect that the banks were weaker as a result of that, poorly diversified.
- 6.51 Now, this contrast between attributing a financial panic to sunspots or self-feeding investor panic versus fundamentals—bad news that gives people a reason to think that the institution is insolvent—this contrast is still with us. It's not just relevant to 19th-century bank failures.
- 6.52 In the Asian financial crisis, I don't know if you remember that, in the late 1990s, I guess it's a while ago now, there were some economists who piped up and said, "There's no reason for people to be attacking these currencies." So the crisis was characterized by the central banks in Thailand, Malaysia, Philippines, having pegged their currency to the dollar, and speculators saying, "Well, they're not going to be able to keep the peg. So we're going to redeem our dollars—sorry, we're going to redeem our local currency for dollars now because they're going to devalue the local currency if we wait." And so, those kind of speculative attacks on pegged exchange rates are very much like bank runs.
- 6.53 Some economists said, "This is just an irrational panic on the part of the hot money. Everybody should just calm down." But others said, "No, no, the fundamentals are bad." People got the signal that these central banks are not going to be able to maintain their pegged exchange rates because they're not committed to the kind of monetary policy, the tight monetary policy, that would be necessary.
- 6.54 For example, in the Philippines, they announced that they were going to bail out all the banks. Well, that means they're going to issue a lot of money. That means the Philippine peso is going to sink in value. They're not going to be able to maintain the pegged exchange rate to the dollar. Time to cash in your Philippine pesos now. So this

debate about what causes these events is still with us. I think the evidence is on the side of the bad news theorists most of the time.

- 6.55** It's very difficult, historically, to find a run on a solvent bank. For the sunspot theory to be true, those should be just as common as runs on insolvent banks, and yet they're very difficult to find. The Silicon Valley Bank was insolvent when it was run upon. Bear Stearns: insolvent. Lehman Brothers: insolvent. So it wasn't a tragedy that they were closed, and it wasn't irrational on the part of their depositors or overnight lenders to stop rolling over their loans.
- 6.56** So, let's dig into it a little more. What is it that would make a bank run-prone? Well, the depositors have to have the option to run, so it has something to do with the contract between the bank and the depositor. What exactly are the terms of the deposit? If there are different kinds of contractual arrangements, you would expect the more robust ones to survive and not the fragile ones. So it's a little puzzling when an economist, or Diamond and Dybvig, or others say, "Here's a fragile bank. Here's a model of a fragile bank, and this explains why banks in the real world are run-prone."
- 6.57** Why would real-world banks be so fragile? Why would fragile banks survive in a competition with more robust banks? What is it that makes a deposit account run-prone? Well, you have to have the right to redeem on demand. Pull your money out now. And there must be a reason for wanting to pull your money out. Let's assume the depositors are rational. There's got to be some greater expected payoff to pulling your money out now than later. So that implies you have a claim that, whose value doesn't depend on the performance of the bank portfolio, by contrast to a mutual fund. I'll talk about more in a second.
- 6.58** You have a debt claim. You have a claim to a fixed number of dollars, and that's typical of bank deposits. You know how many dollars are in your account. You have a claim to a fixed number of dollars. You can pull it out whenever you want. It's unconditionally redeemable on demand. And that practically implies, since the bank has to pay you when you show up, that it's going to have to pay people in the order they show up. It's first come, first served. And then third, this almost goes without saying, it's likely that the bank is going to default on the last customer in line. That's going to be true whether the bank is already insolvent or whether the run makes it insolvent.
- 6.59** What made the Diamond-Dybvig model famous is that it has all these features. The Nobel Prize they got is a testament to the influential nature of the model, the influence it has had on other economists. I want to explain the model. This, I hope, is not too technical to be interesting, because this is the number one rationale offered when you ask people, "Why do we need deposit insurance?"
- 6.60** In the Diamond-Dybvig model, they're trying to capture the idea that depositors have a need to get their money out, and that may happen at a time when it's inconvenient to the bank. So the deposits are more liquid, you might say, than the bank's assets. The

bank can't sell off, at least not all of its assets, immediately, at low cost. So how do they try to capture that? They say, let's strip the bank down to an arrangement in which people who face a risk of needing resources soon rather than later pool their investments with other people, so that some of them can enjoy early withdrawal and earn something on their account, more than they would earn if they just managed their own investments.

- 6.61** This is going to sound bizarre, but this is for the structure of the model. We've got an economy where everybody is growing corn, and you can plant the corn in your own backyard. If you leave it in your backyard for two periods, then you get new corn and you earn a positive return. You get more corn than you planted. But if you dig up the corn halfway through, you just get back the kernels you already planted, let's imagine. It hasn't grown any.
- 6.62** So you ask yourself, "How much corn should I plant?" Well, there's a risk that I might need some corn in the middle period. Then that's going to discourage me from investing. It's a risk. It'd be nice if I had some insurance against this risk. How do you get insurance against risks? You pool with other people who face the same risk, and then only those of you who need the corn in the intermediate period get paid, and others who are able to wait for the corn to come to fruition, they'll get the full payout of what's left.
- 6.63** They'll get a lower payout than if they had planted it in their backyard and waited, because the bank has to pay some of the early withdrawers, and that comes out of what's available to pay the people who wait till the end. Why would I agree to that? Because there's a risk I might need to withdraw early, and I get offered a little bit more if I withdraw early under this plan. So it provides insurance, and insurance is costly. If you never place a claim, you didn't get anything from your insurance.
- 6.64** What they're calling a bank is pooling these deposits of corn and doing the planting for people. It's promising them a better return in the intermediate period, call that period two, than they would get if they planted their own corn and interrupted the investment, because then you'd only get one back for each one you planted.
- 6.65** If corn is planted and kept in the ground for two periods, let's put some numbers on it. Let's say you get 20% return. The bank could offer people in the intermediate period 6% return. But since the corn hasn't grown, that's going to have to come out of what's left over to pay the people who wait till the end. But people who are risk-averse, who want insurance against having to withdraw in the first period— sorry, in the second period, in the middle period—they're going to agree to that.
- 6.66** So suppose we've got 100 depositors; they bring in one unit of corn each. The bank promises 6% return if you withdraw early, and we divide up all the corn at the end among those who waited. So this makes it a peculiar institution. It's a debt claim. You bring the corn in in period one, you have a debt claim in period two. You're entitled to

1.06 per unit you brought. In this third period, when the corn matures, you don't know exactly what you're going to get because you don't know exactly how many people are going to withdraw in the second period. So you have an equity claim. You have a share of the output. Banks don't actually operate that way. They just issue debt claims. In the second period, this becomes a mutual fund.

6.67 Continuing the example, suppose 100 people deposit corn, 100 units of corn, 50 people decide or discover this is a risk, and the information is revealed to them that they need to withdraw in period two, before the corn has matured. If they are each paid 1.06, multiply that times 50, that's 53 units that have to be dug up and distributed to the early withdrawers. That leaves 47 units in the ground. Each of those grows by 20%, so the total yield is going to be 56, 56.4. That's 47 times 1.2. We divide that among the 50 remaining depositors. They each get 1.128. They get a higher return, 12.8%, than the 6% that the people who withdrew early. So they were happy they waited, and everything's cool. The insurance succeeded.

6.68 But here's the problem. Suppose 75 people decide to withdraw in period one, either because they get the information that they need to withdraw, or—it's assumed the bank can't tell these kind of people apart—or because they decide to get out because they don't think there's going to be enough to go around. They think that if they wait, they're going to get a lower payout than if they withdraw now. So, suppose 75 people withdraw, multiply that by 1.06, that's 79.5 that has to be paid to the 75 people who withdraw. That leaves only 20.5 in the ground. Multiply that by 1.2, that's going to yield 24.6. Now we've got to divide 24.6 among 25 people. That's less than 1. So they get a worse payout than the people who withdraw early.

6.69 If you suspect that's going to be the case, you want to withdraw early. So that's going to be a run, and the run ruins the bank. So there are two possible scenarios, one where it succeeds and one where it fails because too many people withdrew. And it may just be that they withdrew out of fear that too many other people were going to withdraw. That's the story.

6.70 Now, how does this provide a rationale for deposit insurance? Diamond and Dybvig say, suppose we keep a list of who withdrew. So call the agency that does this, the Federal Diamond and Dybvig Insurance Corporation, the FDDIC. The FDDIC keeps a list of who withdrew and got 1.06. If so many people withdraw that there's not enough to pay everybody at least 1.06, then we go back and take away, claw back, the extra 0.06 that the early withdrawers got. We redistribute that so that everybody gets at least one. We know there's at least 100 units of corn.

6.71 In that case, they're guaranteeing that the people who wait will get at least as much as they will get if they withdraw early. Then there's no longer an incentive to run. There's no longer a fear that you'll get less if you wait than other people are going to get. Waiting to see now dominates. So there won't be runs for no good reason. The bank will be saved, and the marvelous part of it is that you never need to use this scheme. All

you need to do is announce that this is what we will do to make sure nobody gets a reduced payout, and then nobody will run out of fear getting a reduced payout because they've been reassured. The mere knowledge that they're protected eliminates the incentive to run. So it's a costless deposit insurance.

- 6.72** Now, there is a catch when we try to apply this to the real world, which is that most bank failures are not caused by bank runs. They're caused by bad loans, asset losses. And that means the FDIC makes payout to depositors even when there hasn't been a run. You have bank failures even without runs, so they're not just preventing runs. And they are taxing banks that are well run to pay off depositors in banks that are poorly run.
- 6.73** Well, the Diamond-Dybvig model has inspired a large discussion among economists, a lot of which has been critical. People have pointed to features of the model that make it less than a reliable guide to the real world, let's say. It doesn't match real-world runs. It doesn't, like the bad news theory does, it doesn't provide us much in the way of useful historical explanation.
- 6.74** But even the policy implication is suspect because the Diamond-Dybvig bank, what they're calling a bank, is very much unlike a real-world bank. It's so unlike a real-world bank that you're entitled to doubt whether what's true of the Diamond-Dybvig bank is true of real-world banks, that the implications may not apply to real-world bank—namely, that they're inherently run-prone because there's not enough to go around if everybody wants to claim. The implication that deposit insurance is costless just has to be announced. And the implication that other things that banks have done historically, like suspend payments when too many people tried to withdraw, are infeasible. In Diamond and Dybvig, if you suspend payments, then people starve because they don't get their corn when they need it to consume.
- 6.75** I mentioned that the claims on the Diamond-Dybvig bank are debt if you try to withdraw in period two, but it's equity if you wait till period three. That's not how real-world banks run. They have separate depositors and shareholders. The shareholders are the residual claimants. They get what's left over. They get returns on their shares of stock; those are not depositors. And the net worth that is owned by the shareholders provides a cushion that protects the depositors against losses. The losses come out of the shareholders before they ever touch the depositors.
- 6.76** So if a bank has a bad quarter, they don't go to the depositors and say, "Well, we promised you 6%, but we can only afford to pay you 4%." No, their contract says 6%; they pay them 6%, it's the shareholders who get lousy dividends that quarter. But the Diamond-Dybvig bank doesn't have any class of shareholders to insulate depositors or that are separate from depositors. In fact, if you look at period two, the intermediate period, the bank has promised everybody 1.06, but by assumption, there's only 100 units available to distribute, so that bank is insolvent.

- 6.77** Every time the bank is insolvent in the intermediate period, and it's only if they're lucky, and depositors don't call them on it, that they survive to the next period. Well, if you told me, "Here's a model of a bank that's insolvent, and therefore people run on it," I would say, "Fine, but what does that tell me about banks that are solvent?" If you give a bank adequate capital, it can survive all the withdrawals. It's not going to be made insolvent just because people are withdrawing in the intermediate period. The shareholders are going to bear the brunt of it, but it doesn't mean the bank is always going to be wiped out.
- 6.78** The service this bank, this Diamond-Dybvig bank, is providing to people is a kind of insurance, as I tried to explain. They're not providing payment services, and they're not providing intermediation. But when I talked about banking, I said that's the core of banking, payment services and intermediation. They're not providing payment services because the claims on the Diamond-Dybvig bank, you can't spend. They're only good for redeeming for corn.
- 6.79** If you change that, if you allow a market where you can sell your claim in the event you need to withdraw early, then there can be a separate class of people who buy these claims and then wait for the corn to mature, and profit by doing that. In which case, you don't need to liquidate the bank in order for some people to get their money out.
- 6.80** And third, there's an inconsistency in the deposit insurance scheme. So the assumption was, remember, the FDDIC keeps a list of who withdrew, and if too many people withdraw, they claw back the extra. Why can't the bank do that? Why can't they make that part of the contract? Diamond and Dybvig assume instead that it's first come, first served. The bank has to pay it all out and they can never claw it back. But if it's feasible for the FDDIC to claw it back, why isn't it feasible for the bank to claw it back? And if it's not feasible for either of them, then the deposit insurance isn't feasible either.
- 6.81** But deposit insurance, as I guess I've already said, is not costless in the real world. Just announcing it doesn't mean it's no longer necessary to invoke it, because banks don't fail on account of runs, typically; they fail on account of bad loans. There are payoffs, even in periods without bank runs.
- 6.82** Socially, deposit insurance has spillover effects on the incentives of banks. It creates a moral hazard for banks, moral hazard meaning an incentive to take on undue risk. And it's a little indirect. The insurance is not on the bank, it's on the deposit. So as a depositor, you don't have to worry about not getting paid back because if the bank fails, today, the FDIC will pay you back. But it does change your incentives as a depositor. You don't need to shop around for a safe bank the way you would in the absence of deposit insurance.
- 6.83** If you don't need to shop around for a safe bank, it means there's no reward to a bank for being safe. Being safe doesn't help them attract depositors. Safe banks don't get to pay lower interest rates on deposits because they're safer than banks that are taking

high-flying risks. All that high-flying-risk bank needs is an FDIC sticker in the window, and the deposits will come in.

- 6.84** There are deposit brokers who, if you have a million dollars to put on deposit, and you want the highest rates from banks, and the insured limit is only a quarter of a million, the broker will find the four banks paying the highest rates and put your money in those four banks. Do you care who those banks are? No, they're insured. So, this is a subsidy to risk-taking, or risk-taking banks are no longer facing the market penalty for taking risks. Namely, it scares away depositors, and depositors are going to demand higher returns before they agree to put their money in a risky bank.
- 6.85** So Diamond and Dybvig have given us a picture of a fragile institution. I grant that, but is it a bank? Not really. Are real-world banks as fragile as the Diamond and Dybvig banks? No, real-world banks have different contractual arrangements than the Diamond-Dybvig bank.
- 6.86** So again, to make the deposit run-prone, you need a debt claim, unconditionally redeemable on demand, default likely on the last claim. How did banks avoid being run upon? Well, they did issue debt claims, but today there's an alternative, namely, money market mutual funds. Those are not run-prone because there's no me-first problem. If you hear that your mutual fund suffered a loss, too late, the value of your share in that fund already went down. So there's no reason to run to withdraw it. So that's something to consider if we did away with deposit insurance, what kind of institutions would take the place of insured banks. Maybe mutual funds would play a bigger role because they're not run-prone.
- 6.87** Most important things, historically, that banks did were to make the withdrawal conditional. And second, to make default less likely. How do you make withdrawal conditional? It used to be typical for a savings bank to have in its contract, "You have to give us 90 days' notice if we ask you to, before you withdraw money." When I was a kid, I got a passbook account which had this text in the inside cover of the book that said, "If we want, you have to give us 90 days' notice to withdraw your money." I said, "I don't want to give you 90 days' notice."
- 6.88** I actually had this conversation with the teller. "I don't want to give you 90 days' notice. The new Batman comic book comes out, I want to be able to buy it. I want to be able to take out my 25 cents right away." The teller said, "Oh, don't worry about that. That's just some kind of legal boilerplate. We'll give you the money whenever you come and ask for it." Then in my fantasy, I said, "Why do you have this legalism in here?" The teller says, "Have you heard of the Diamond-Dybvig bank model?"
- 6.89** Now, she wouldn't have said that because this happened before 1983. But if she were historically informed, the teller would have said, "Well, in the panic of 1907, these kind of clauses proved very useful for savings banks and trust companies, to tell people to wait before they could withdraw. Normally, we don't make you wait. But if too many

people are trying to withdraw, and we would suffer losses from hasty asset liquidation, we can ask you to wait, and thereby we have enough time to sell off our assets without suffering losses, so we're better able to give you a full payout."

- 6.90 The customers of trust companies and savings banks that had these clauses understood that this was in their own interest, to keep the bank from being emptied out by other depositors.
- 6.91 The most important thing banks did, though, was to make customers assured that they weren't insolvent. This is why it's important that it's the bad news theory that is historically applicable and not the sunspot theory. If it's bad news that makes people run, assure them that bad news is not going to turn the bank insolvent. How do you do that? You have enough capital to absorb asset losses if some of your loans go bad.
- 6.92 Banks used to advertise how much capital they had. They used to put it in the newspaper, "Bring your money to us because we have this much capital to safeguard your deposit." Some banks actually painted in the window in gold paint, "This bank has \$2 million in capital." When federal deposit insurance came along, they hired somebody to scrape that off the window and just put in the sticker, "Protected by FDIC," because that's all the depositor needs to know now.
- 6.93 Banks had a way of helping each other out, if a bank was subject to a random large withdrawal but was still solvent—namely, they would lend to each other. And the clearinghouses at which banks got together to exchange checks and banknotes would facilitate these interbank loans because the clearinghouse was checking up on all the banks to make sure they were solvent. It was in the interest of every member bank to make sure all the other members were not going to default at the clearinghouse. So the clearinghouse had timely information and could arrange these last-resort loans. So they were an unofficial, if you like, lender of last resort.
- 6.94 So historically, we don't see runs and panics being a problem in every banking system, just in weak banking systems. Weakness is mostly a creature of legal restrictions, like branching restrictions and restrictions on the assets banks have to hold to back their banknotes.
- 6.95 Why was the U.S. in particular weak in the 19th century? Because of these legal restrictions. Branch banking limited diversification. In issuing banknotes, banks were required to first buy government bonds, send them to the Comptroller of the Currency, get the blank banknotes. It was very time-consuming. So they weren't able to meet demands for currency on short notice. That meant they were more subject to depositors saying, "Well, if I can't get banknotes, I'll take gold out of the vault." And that's a run on the bank's reserves.
- 6.96 To solve that problem, some American bankers noticed that Canada wasn't having these problems. We should be more like Canada. The American Bankers Association actually made a proposal to make the U.S. system more like Canadian banking, which

meant what? Allow nationwide branch banking and eliminate the specific collateral restrictions on issuing banknotes. They didn't have those in Canada either.

6.97 But the small banks didn't like that idea of nationwide branch banking, so they killed it politically. And what we got instead, as a kind of second-best way of preserving the banking system against panics, was the Federal Reserve Act. So the Federal Reserve Act wasn't necessary to remedy the panics, but it was the political compromise that we arrived at. And the function, the purpose of the Federal Reserve System, was basically to be on standby for when a panic occurred and then to spring into action, to issue more currency and to provide more bank reserves.

6.98 The idea that the Fed should be running monetary policy, that was not part of the Federal Reserve Act. That came later because, when the Federal Reserve Act was passed in 1913, we were on a gold standard. The gold standard was supposed to run the monetary system. When we lapsed into a fiat money system, that's when the Fed became responsible for monetary policy.

Lecture 7: Money Making

At a glance — Money making and the causes of inflation.

- 7.1 We've been talking about the causes of inflation, and I said, put the blame on money growth, but that pushes the question, or raises the question: what's responsible for money growth? Why are central banks printing more money than is consistent with the stable purchasing power of the fiat money? When I say printing, I don't mean it literally. They do print Federal Reserve notes at the Bureau of Engraving and Printing, but mostly it's electronic, mostly it's on balance sheets.
- 7.2 What is the objective of issuing more money? One objective, which is more important in some countries than others, is the profit from printing money. It's pretty simple to understand, if the President-for-Life of some small republic wants to go on a spending spree, he calls up the central bank and says, "Send me over an armored car with 50,000 monetary units in it, in currency," and they say, "Yes, sir, Mr. President-for-Life," that's government spending financed by issuing new money. That's pretty direct.
- 7.3 We do it in a more subtle way in the United States. I'll get to that shortly. This idea of a profit from issuing money has the technical name of seigniorage. The name comes from the French term "seigneur," who was, in the Middle Ages, the lord of the manor. It was the prerogative of the lord of the manor to operate a mint, and the profit from operating that mint was the seigniorage.
- 7.4 Now, in a competitive coining system, as we've talked about, there isn't any excess profit. There's just a normal rate of return in running a mint. You get paid for the services you provide, but you don't earn more than other industries because competition will erode any excess profit by new firms entering to compete for that profit. But if you have a monopoly of the mint, then you can earn a higher-than-normal profit, a higher-than-competitive profit, and you can maintain that profit indefinitely without anybody eroding it away because you've barred them from competing with you.
- 7.5 So, to make it a durable source of revenue under— let's go back to the silver and gold standards—the monarch, the autocrat, would monopolize the mint. That is, nobody else was allowed to operate a mint. Then there were two ways to make an excess profit. One is just charge a high price for the service of turning raw metal into coins. Competitive mints will be forced by competition just to charge what it actually costs them to operate the mint. But a monopolist can charge more than that. And the coins being produced could be perfectly honest, but there's still a profit from the high fees being charged.
- 7.6 Some royal mints, in order to increase the flow of metal into the mint, made it a crime to take your gold or silver to any other mint. So the Spanish mint had that rule. All the gold and silver that came from the New World, the mines, to begin with, were owned by the Crown, but all the gold and silver had to come to the royal mint to be coined,

even though they were charging a high fee. And the form the high fee took was, they paid less per ounce of silver or gold than you could get somewhere else, if you were able to divert it. And so diversion and smuggling was a problem because they were creating an incentive to do that. But still, they could issue full-weight, honest coins and still make a profit.

- 7.7** But there's a second way to make a profit, which is a little sneakier, and that is to overvalue the coin. Claim that the coin has an ounce of gold in it when it has less than an ounce. Actually, that sort of tactic was less common with gold coins because there was a more competitive international market. But for silver coins, claim there's an ounce of silver in the coin, put less than an ounce of silver in it—this is called debasing the coin, or debasement. It's not the floor under the house. It's reducing the metallic content of the coin. If you can do it surreptitiously, without announcing it, you can get people to continue to accept the coin as though it's full weight.
- 7.8** Once people catch on, they're going to knock down the value they place on the coin. Then you might need to bring out some muscle, and in some countries, it was illegal to charge a different price depending on whether the customer paid in full-weight coin, old coins, or lightweight coins, new coins. When that happens, of course, people are only going to pay in lightweight coins, because prices are going to rise to the number of lightweight coins it takes.
- 7.9** And as we've discussed before, the heavyweight coins, which are being declared by law to be equal in value when they aren't, they're going to disappear. People would be foolish to spend them. They're going to melt them down and export them. So the track record, as I mentioned before, the track record of the sovereign mints is pretty bad, the coins were just continuously debased.
- 7.10** Well, this is a footnote. There were occasions where they fortified the coinage, that is, restored a little bit of the silver content. And that was mostly after complaints by the people who were being impoverished by this inflation, namely, people who had long-term contracts. Landlords and lenders would say, "No, you're letting people pay us in coin that's worth less and less. Bring it back to the value it had before." So, the amount of silver in, say, a Roman denarius, there's a clear downward trend, but there are a few occasions where it goes back up a little bit. So debasement gives a bigger profit per coin by putting less precious metal into the coin. It's a cheaper way to produce the coin of a given face value.
- 7.11** Some medieval mints didn't have a face value on the coin. There was no number on the coin. Most members of the public were illiterate and innumerate. But there was a picture, somebody's face or some other design, that allowed people to identify which size coin it was. One way to debase the coinage without tampering with the metal in the coin was simply to declare that this coin that used to be worth five units is now worth 10 units. That's called "crying up the coinage." That has a similar effect of increasing the profit on each coin that's being produced.

- 7.12** To continuously sustainably derive profit from debasement takes a little bit of deceit or a little bit of compulsion. So as I said, if you can issue the new coin and not tell people that there's less silver in it, they will treat it as though it's full weight. That makes it more profitable. The Royal Mint can buy more things. The king can hire more soldiers with a given amount of coin. So it pays to disguise when there's a debasement, and the mints typically use the same coin dies. So the new coins looked exactly the same as the old ones. It's just that the metal in the coin was a lower percent silver and a higher percent something else: tin or nickel or copper, something cheaper.
- 7.13** Once the public wises up, they're going to start discounting the new coin against the old coin or the heavier-weight coin. You can pass laws that force them to treat the two as the same. These are legal tender laws that say the new coin is legal tender for all debts, even though it's contracted in the old coin. This is to the advantage of debtors. They now get to repay their debts in cheapened coin. It's to the disadvantage of the creditors, of course. Who's the biggest debtor in these economies? Typically, it's the king. Kings are borrowing money to fight wars. So by debasing the coinage and declaring it of full value for the purposes of paying debts, the kings are defrauding their creditors. So this is a continual problem with the royal mints.. Until there is a reform movement, and eventually, once modern governments can derive revenue from other sources, they're less reliant on debasing the coinage.. And to be clear, the reliance on seigniorage is pretty big. It's like a third or a half of some government's budgets. And in times of war, when they need a big infusion of money, they would call in all the existing coins and recoin them with less precious metal in them. So if you call in 1,000 coins, you can reissue 2,000 coins with half the silver in each coin. There's a big profit right there. And so, sometimes that was more than half of the royal revenue for that period.
- 7.14** We're not on metallic standards anymore. We're on fiat money standards. So the concept of seigniorage is going to be a little different. You can't take any more silver out of the U.S. dollar. It's all gone. The U.S. dollar has no fixed value, no defined value in terms of silver. So what is the profit on issuing a \$100 bill? Well, it's a difference between what \$100 is worth and the cost of producing the \$100 bill, which is about six cents for the paper and ink and watermarking. So, as a first approximation, issue a \$100 bill, make \$100 in profit. It's really \$99.94, but we'll round it. The profit is equal to the amount of money issued.
- 7.15** Now, today, that's only a first approximation because a lot of the liabilities of the Federal Reserve pay interest now. There's interest on reserves. So the bank balances that are on the balance sheet of the Federal Reserve, those are paying interest; so it's more like borrowing money than it is like issuing money. The Fed is paying banks interest to hold Federal Reserve liabilities. It's not that much different from the Treasury paying people to hold Treasury bills.

- 7.16** But in a fiat standard, you're not deceiving people about the silver content of the dollars. Everybody knows it's zero. So that kind of deceit is not really necessary. And the compulsion is much more subtle. You don't have to pass a law saying you have to treat new \$100 bills the same as old ones. They are the same. They have the same metallic content, which is zero. People don't really have to be coerced into accepting newly issued Federal Reserve notes because they are interchangeable with the old ones in every way. They spend exactly the same.
- 7.17** In the background, there are the measures that created the system we're living under. So, in the transition, when fiat money was declared legal tender for debts that had been contracted in gold dollars, when the government said, and the courts ratified this, that if you get \$100 in Federal Reserve notes, even though you were looking for \$100 in gold coins, that was in the contract, you've been paid, sorry. That's coercion. That's imposing on people something other than what they contracted for.
- 7.18** But once everybody knows we're on a fiat standard, everybody knows that their debts are going to be paid in fiat money, there's really no coercion necessary. So the seigniorage is derived simply by diluting the value of the existing money. Issue more, you're going to water down the value of the existing dollars. We talked about the quantity theory of money, which tells you the dollar drops in value in proportion to the number of new dollars that are introduced.
- 7.19** There's an accounting, well, it's not exactly an identity. There's a way to describe the government's budget, which goes as follows. Every dollar the government spends has to come from somewhere. Most of it comes from ordinary tax revenues. I'm leaving out seigniorage for now. What doesn't come from tax revenues, the difference between spending and tax revenues, is the budget deficit. And the budget deficit is mostly financed by borrowing. The government has to borrow the money to pay the recipients of government spending, when they don't have it coming in in taxes.
- 7.20** And how do they borrow it? They issue treasury bonds in the U.S. So, they introduce new debt into the economy. But there's a small share, and it varies from country to country, that's paid for, instead of by issuing interest-bearing debt, that's paid for by issuing base money, issuing new liabilities against the Federal Reserve that don't pay interest. So that's Federal Reserve notes, basically. And it works, as I described the President-for-Life financing his spending spree, it works by introducing new money into circulation.
- 7.21** So, monetary expansion generates some revenue for the government. And that means, getting back to the question of what's motivating central banks to issue more money, there may be fiscal pressure. The treasury may come to the Fed and say—I'm sure these communications don't have to be overt—and say, "Look, we need to borrow a lot of money. It would be helpful if you would buy some of the debt, instead of our having to flog it all in the market." But for the Fed to buy the debt means the Fed is going to create new money because that's how the Fed buys assets. It just writes up a claim on

itself, puts that on the balance sheet, and puts the treasury bonds on the asset side of the balance sheet.

- 7.22** I mentioned it's a little more subtle in the U.S., the way new money is introduced, than simply printing up the bills and giving them to the President. I'll give you an example of actually where the direct spending took place. During the pandemic, there were programs in which the Federal Reserve itself lent money to business firms and municipalities, to people they had never lent money to before. And that lending, if you consider it part of government spending, was directly financed by creating new monetary liabilities. So that's government spending being increased, not by raising taxes, not by issuing interest-bearing debt, but just by issuing more Federal Reserve liabilities. Well, it's some mix of interest-bearing Federal Reserve debt and currency.
- 7.23** But usually when the Treasury is spending, it's the treasury that's spending, not the Federal Reserve. And for the treasury to spend more, they go into debt, they borrow more, they issue more T-bills. If the T-bills they issue, suppose they issue \$10 million in treasury bills, if the Fed then buys those \$10 million in treasury bills, the debt held by the public goes back down to where it started. That debt is now being held by the Fed. And that means that what's financing the spending is now an increase in Federal Reserve liabilities. The public is not financing it because they don't hold those bonds anymore. The Fed holds them.
- 7.24** So, in those two steps—the Treasury issues bonds, and the Fed buys them— put those two together, the change in the debt held by the public cancels out, and you've got more government spending financed by an expansion in the Federal Reserve's balance sheet, so government spending financed by an expansion of base money.
- 7.25** Now, for a government that relies on seigniorage for a big share of its budget, they have to think about the following. If you try to raise more and more revenue through this method, you're going to run into increasing reluctance to take your money. This is like what happens with any tax. With any tax, if you raise the tax per unit of, let's say you have a tax on liquor of a dollar a bottle, if you raise it to \$2 a bottle, it's probably going to increase your revenue, but it's not going to double your revenue, because you're adding to the price of a bottle of liquor. At the higher price, fewer units are going to be sold. And so, you have to worry about how sensitive the demand for the thing you're taxing is, with respect to the price.
- 7.26** There's a well-known proposition known as the Laffer Curve, because an economist named Arthur Laffer scribbled it on a napkin at some cocktail party, which says that if you raise the tax rate too high, you can shrink the market. You can shrink the sales of the things being taxed so much that revenue starts to go down. Beyond some point, you're killing the goose that's laying the golden eggs. So if you raise the tax on bottles of alcohol to \$20 a bottle, you're going to suppress sales of taxed alcohol. People are going to move to substitutes, namely, illicit alcohol, untaxed alcohol, but other kinds of intoxicants.

- 7.27 The same thing applies to money. If you try to tax money too heavily through seigniorage, at some point, people are going to start refusing that money or choose to hold very little of it, so that the thing being taxed by dilution is so small, the value of the entire money stock in real terms and purchasing power is so small, that even increasing it 10% or 20% doesn't buy very much. This is a phenomenon you see in countries with hyperinflation. So in a hyperinflation, a government is printing a lot of money to try to finance government spending, but the demand to hold that money becomes so small in real terms that the revenue isn't all that great. They've gone too far.
- 7.28 Long before Laffer drew the Laffer curve, an economist named Martin Bailey drew a curve relating seigniorage to the rate of monetary expansion. So we really should call the Laffer curve the non-monetary Bailey curve. This was back in 1956, Bailey plotted this out. So, at higher rates of monetary expansion, you get higher inflation. At higher inflation, you get higher expected inflation. People who expect higher inflation don't want to hold as much non-interest-bearing money. It's too costly.
- 7.29 So, the real demand for that money is going to dwindle, and at some point, it's going to be dwindling faster than the expansion is proceeding. So if you try to tax money too heavily, you actually get less revenue. The real tax base for this tax, which is the real stock of base money, shrinks. Now, governments that rely on seigniorage have some tactics for preventing this shrinkage in demand, for trying to resist it, trying to enhance the holding of their money.
- 7.30 One of the most famous is to force banks to hold more non-interest-bearing central bank liabilities through reserve requirements. Suppose a bank only needs 5% reserves to clear all its checks and to be safe against running out of reserves, but the government says you have to hold 20% reserves in the following form: non-interest-bearing liabilities of the central bank. And that was the routine in the U.S., up until we started paying interest on reserves. Reserve requirements were a tax on the banks by forcing them to hold non-interest-bearing Federal Reserve liabilities when they could have been holding interest-bearing assets. So reserve requirements corral the banking system into holding the base money.
- 7.31 An example where this was pretty explicit was when Italy decided to fight inflation by pegging the lira to the German mark. This was back in the '80s or '90s. And they understood that, to peg to the mark, the lira can't have an inflation rate higher than the mark's inflation rate. So they were going to have to tighten money growth quite a bit. They had been generating seigniorage by printing liras at a rapid rate. But now they're trying to fight inflation. So they're going to turn down the rate of growth of the money supply.
- 7.32 But what are they going to do to replace the revenue that they were getting by printing money? They said, "We'll tax the banks by imposing reserve requirements." So on the very day they pegged to the Deutsche Mark, they imposed much higher reserve requirements on the banks to offset that loss of revenue.

- 7.33** Other things governments can do to enhance the demand for their liabilities and make them less sensitive to inflation—any item being taxed, the sensitivity of demand to the price is what you need to worry about if you're the tax authority; that's called the elasticity of demand— and the way you limit elasticity of demand is by limiting substitutes. If you have close substitutes, people are going to switch quickly out of the thing being taxed. If you eliminate those substitutes, you can tax it, get more revenue by taxing it.
- 7.34** So what are the leading substitutes for holding currency, non-interest-bearing currency? Well, in many countries, especially high-inflation countries, people hold their savings in foreign currency. So if a government wants more seigniorage, it can restrict that. This is known as exchange controls. You're not allowed to go and get as many dollars as you want. You need permission from the government to get dollars, and that permission is only given to favorites, to friends of the rulers.
- 7.35** Bank-issued money, that is, deposits and banknotes, are substitutes for government-issued currency. But most countries, maybe not for this reason, but most countries have banned private banknotes, so that's off the table. To restrict the usefulness of switching into deposits, the government can restrict the interest rate that deposits can pay, so that it doesn't rise with inflation, and therefore, it doesn't detract people away from the government's money. So deposit interest ceilings can be used for that purpose.
- 7.36** In some countries, in some developing countries, the only commercial banks are owned by the government. So there, it's not that they're imposing this on a private bank; they're just sending a memo to the national bank, "Pay less interest on deposits. Don't raise the interest on deposits to keep up with inflation." And then there's no use moving your money from currency into deposits. It doesn't reduce your cost.
- 7.37** What's another thing people substitute into, to get away from inflating currency? Gold. If you want to restrict that avenue of substitution, put restrictions on the importing of gold, put taxes on people's holdings of gold, capital gains taxes are common. The government of India has a special import tax on gold now, because they were afraid that people switching into gold would reduce the demand for rupees and reduce the value of rupees. I think it's pretty pointless for that purpose, but nonetheless, that was their motivation.
- 7.38** Of course, today, one of the substitutes for fiat money is cryptocurrency. Governments that rely on seigniorage have been the most strict about banning Bitcoin or putting such onerous reporting requirements on it and taxes on it to discourage substitution from fiat money into cryptocurrency.
- 7.39** Now, if you look at actual rates of seigniorage, what share of the government budget is money-printing financing? It's pretty small in rich countries. In the eurozone, in the United States, it's typically been less than 2% of government spending is financed that

way. There are some years, like in the pandemic, where it was a little bigger than that. But at the other end, if you make a table of all countries and rank them by how much they rely on seigniorage, when you get to the countries that rely heavily on seigniorage, there have been cases where 100% of the government's budget was financed by seigniorage.

- 7.40 Or actually, the way it's measured, Yugoslavia financed 150% of the budget with printing money. I scratch my head over that one. What are they doing with the other 50%? It turns out, of course, Yugoslavia had a federal system where total spending, including the spending in all the parts of Yugoslavia, the regional governments, Croatia and Serbia and so on, was being counted in government spending. So it was 150% of the federal budget, but not more than 100% of total government spending at all levels.
- 7.41 Israel had a period where they relied heavily on seigniorage, and as a result, had inflation in triple digits for a while. Mostly, the countries that rely heavily on seigniorage, that is, from, say, 10% or more of their government budgets, are poorer countries—countries that have trouble collecting other taxes, have trouble collecting income taxes because, say, a lot of the economy is subsistence farming. It's hard to tax that. But you can always tax the currency that people use by issuing more of it.
- 7.42 It's important to understand seigniorage to be able to explain the cases of hyperinflation that we've seen in some countries under fiat money. If a government wants to stay in power, it doesn't want to destroy the value of the currency. It doesn't want to cause hyperinflation. And yet, that can happen if the government aims for something close to the maximum seigniorage they can get, and they miscalculate. They try to get more seigniorage than they can get in the long run, when people respond to the inflation by shrinking their demand to hold the money.
- 7.43 So the story goes something like this. The maximum the government can get, given the Laffer curve or the Bailey curve, let's say, is 10% of GDP by printing money, but they aim for 12%. Well, in the short run, when the inflation that's being created by rapid money-printing is unexpected, they can get 12%. They can buy up enough goods at existing prices to get 12% of GDP. But in response to this rise in inflation, people are going to shrink their real demand for money, which means that if you tax the existing real monetary base now 12%, you're not going to get as much revenue in real terms or as a percent of GDP.
- 7.44 So suppose the government responds to that by cranking up the expansion rate beyond what people expected when they shrank their demand. Well, then again, you can get more than is consistent with the long run, where people are correctly adapted to the inflation rate going on. So through a series of steps, you can get the inflation rate continually ratcheted up to get more seigniorage. And if you're aiming at seigniorage, which is infeasible when people correctly expect inflation, that is, you have to outrun their expectations to get that much, you're going to end up in hyperinflation. The inflation rate is going to keep accelerating in order to get more and more seigniorage.

- 7.45 Well, in order to preserve the amount of seigniorage you were trying to aim at initially, as the tax base shrinks, as the demand for the real money shrinks, you have to tax at higher and higher rates, and eventually the thing explodes. It doesn't converge to a stable equilibrium. It can't, because you're aiming for more than you can get in the long run.
- 7.46 Now, that's not the typical case in a fiat money economy. Most governments don't aim at maximum seigniorage and don't exceed the seigniorage-maximizing rate of monetary expansion, which has been estimated at being something like 200% a year. Most countries don't have inflation of 200% a year. So you might ask: why aren't they maximizing the take from seigniorage?
- 7.47 I think you could ask the same question about any tax. Income tax, I think, is not generating as much revenue as it could if rates were raised. But there would, of course, be pushback; and likewise, there's pushback against the inflation rates necessary to maximize the take from seigniorage. People rebel when inflation gets to 200%. They replace the government that's trying to get that much seigniorage if they can, so I think that's the practical limit. Getting that much seigniorage is very unpopular because it causes inflation rates that are very unpopular.
- 7.48 Then how do you explain hyperinflation, where they have cranked up the inflation rate to an intolerable level? Those seem to be cases where the government is not worried about the long run, not worried about creating a stationary flow of revenue, but they just need to grab as much as they can in order to stay in office. So, when there's some kind of crisis, like a rebellion or an invasion, they may say, "We can't worry about next year. We need this much revenue now."
- 7.49 Early in my career, I was invited to—so this was many years ago—I was invited to a conference in Argentina, in Buenos Aires. The conference was about privatizing government-owned industries. I was giving a talk about banking. But somebody who organized the conference arranged for me to have an audience with the head of the central bank of Argentina. I don't know why he wanted to talk to me. I was just an assistant professor. But I talked to him. I went to his office, talked to him. His name was Gonzalez Fraga.
- 7.50 At this time, Argentina had very high inflation. When I arrived for the conference, it was 600 pesos to the dollar, and a week later, it was 900. So the peso was dropping pretty rapidly. And the central banker was very frank about it. He says, "I know we're printing too much money to be consistent with any reasonable rate of inflation, but what can I do? I sit here at the central bank. I get a memo from the secretary, from the finance minister, and it says, 'Here's the payroll we need to make. Please print this many pesos.' Why are those—not literally print, but 'put that many pesos in our account so that we can pay government workers or pay pensions.' What can I do?" That's a case in which the fiscal authority was really calling the shots, and they thought they needed this many pesos, and they weren't worried about the long-run consequences.

- 7.51 I once had a student write a dissertation on the civil war in Lebanon back in the '80s. That was a case in which the central government of Lebanon was enjoying a certain amount of seigniorage. But when the civil war happened, suddenly the demand for their currency shrank, but they still needed the revenue. In fact, they needed more revenue to fight against the factionalization of the country.
- 7.52 So, by continuing to aim for the same seigniorage, when the demand to hold their money had shrunk dramatically, they generated hyperinflation, but they had no other means of raising revenue rapidly. Their tax base on all kinds of taxes was shrinking. So there's usually a fiscal, I think maybe always, a fiscal motive behind hyperinflations that accounts for the willingness to print money like there's no tomorrow.
- 7.53 There's a well-known book by Thomas Sargent, who I've mentioned, called "The Ends of Four Big Inflations." If you want to come out of a hyperinflation, you need to make a credible commitment, the government needs to make a credible commitment, not to keep up those rates of monetary expansion, but to make it credible, to make people believe it, and thereby lower their expectations of inflation and restock their real money balances; there has to be a fiscal reform. To convince people they don't need to print money, the government has to convince them, "We've got other sources of revenue. We can raise revenue from other methods than printing money."
- 7.54 Often these days, a country that's trying to come out of a hyperinflation will go to the IMF and say, "If we're going to turn off the printing press, we need to replace that revenue somehow; and the economy is not going to rebound immediately, and our income tax revenues are not going to rebound immediately, so we need a bridge loan." And that's the case they make. "If you get us through this patch until we can restore people's confidence in the currency, we're not going to make it."
- 7.55 Now, how big is seigniorage in the United States? I mentioned it's typically about 1-2% of the federal spending. In the early 2000s, seigniorage was running about \$100 billion a year; that is, the Federal Reserve earns interest on the assets it buys with its newly printed money. When they weren't paying any interest to banks on their reserves, that's a pretty profitable business. You earn interest on treasury bills, which you can buy at will. You don't pay any interest to the people who are holding your liabilities. Result: profit.
- 7.56 But in the financial crisis in 2008, the Fed asked—well, Congress had authorized the payment of interest on reserves. The Fed asked them to speed that up. This is under Ben Bernanke. The reason they wanted to do that was so that they could expand their balance sheet enormously, make all kinds of loans to people they thought needed them to get through the financial crisis, without causing inflation in proportion to this expansion. So they more than doubled the monetary base. If nothing else had happened, that would have doubled the price level. It would have had very high rates of inflation.

- 7.57** They said, "We will bottle up the potential inflation by paying interest on reserves, and that will convince the banks to hold the reserves, rather than to lend them out and generate more spending in the economy." It sounds paradoxical, but as part of the recovery program, the Fed was paying banks not to lend, and it's still paying banks not to lend. But that corresponds to the Fed itself doing the lending through the expansion of its balance sheet.
- 7.58** Before the financial crisis, the Fed had about \$800 billion in assets. Today, it has over \$6 trillion. So there's been enormous expansion in the last 20 years, the last 18 years. So the Fed began paying interest on reserves in 2008 and expanded its balance sheet. And for a long time, interest was close to 0%. The Fed was holding interest rates close to 0%.
- 7.59** Sorry, I have the numbers here. Between 2011 and 2019, the Fed issued about \$80 billion a year in new Federal Reserve notes. So that's pure profit, because they're not paying any interest on those. This peaked during the pandemic, at about \$200 billion, it has since dropped back down to about \$40 billion. Last year, it was \$42 billion. So that's the current seigniorage. That's a small drop in a \$7-trillion federal budget. That's much less than 1%.
- 7.60** If the treasury is not getting as much seigniorage as a share of its spending than it used to, where is it getting the rest? Debt, and it's borrowing it. In a sense, interest on reserves is another way of borrowing. The Fed is selling its T-bills to the treasury, but the treasury, in order to hold those T-bills, has to create reserves. For those reserves not to be inflationary, the Fed has to take the interest on the T-bills and pay it to the banks. It's a kind of indirect way of getting the banks to hold treasury bills, to finance the additional spending.
- 7.61** So, what do we see? In recent years, we see a growing ratio of federal debt to GDP. The share of government borrowing to total income has gone up. Before the financial crisis, it was fairly steady at about 60%, the ratio of public debt to GDP. Public debt, that is, the outstanding stock of treasury bills, represents the accumulation of all the previous deficits, minus any periods in which there were surpluses; in a surplus, you buy back some of the debt. There have been very few of those periods. So the debt-to-GDP ratio is about 60%.
- 7.62** And for a household, the amount of debt you can carry depends on your income. If your debt is not too high relative to your income, you can service the debt. Now, for a household, 60% debt to income would be a very high ratio. But for a national government, it's sustainable. It's not ideal, but it's sustainable. During the financial crisis, borrowing ramped up enormously, and the new plateau was 100%. So from about 2012 to 2020, federal debt was about 100% of GDP. In the pandemic, it shot up again, and the new plateau was about 120%. So total federal debt is about 120% of GDP.

- 7.63** And why does that matter? Because to service the debt, to pay the interest on the debt, and roll over the T-bills as they mature, the federal government needs revenue. And to get revenue from taxes, it needs to have a growing economy. So, the way we measure the sustainability of the debt is to ask: how big is it relative to the ability to pay, which is proxied by GDP? And so we're now, compared to 2006, we now have double the debt burden per dollar of GDP. So the debt the federal government has taken out on your behalf has doubled.
- 7.64** And what that is going to mean is that the cost of servicing the debt is going to double. The cost of servicing the debt, of course, depends on interest rates. When interest rates were being held close to 0%, it seemed like this was a free lunch. We can carry this debt without having to pay out anything. But as interest rates return to normal, it should be obvious that the more debt you have, the higher your debt service is going to be, the more interest you have to pay out every year.
- 7.65** If we run a deficit, say, of 10% of GDP, that's going to add to the size of the debt, 10% of GDP. That's going to consume a big chunk of private savings, that is, savings of U.S. households. In order to spread the burden, or in order to sell the bonds at favorable prices, of course, the treasury sells them to anybody in the world. And a lot of U.S. debt is held abroad, nearly half of it. The biggest places it's held are Japan and China. And as a taxpayer, you should be glad that people in the rest of the world are willing to buy this debt, because if they weren't, the interest rates would be higher, and the burden on U.S. taxpayers would be higher.
- 7.66** But no matter who buys the bonds, the debt service is going to have to come from U.S. taxpayers, so future taxpayers; they're going to have to service the debt. If the debt gets too high to service with taxes that people are willing to pay, then there's the threat that the next resort is printing money.
- 7.67** A country that got into very heavy debt service in Europe was Spain. As a result of the financial crisis, there were sovereign debt crises throughout Europe, and especially Greece, Italy, Portugal, Spain, and Ireland: G-I-P-S-I, the GIPSI countries. Some people rearrange the letters and call them the PIGS, but I think that's a little too derogatory. But in Spain, in 2013, one-third of the government budget was debt service. In the U.S., that's not that high, but it's growing.
- 7.68** Now, there is a school of thought, which likes to call itself modern monetary theory, which says, "Don't worry. Issuing more money won't cause inflation until the economy is completely at capacity. Until we've reached the absolute limit of what we can produce, people won't raise prices when there's more demand for goods. They'll just produce more output. You'll put to work unemployed workers and machines."
- 7.69** This is a kind of wishful thinking, or maybe fantasy, that has been inherited from the Keynesian economics of the 1960s. This is the idea of the multiplier effect of government spending. So the wishful thought is that government spending will pay for

itself by stimulating so much economic activity that tax revenues will rise by enough or more than enough to pay for the spending.

- 7.70** My freshman economics instructor was an economist named Otto Eckstein. We had to read Eckstein's book on fiscal policy, and he makes the following statement. He doesn't call it deficit spending. He calls it debt, which is admirably frank. He says, "What about debt incurred during periods of unemployment? Resources would have been idle, so no other outputs are foregone," meaning the government can spend to buy things, and it doesn't cost the economy anything, because those things are going to be produced by workers and machines that were otherwise just sitting there idle. So there's no cost to using them. We just get more output. We won't see any rise in prices.
- 7.71** In fact, he says, output is going to be increased by a multiple of the initial spending because the recipients of the government's spending are going to turn around and spend on other things and raise demand for other goods and services. So he says, "The creation of debt in this situation raises output and is likely to raise investment and the total growth of the economy." So, financing your spending with debt in a less-than-fully employed economy is a free lunch. It's better than a free lunch if you have a multiplier effect. It's free lunch with a free dessert.
- 7.72** But this is economics without scarcity. That's why I'm calling it wishful thinking— the idea that there are unemployed resources of all kinds that can be called upon at existing prices. You don't have to pay the workers any more; you don't have to pay the machine owners anymore than they were getting, because they're sitting around doing nothing; they'll be happy to take the job. This is wishful thinking.
- 7.73** In fact, when governments spend on shovel-ready construction projects, a big share of the workers they hire, they hire away from other employments. And a big share of the machinery they hire, they hire away from other employments. So there is a sacrifice of other outputs.
- 7.74** But if your view is that there's no cost to government spending, well, then you don't need to be worried about the debt being incurred, because what you really want to worry about is the ratio of debt to GDP, and this spending is going to increase GDP by as much or more than the debt is being increased, due to the multiplier effect. So don't worry about the debt-to-GDP ratio.
- 7.75** Now, this idea that incurring debt makes the economy behave as though it's wealthier —people are going to spend more, consume more— was challenged by an economist named Robert Barro in a paper called "Are Government Bonds Net Wealth?" And the implicit answer was no, it's a debt from some members of the economy to others.
- 7.76** Barro recalled a much earlier argument made in the 19th century by the economist David Ricardo, who said, "If government borrows to finance spending and services that debt with future tax revenues, it's the same to the taxpayer as if they taxed him the full amount because, if he's got money in the bank, they're now taxing basically away the

interest on that money in the bank. That's the same as just taking the money up front." In present value terms, the two streams are equivalent: either taxes today or a stream of taxes tomorrow, which, with interest, is equivalent to the tax today. So that's called accounting equivalence. That's Ricardian accounting equivalence. And that's just good bookkeeping. That's not really controversial.

7.77 Barro made a stronger claim, though, which is that taxpayers will recognize and respond to this increase in the deficit. So if the government switches from taxing them to borrowing, holding government spending constant, taxpayers will respond by saying, "Oh, we've got future tax liabilities now coming that we didn't have coming before. We need to save in order to pay those taxes." So the story assumes that the public will buy all the treasury bonds in order to finance the spending.

7.78 And then there's no effect of the increased deficit on drying up the pool of savings available for private investment. There's no increase in the interest rate. There's no crowding-out of business borrowing. There's no reduction in capital formation. And therefore, there's no pressure to monetize the new debt. That, I think, is maybe a little bit optimistic, because we do see governments that are heavily indebted facing higher interest rates, the debt service crowding out other government spending. And so, there is pressure from the fiscal authorities to the monetary authorities, if there isn't any constitutional barrier against it, to monetize some or all of the deficit; and by monetize, I mean print the money to cover it. So in the absence of Ricardian behavioral equivalence, that is, in the absence of a public internalizing the effect of deficit spending by saving more, there is going to be pressure on this scarce pool of loanable funds. It is going to crowd out private borrowing because there's only so much savings to go around, and that can put pressure on the Fed to monetize the deficit, to buy more of the treasury's debt, in order to help finance the spending without the treasury debt increase, raising interest rates, and crowding out private borrowing.

7.79 So the printing of money to finance the deficit will, of course, be inflationary. And by causing higher inflation, it swindles the holders of existing government bonds because now they're going to be repaid in dollars that don't buy as much as the dollars they lent, as the dollars they expected to get back when they bought the bonds.

7.80 So we have to define it relative to expectations. It's surprise inflation that favors the debtor—the big debtor here is the U.S. government—and impoverishes the creditors—here, the creditors are anybody who holds treasury bills. And a lot of us do hold treasury bills without realizing it, because we have pension funds that are holding treasury bills. If we don't take into account these future debt obligations, it becomes very tempting for Congress to say, "Yeah, let's finance the spending by debt. And we'll count on the Fed to make sure that the debt doesn't squash the private economy." So they're tolerating faster money growth, higher inflation, to stimulate growth or to keep

interest rates low. Of course, the Federal Reserve right now is subject to enormous political pressure to keep interest rates low.

- 7.81** But this is short-sighted because faster money growth and higher inflation, those are going to raise nominal interest rates. When borrowers and lenders come to anticipate higher inflation, they're going to agree to higher nominal interest rates, so that the real terms of exchange in purchasing power terms are preserved. Lenders are not going to keep lending at the same interest rate when they know they're going to get paid back in dollars that are worth less than they previously expected.
- 7.82** This phenomenon of debts being repaid by reducing the value of the currency, today it's done by diluting the existing currency, issuing more money and driving down the purchasing power per dollar. This same phenomenon was seen centuries ago. Adam Smith commented on it. He says, "All over the world, you see princes and sovereign states who have debts, who repay them by diminishing the quantity of metal in their coins and claiming the new coins are just as valuable and forcing the creditors to accept them as full payment."
- 7.83** So, in appearance, they pay their debts, but they're doing it with a smaller quantity of silver than would have been an honest repayment of the debt. So the creditors were really defrauded, he says, of a part of what was due to them. So debt is being repudiated through inflation.
- 7.84** We had a similar phenomenon under our fiat dollar standard. If you look at the debt the U.S. had at the end of World War II, that's the last time we had debt above 120% of GDP; it was about 130% in 1946. By 1970, the debt-to-GDP ratio had been reduced to 40% of GDP. That would be great if all of it was due to growth in real GDP, but about half of it was due to growth in real GDP, and the other half was due to inflation. Inflation raises prices; it raises nominal GDP. If we're measuring debt in dollars and GDP in dollars, that's going to raise the GDP and the denominator. If you bought a 30-year treasury bond in 1946, the dollars that were used to pay you back at the end were worth about half of what the dollars had been worth in 1946. So about half of the debt was inflated away.
- 7.85** Now, the U.S. is not in a condition right now where the Fed is responding to requests from the Treasury to print money to pay bills. That is a problem in some countries. It's not yet a problem in the U.S., but it can become a problem if debt becomes so large that it's not feasible to finance it any other way. And this scenario goes under the name of "unpleasant monetarist arithmetic."
- 7.86** The economists Thomas Sargent and Neil Wallace wrote about this back in the '80s when it was a problem far on the horizon. Because the debt-to-GDP ratio was in the range of 40-50%, it didn't seem like this was a pressing concern. But we're now at 120% of GDP. It's becoming a little more plausible that debt service is going to start to eat up

more of the budget, and that's going to put pressure on the Fed to monetize more of the debt.

- 7.87** And there's a limiting case where there's no other choice; well, your choices are limited to either inflating away the debt or monetizing the debt or defaulting. So if debt gets so high that you saturate the market for your treasury's bonds, you get to a point where you can't issue any more debt without the interest rate on that debt going up by so much, because the market is reluctant to buy any more of your debt, going up by so much that it negates the amount you borrowed. Your debt service rises by the amount you borrowed, so you don't get any more resources on net. This is maximizing the amount of financing you can do with bonds. If you try to go beyond that, there's no way to finance additional spending with debt. Something has to give; either you default on your debt or you try to inflate it away.
- 7.88** If we look at the countries that have had debt crises in recent years, well, in the 2010s, we look at Greece, for example. They couldn't inflate away their own debt because Greece is part of the eurozone; they don't issue euros themselves. The ECB controls that, the European Central Bank. So what were they going to do when they couldn't pay their debts? They appealed to the ECB to buy the debt from them at a concessionary rate. They wanted transfers or loans at low interest rates to try to refinance the debt. Eventually, they defaulted.
- 7.89** I remember the week they defaulted, because I was listening to NPR, and a commentator came on and said, "We can't let Greece default, because if Greece defaults, Greece has to leave the eurozone. If Greece leaves the eurozone, the eurozone falls apart. If the eurozone falls apart, the European Union falls apart. If the European Union falls apart, World War III." So, this kind of doomsday scenario was to encourage all the governments in Europe to bail Greece out. They finally sat down and said, "We don't want to bail Greece out. There are some banks that have a lot of Greek debt, so let's just rescue those banks, keep them from failing, but let the other creditors to Greece take a haircut." And Greece did default on its debt.
- 7.90** But in the U.S., we never have to default on our debt. It's in dollars, and we can print as many dollars as we want, to pay what is legally owed, which is not dollars of a particular purchasing power, but just dollars. Deficits in a country that doesn't need to default or doesn't want to default, when they get too high, when the burden of debt gets too high, it unleashes more reliance on seigniorage, more money-printing. That's the unpleasant arithmetic.
- 7.91** How do you stop that? Well, having an independent central bank is kind of a first line of defense, but it's really not enough. You have to have a system in which the monetary authorities don't have to, or the government as a whole doesn't have to, resort to money-printing, because the debt doesn't get that high, because the fiscal side of the picture is under control. If you have a really binding commitment not to inflate, then

you force the fiscal authorities to behave. But the question is: how can you write that kind of binding commitment? And we'll talk about that in the next lecture.

- 7.92 Now, in the U.S., we haven't reached the point yet where additional debt will raise interest rates so much that it negates the borrowing; the rest of the world seems pretty willing to keep accumulating treasury bonds, but that can't go on forever. At some point, we're going to exhaust the willingness of the world to buy U.S. debt.
- 7.93 So it's hard to say when that's going to happen. We've raised the debt-to-GDP ratio. How high does it have to go before lenders in Japan and China and Europe say, "We think there's a risk here in buying your treasury debt—not a risk of default. We know we'll get the dollars, but a risk that the dollars won't be worth anything when we get them." So an inflation premium will get put on treasury debt that will make it very expensive to finance.
- 7.94 Borrowing in your own currency, which is a privilege the U.S. has—a lot of countries can't do that; nobody will lend them money in their own currency because they are aware that it can be inflated away if they do that. Borrowing your own currency means you don't have to default, but on the other hand, there's a risk of inflation. If we want to avoid that risk, we have to keep our finances on a sounder basis, so that we're not threatened with an avalanche of debt that forces monetization of that debt.
- 7.95 It's not clear when we're going to reach that point, but we have doubled the amount of debt. Then between 2007 and 2018, we tripled the amount of treasury bonds in dollars. We've since increased it. We've doubled it again since then. We got more than \$30 trillion in debt out there in the market, whereas back in 2007, it was less than \$5 trillion. That's not encouraging. That's concerning.
- 7.96 Well, on that happy note, but this raises the question that we'll talk about in the next lecture, which is: how do we constrain monetary authorities not give into this pressure? How do we commit them to a low inflation policy?

Lecture 8: How to Secure Better Money

At a glance — How to secure better money, in the finale.

- 8.1** Welcome to the eighth lecture. Thanks for holding on so long. So we ended the seventh lecture on a little bit of gloom and doom. Basically, where we arrived at was that we have a fiat monetary system where the control of the quantity of money is in the hands of a political agency or an agency that's subject to political pressure. There's a danger that problems in the fiscal side of the federal government are going to spill over into a threat to the soundness, the integrity, the continued purchasing power of the dollar. We have to worry about the road to the devaluation of the dollar from the inability to get our fiscal house in order.
- 8.2** But I said one way to try to fight that was to strengthen the commitment of the central bank to maintain low inflation. So are there feasible ways to do that? Are they credible? Will they really work? That's what we're after: some way to secure sound money, prevent our money from being the victim of fiscal problems or other reasons for wanting to print lots more money. Economists have been concerned about this issue of maintaining or ensuring sound money by giving better instructions to the central bank or by giving more resistance to the central bank, the ability to resist fiscal demands.
- 8.3** The discussion has focused on what are called rules for central banks. So there's a famous article entitled "Rules Rather than Discretion" by Kydland and Prescott, who got a Nobel Prize a few years ago, basically, partly for this article, at which they make the point that the lack of rules, the lack of a pre-commitment—this is language they introduced—the lack of a firm pre-commitment not to inflate is something that economists thought was a good idea. They thought discretion was a good idea. The central bank can always pick the best policy, so why would you want to tie its hands in advance? Why would you want to say to it, "No, you're not allowed to expand money more than this or that, or have some other formula?"
- 8.4** They made the point that this lack of pre-commitment is actually a weakness. It means that people can't trust the central bank to do something in particular. They can't form expectations of a low inflation policy. If the central bank faces temptations to print money faster, and they're not firmly committed not to do so, that can ruin the arena they're playing in.
- 8.5** In particular, if the public has higher inflation expectations, then the central bank is in a difficult position. Either they provide the inflation that people are expecting so that people's pricing decisions and wage decisions are correct, or if they inflate less than that, they're going to disappoint everybody. There's going to be less demand for goods than was expected. There's going to be less demand for labor in dollar terms at the prices people are charging for goods and labor. That's a formula for a recession.

- 8.6** So the central bank becomes trapped in ratifying high inflation expectations. It's a game being played between the public and the central bank, which has an outcome which nobody wants: high inflation, but no impact on the real economy other than the cost of higher inflation.
- 8.7** Now, let me step back a minute and ask, why am I focused on inflation? Can't we have a more ambitious goal for central bank policy in a fiat money system? Can't we ask them to promote real economic growth? Around the world, there are some central banks that are also supposed to act as development banks. They're supposed to choose places for the economy to invest. Can't we ask the central bank to lower real interest rates? Can't we ask it to stabilize the economy against business cycles?
- 8.8** For any goal like this, we need to ask: Is it really feasible? Can the central bank do it, given that its tools are limited to controlling the growth in the quantity of money? Now, we know growth in the quantity of money, control over that, is sufficient to control inflation. But what about these other more ambitious goals, especially goals defined in terms of real variables?
- 8.9** First, we want to ask, are these other goals really feasible? And for the ones I just mentioned, they're not. Monetary policy cannot permanently raise real income. It can't even permanently lower the unemployment rate. And I mention that because the Federal Reserve Act, as amended, charges the Fed with responsibility for maintaining what's called maximum employment, which means they should be concerned about not letting the inflation rate rise too high.
- 8.10** But what we've learned at this point in the course is that faster money growth is going to get you higher inflation, but it's not going to raise the real productive output of the economy. That requires real resources, not more zeros on bank accounts and currency. This message is sometimes hard to get across because there is a short-run effect where unexpectedly expansionary monetary policy, putting more money into people's pockets, generating more spending, can give a temporary goose to the economy. It can boost real output, reduce unemployment.
- 8.11** When business owners see more demand coming in their doors, they're not 100% sure, "This is just more inflation, and so I need to raise my prices by the full amount of the increase in demand." Instead, each producer says, "Well, it looks like what I'm selling is more popular than it was. So I should raise my prices a little bit to take advantage of that, but I should also increase output to sell more. So I'm going to send more orders to the warehouse, and the warehouse is going to send more orders to the factory to crank up output." So you can get a temporary boost in output.
- 8.12** But we've learned that these temporary boosts in output don't last. It's a boom followed by a bust, a return to normalcy. In fact, resources are wasted in the investments that are initiated during the boom that can't be sustained. People make

commitments by sinking capital into construction projects, and then they end up with houses they can't sell. That was the housing crisis in 2007, 2008.

- 8.13** So, printing more money doesn't increase the economy's real resources. It doesn't improve the skills of the labor force. It doesn't increase the amount of machines or raw material or land. It doesn't improve our institutions and so on.
- 8.14** So should unemployment be one of the Fed's policy targets? I don't think so, because what I said about output also applies to unemployment. A monetary policy stimulus can temporarily reduce unemployment or raise employment, if you want to put it that way, but that effect doesn't last. People will take jobs sooner if wage offers go up more than they expected.
- 8.15** When people are searching for a job, they have some estimate of what their labor is worth. That estimate in dollar terms is going to incorporate their expectation of how much wages in their field have been rising. So it incorporates an estimate of inflation. If there's more inflation, and so dollar wage offers are higher than people expected, they'll meet an acceptable wage offer sooner in their search for a job. And so that can reduce unemployment: people spend less time looking for a job. The unemployment rate, which is the ratio of people looking for a job to the total labor force, that ratio goes down because people find jobs sooner.
- 8.16** But that's based on an illusion. It's based on fooling the job seekers into taking jobs sooner than they otherwise would. On the other side, employers are hiring more people because they think their business is more profitable. But that's based on not realizing that this is just an increase in nominal demand, thinking that there's a shift in demand toward what they're producing, so they ought to increase output in real units.
- 8.17** The temporary nature of the boost to employment or the reduction in unemployment is illustrated by the breakdown of what was called or is called the Phillips curve. So Phillips plotted a relationship between the rate of change of wages and the unemployment rate in Great Britain under the gold standard. And he found that in years where there was a tight labor market, that is, higher than usual demand for labor, unemployment would go down—makes sense—and the rate of rise of wages would go up. So employers are bidding up wage rates and hiring people out of the pool of the unemployed.
- 8.18** In Keynesian economics, two Nobel Laureates, Paul Samuelson and Robert Solow, converted this into a theory of inflation. They said, "Let's put the rate of inflation on the vertical axis instead of the rate of change of wages." Now we have a relationship between inflation and unemployment. It's a trade-off. If you want less inflation, you have to be willing to tolerate more unemployment. If you want lower unemployment, you have to be willing to tolerate higher inflation.
- 8.19** This is the choice policymakers face. This is the menu of options they have. If you want to lower unemployment, you got to crank up inflation a little bit and vice versa. The

claim is that the Fed can lower the unemployment rate by expansionary policy. You do have to tolerate higher inflation, but they make it look like this reduction in unemployment is permanent. There's no talk about it being temporary. This is a permanent policy trade-off.

- 8.20** So the wise policymaker chooses the least bad combination of inflation and unemployment. That is Phillips curve thinking, and it's still with us. You find it in the Fed's press releases and policy statements that we're worried about balancing the risks of inflation with the risks of unemployment. But this is only a temporary trade-off, and it only lasts as long as people don't perceive the inflation that's taking place. But people have become more sensitive to forecasting inflation.
- 8.21** When Samuelson and Solow wrote in 1960, it was empirically plausible. You couldn't point to evidence that this relationship was going to break down. If you plot all the combinations of inflation and unemployment, it looked like there was a trade-off in the 1960s, in the late '50s through the late '60s. So it looked like there was a stable Phillips curve.
- 8.22** But then all hell broke loose. In the 1970s, we got higher inflation combined with higher unemployment. So we were way above the original Phillips curve, more of both bad things. In the early '70s, we were a little bit above. In the late '70s, we were way above. So the combinations in the '60s were like 4% inflation with 3% unemployment. But in 1980, we had 12% inflation with 7% unemployment, not in the range that the Phillips curve captured.
- 8.23** So what happened? What happened is that people stopped expecting zero inflation, and so increases in inflation stopped fooling them into taking jobs sooner and stopped fooling employers into expanding output just because nominal demand was going up.
- 8.24** The Federal Reserve has an economic database where they keep track of a long list of variables, including the unemployment rate. If you look at a plot of the unemployment rate, it shows a great deal of cyclicity. In fact, almost by definition, a recession is when unemployment is high. Recession is technically defined by the behavior of real output. But when real output is declining for consecutive quarters, you can be pretty sure that unemployment is going up because real output declining means that factories are closing and workers are being laid off.
- 8.25** But if you take a longer run perspective, you can see that the unemployment rate keeps coming back down after a recession. What does it come back down to? That's sometimes called the natural rate of unemployment. The Fed calls it the non-cyclical rate of unemployment. That is, if we take out the business cycle, here's what the unemployment rate we should expect. So this is not directly observable. This is inferred from the data. They call it the rate of unemployment arising from all sources except fluctuations in aggregate demand.

- 8.26** But what are these other sources? Well, just normal frictional unemployment, that is, you graduate from college—I used to advise my students this—you walk across the street to where your parents have parked. You leave the graduation ceremony. You walk across the street to the shopping center where your parents have parked the car illegally. You pass the Wendy's that has a sign in the window that says, "Now Hiring." If you just got a BA, you don't take that job. Why don't you take it? Because you have a degree in economics, and the Job Office, the Career Planning Office, told you, "Here's the salary you can expect to earn with your degree." And that's what you should hold out for. Don't take the first job that comes along.
- 8.27** So the natural rate of unemployment is people searching for jobs in a normal fashion. They've left school, now they're looking for a job. They've quit their previous job, they're looking for a job. They've been laid off because some industry is in relative decline, not the whole economy, but some industries become obsolete, and have to lay off workers. Those people have to look for jobs and so on. There's a certain amount of churn or turnover in the labor market. That's what's being estimated.
- 8.28** Estimates of this natural rate of unemployment are currently around 4.3 %. The U.S. economy is currently around 4.3% unemployment. So we're not in a recession. We're at a normal rate of unemployment. When you're at a normal rate of unemployment, you can get unemployment below that. So it's not a lower bound. In a boom period stimulated by easy monetary policy, unexpectedly easy monetary policy, wage offers are going to be better than expected. People are going to take jobs sooner than they usually would. That's going to shrink the pool of people looking for jobs. That reduces the unemployment rate.
- 8.29** So, that's adding expectations to the analysis of unemployment, which is a big improvement because then you know that high inflation is not going to permanently reduce unemployment. People are going to build that inflation into the wage they hold out for. And so if they're accurately predicting inflation, high inflation is not going to make them take a job sooner. It's not going to fool them. So there's no relationship in the long run between the unemployment rate and the inflation rate.
- 8.30** This is sometimes described as having a vertical long-run Phillips curve: no relationship between inflation and unemployment. So if all inflation was perfectly anticipated, you wouldn't see expansionary monetary policy reducing unemployment at all. It's only when it's unexpected that it has that effect. But in a sense, you're fooling people into taking jobs sooner than they otherwise would, which isn't efficient. People are taking jobs that maybe they should have held out for a better job. So, permanently reducing unemployment is not within the powers of monetary policy. It's futile, actually worse than futile; it's counterproductive to assign that task to a central bank to keep the unemployment rate down.
- 8.31** A similar argument applies to the real interest rate, that is, the interest rate after you subtract inflation from it, the real terms of trade between present and future. That's

going to be determined by how willing savers are to give up present consumption in exchange for future consumption, and how willing borrowers are to give up future consumption when they repay the loan to get the present resources they can buy if they borrow money. So the supply is thrift. The demand is expected investment returns, mostly.

- 8.32** Those are real variables. There's nothing monetary about them. But an increase in money growth can disturb interest rates because it can put more money on the loanable funds market before people have increased their expectations of inflation correctly, and thereby, in the short run, drive down the real interest rate.
- 8.33** This is much like driving down the unemployment rate. It's only temporary until people catch on to the inflation that's coming down the pipe. When they catch on, the relationship is—which we discussed in, I think, Lecture 4— higher money growth, higher inflation, higher inflation, higher expected inflation, higher expected inflation, higher nominal interest rates. As I mentioned, countries with chronically high inflation and rapid money growth don't have low interest rates; they have high interest rates. The nominal interest rate is the sum of the real interest rate, approximately, and the expected inflation rate.
- 8.34** What about fighting business cycles? Can we use monetary policy to stabilize the real output of the economy? Under certain circumstances, yes. I mean, that may surprise you that I say monetary policy has any capability to do good. But here's a case where they can do good, but it's very difficult in practice to do good because you have to get the timing right, and you have to get the magnitude of the change in money growth right. To get those right requires a degree of foresight at the central bank better than private forecasters, which is unlikely because private forecasters pay better than the Fed does.
- 8.35** If you could, if the central bank could do that, it would be great. So, when is it called for to increase the growth rate of the money supply? If people are trying to accumulate money balances or hoarding to give it a negative connotation. But if people are raising their desired ratio of money balances to spending, providing them with more money balances avoids them having to reduce their spending in order to build up their money balances. And so you avoid a rise in interest rates due to a scarcity of money. You help keep the output of the economy near its potential output by not putting the economy through a recession due to unsatisfied demand for money.
- 8.36** A recent example of that was at the beginning of the pandemic. A lot of things were going on, but one thing that was going on was households were not spending as much. They were building up their money balances. People who were still employed, still being paid, their bank accounts were starting to swell because they were not spending as much. They were not spending as much because restaurants were closed, flights were being canceled, and so on. Under those circumstances, it helps to stabilize total

demand for goods in the economy to provide people with the additional money balances they want to hold.

- 8.37** But when spending starts to return to normal, when people start to spend off those additional money balances they accumulated, the central bank, the Fed, has to be prepared to pull the extra money out of the economy, or you're going to get an excess supply of money. Instead of satisfying an increased demand, now you've got money in circulation that's not being demanded at current prices that's going to drive prices up. That's why we got 9% inflation in the U.S. in 2022. The Fed failed to pull out the extra money it had provided as spending returned to normal, as the ratio of money balances to spending returned to normal.
- 8.38** So it would be desirable if the Fed could anticipate changes in the public's desire to hold money relative to spending, or, in shorthand, changes in the velocity of money, anticipate them and meet them with the right size change in the quantity of money. Track record is not good. So it would be desirable if it could be achieved, but the track record in practice is not good.
- 8.39** What are the problems? It's hard to anticipate. Nobody anticipated the pandemic, of course. Typically, the Fed responds too late to an economy that's entering a recession because of a scarcity of money. And they react too late in both directions. They're too late to increase the quantity of money and too late to reduce the quantity of money when that's appropriate.
- 8.40** To improve the performance of the private economy—some economists have shown using various very abstract formal models—the Fed would have to better anticipate the path of the economy than people in the private economy anticipate it. But that's, as I said a minute ago, that's unlikely. In practice, the Fed tends to react too late and to overreact, making the swings in the economy bigger.
- 8.41** This is a persistent theme in Milton Friedman and Anna Schwartz's "A Monetary History of the U.S." They have a simple test for whether the Fed is helping stabilize or destabilize. If they're helping stabilize, then the money supply should be going up when velocity is going down: total spending is M times V , the number of units of money times how many times each unit turns over per year. That gives you total spending. So, you could stabilize total spending if the money supply goes up when velocity is going down and vice versa. Then the product of the two remains relatively stable.
- 8.42** And what they found historically was that when velocity was going down, the Fed was also shrinking the money supply. And when velocity was going up, the Fed was expanding the money supply more often than not. So more often than not, the Fed was amplifying movements of the economy away from a steady growth path. That's known as the monetarist critique of stabilization policy. The central bank policy moves are poorly timed and poorly sized in practice. If we had an omniscient Fed, that would be different, but we don't. In practice, this policy of leaning against the wind or trying to

counter the business cycle has generated more variability, not less, in the real output of the economy.

- 8.43** To make things worse, when the public realizes that the monetary policy is like that, it's not stabilizing inflation at a low figure, but it's variable, it's unanchored, people's expectations of inflation become a little more volatile. When their expectations of inflation go up—this is sometimes called an inflation scare— people will try to reduce their ratio of money balances to spending. They want to hold smaller money balances if the cost of holding has gone up because expected inflation has gone up.
- 8.44** But that itself, people spending off what are now excess money balances, will drive up spending, drive up inflation, and so the expectations of inflation being unanchored will introduce more volatility to inflation and to real output. And that's what we see since the last constraint on monetary policy in the form of the commitment to redeem the dollar for gold at \$35 an ounce: it disappeared.
- 8.45** Now, we have a new constraint, which is that the Fed has adopted on its own accord, at its own initiative, a commitment to—a commitment is maybe too strong—a target of 2% inflation. They say, "We want to keep inflation from rising above 2%." It remains to be seen whether that is going to stabilize inflation expectations. That's the hope. If the Fed actually behaves in a way consistent with maintaining 2% inflation, then it should bring inflation expectations down. This commitment means that the Fed is not looking at unemployment and not trying to vary monetary policy to dampen swings in unemployment or swings in real output.
- 8.46** If the Fed's not committed to a certain inflation rate or committed not to try to counteract unemployment or recessions, the public knows that, they're going to raise their inflation expectations because they're going to worry— this is the Kydland and Prescott story—they're going to worry that the Fed is going to be tempted to inflate, to try to pull inflation up. Even when inflation is at the normal rate, the Fed may prefer inflation above that.
- 8.47** So what can monetary policy do usefully? It can stabilize the path of the price level. It can keep inflation down. Now, by controlling money growth, because we've got two other variables in the equation of exchange, velocity and real income, variations in velocity that are unanticipated are going to change the inflation rate from the target path. Changes in real output for real reasons, like a pandemic, are going to change real output. The Fed can't perfectly hit an inflation target or a price-level path target, but it can keep bringing the price level back to the target path within a few quarters. So it can offset any real disturbances in the economy that are affecting the realized inflation rate.
- 8.48** Compared to an unpredictable price level, this is an improvement. Making the price level path predictable will help people in drawing up long-term contracts. It reduces risks involved in lending money, other long-term contracts, rental contracts, labor

contracts. It reduces the risk of holding Federal Reserve notes. So there is a case compared to no commitment at all for having a commitment to a price-level path, or it's often called an inflation target.

8.49 But there are other feasible targets, too. Going back to the distinction between nominal and real variables, the other feasible targets are also defined over nominal variables. Price level is a nominal variable, so it's feasible to target that. But so is total spending in the economy. There are pretty strong arguments that for the sake of minimizing business cycles, stabilizing nominal spending rather than the inflation rate would be preferable.

8.50 Nominal GDP is what we're talking about. This is sometimes called a nominal GDP target. The Fed can hit that target by controlling money growth because nominal GDP growth is the sum of inflation and real output growth. Fed doesn't affect real output growth in the long run, but it can affect the inflation rate in the long run. So it can hit either an inflation target or a nominal income target.

8.51 Inflation targets are more popular. Lots of countries have adopted them. The Fed was not the first. The Bank of New Zealand was given an inflation target. They didn't impose it on themselves the way the Fed did. They were given an inflation target by the Parliament in New Zealand. Here is the problem setting: inflation in New Zealand was running in double digits. A newly elected government said, "We want to bring that down, but we don't want to bring it down in a way that throws the economy into a recession." If you just tighten money without people believing that you're committed to bringing down inflation, they're not going to start moderating their wage demands or start bringing down the rate at which they're raising prices. So, you're going to have unsold labor because it's overpriced. You're going to have unsold goods because they're overpriced.

8.52 To have a disinflation, to reduce the inflation rate from double digits to—well, their target was 2%—without tears, you need to make a credible commitment so that people change their pricing behavior. How did they do that in New Zealand? They said, "If the central bank doesn't meet the target, we're going to call the central banker on the rug at Parliament, and he has to explain why he missed the target." He's subject to dismissal for failure to meet the target. We don't have that in the U.S. We don't have dismissal of the central bank for tolerating 9% inflation.

8.53 This one was not as publicly known, but the central banker in New Zealand at the time has talked about it. They told the central bank, "Here's your budget. The higher inflation is, the less your budget will buy. So in your own interest, if you want to keep the purchasing power of your wages, you want to keep inflation down. You're just reducing your own real wage if you tolerate inflation." So that provides them with a little incentive beyond just being publicly embarrassed.

- 8.54** The Fed's inflation target is different and much less strict because it was adopted by the Fed. Bernanke announced it in 2012. He said, "We're going to have a 2% inflation target." They put it into their formal statement of goals and monetary policy strategy. As initially announced, the target was symmetric, meaning if we're below 2%, we're going to bring it back up to 2%. If it's above 2%, we're going to bring it back down, and presumably at the same speed.
- 8.55** But in 2020, the Fed moved to what it called flexible average inflation targeting. They had been slightly undershooting 2% inflation. Inflation had been running like 1.5%. And the Fed said, "We need to make up for the undershooting of inflation; we think that's keeping the economy back," which, of course, it doesn't if people expect 1.5% inflation and the Fed delivers it. The Fed said, "We need to offset the undershooting by tolerating inflation above 2% for a while." So that's the average inflation targeting. "We want the average inflation rate to be 2%, and if it was below, then we need to put it above." They were a little vague about for how long.
- 8.56** Well, 2022 comes along and we hit 9% inflation because the Fed's accommodation for the pandemic wasn't withdrawn promptly. And people ask Jerome Powell, "So you said that when inflation was below target, you're going to tolerate higher inflation to bring the average down to 2%. Now that you've hit way above 2%, are you going to shoot for inflation below 2%?" So if you were seven percentage points above, are you going to put it one percentage point below for the next seven years to bring the average down to 2%? And Powell said, "Don't be ridiculous. We're going to take our time bringing inflation down so that we don't cause a recession."
- 8.57** But that means the target is no longer symmetrical, and in fact, it's no longer averaging 2%. If it's below 2%, you promptly bring it back up. But if it's above 2%, you take your time bringing it back down, then your average inflation rate is going to be above 2%. You're going to spend more time above 2% than below. And that means expecting 2% inflation? No, you shouldn't do that. You should expect more. How much more? Well, it depends on your guess about how quickly they're going to correct the path when inflation gets off target. Currently, bond markets are suggesting an inflation expectation over the next 5 to 10 years of 2.5% to 3%. So bond markets have learned not to predict 2%.
- 8.58** Is having a low inflation target desirable? Compared to a high inflation rate, yes. For the reasons we've talked about, inflation makes it more costly to hold money. We talked about it in the context of the optimum quantity of money argument. Lower inflation lowers the cost to households of having enough money to make the transactions they want to make without resorting to lots of financial transactions in the meantime or incurring costs to keep their money balances lower.
- 8.59** We've talked about seigniorage in the last lecture. The monetary expansion that generates higher inflation acts like a tax, and it acts like a tax in both senses. It transfers money from the public to the government—the monetary expansion transfers real

resources by letting the government spend the new money. But it also is a tax in the sense that there's a cost, a deadweight cost of the activity discouraged by the tax.

- 8.60** In the case of holding money, what's discouraged is holding adequate money balances. If you try to keep your money balances low, it means you're trying to speed up your receipt of payments, and you're trying to slow down outgoing payments—sorry, that's at high inflation rates people incur those costs. Those are sometimes called shoe leather costs. If people make more trips to the bank, it wears out their shoes. But that's a metaphor for all the costs people incur to manage their money balances more closely. If you can count on inflation being lower, you don't bother with those things. We saw in the optimum quantity of money argument that the policy that minimizes the deadweight costs is inflation slightly less than zero.
- 8.61** There's a second loss from inflation that's often overlooked, even though there are some important empirical studies by an economist named Martin Feldstein trying to measure the size of this, and that's the interaction of inflation with our tax code. Income taxes are now indexed to the price level. So the size of the standard deduction and the dollar income at which various tax brackets begin goes up as the price level goes up. It's adjusted every year. So there's no real boost to tax revenue from that.
- 8.62** But not all of our taxes are indexed. In particular, taxes on interest income don't just tax the real income after inflation. They tax the entire income, including the part that's just compensating for inflation. So, having the tax in that way discourages savings. We also tax capital gains at a rate that's not indexed to inflation. So, if you buy, let's say, gold and the price of gold just keeps up with the price level, you haven't really made any profit in real terms, but you're going to be taxed on the nominal appreciation of your gold at the rate of inflation. So that discourages investments, including investments in inventories and productive equipment. Any investment in real stuff is taxed at the capital gains rate.
- 8.63** Feldstein's estimate was that this discourages savings and investment to such an extent that it reduces national income by about 1% because we have a smaller stock of productive equipment as a result of people responding to these tax rates. One way to put it is that the unindexed tax on interest income and on capital gains is a tax on saving for your retirement. People will save less for their retirement, and they will end up with less consumption in retirement.
- 8.64** So far, I've only talked about predictable inflation. Even if you anticipate inflation, you're going to incur the costs of minimizing your cash balances, and you're going to have these tax effects. The costs of unpredictable inflation or unexpected inflation are much worse. So when the inflation rate is hard to predict, then firms don't know at what rate to raise their prices, and workers don't know at what rate to raise their reservation wage demands, you're going to get more resource misallocation. You're going to get a noisier price system. It's not going to steer resources into their most

valued uses as effectively. The price system communicates information. Inflation, especially at an unpredictable rate, is going to jam those signals.

- 8.65** Economic growth suffers. We know this from cross-country studies: countries with higher inflation have slower economic growth. The stock market is going to suffer. Stock prices don't quite keep up with inflation because of the noisiness of the information that companies are announcing: did they really make a profit, or is that just a gain because there was inflation this quarter and they raised their prices ahead of other people? It discourages investment in the stock market.
- 8.66** We all know about the redistribution of wealth from debtors to creditors or creditors to debtors, depending on whether inflation is underanticipated or overanticipated. If inflation is higher than anticipated, people get to repay their debts in dollars that are worth less than was anticipated at the time that the contract was made. So the creditor is getting a lower return than anticipated. And the risk attached to not being able to know what the inflation rate is going to be means that fewer contracts are going to be made. Borrowers are going to be reluctant to lend. Bond buyers are going to be reluctant to buy bonds. It's going to discourage long-term financial contracts.
- 8.67** You can see this if you look across countries. In countries with high and variable inflation, it's hard to get a long-term mortgage. You're lucky if you can get a five-year mortgage, which you then have to roll over at whatever is the current rate outside the United States, outside countries—well, the United States, the Eurozone, countries with more dependable inflation rates. So you have smaller financial markets in countries with variable inflation because it's just too risky to make long-term commitments.
- 8.68** Now, the Fed announced an inflation target, as I mentioned, of 2%, but I said the optimum quantity of money argument says lower inflation is better all the way down to -1%. So, why is the target 2% rather than -1%? Well, maybe -1% is going too far. Zero percent inflation, price stability has, over the years, attracted a lot of supporters because it's simple. You can put it on a bumper sticker. You can try to get a consensus of opinion that no inflation is better than anything higher.
- 8.69** Ben Bernanke, when he announced the 2% inflation target, was asked by a reporter, "Why is it 2% rather than 0%?" He gave three reasons, and none of them is very strong. He said, "There are measurement biases so that when we say there's 2% inflation, we're not counting all the improvements in product quality, and so we're overstating inflation." Well, people who've seriously tried to measure the overestimate of the inflation numbers have said maybe it's a 0.5%. So that doesn't account for a difference between 0% and 2%. That would account for having a target of 0.5% instead of 0%, not 2% rather than 0%.
- 8.70** Secondly, Bernanke has a fear of deflation. Partly this is because he studied the Great Depression and hasn't studied other episodes where deflation was not associated with recession. To quote what Bernanke said at this press conference, he said, "Deflation

can be bad for economic performance and bad for employment, and therefore, setting an inflation target at zero would not be consistent with the other part of our mandate, meaning full employment," because we'd be spending half of our time below zero just through random fluctuations."

- 8.71** He likened being below zero to falling off a cliff, in so many words, but it's not like that. When deflation is unanticipated, when there's not as much spending as businesses were anticipating when they produced and priced their products, then, yes, you get unsold inventories piling up on the shelves. That's the Great Depression. But that was due to a shrinkage in the money supply. That unanticipated deflation, yes, is very harmful.
- 8.72** But what about a deflation that is anticipated? If you've made it your policy target to have 0% inflation, and people know we're going to fluctuate a little bit above and a little bit below, it's not going to have the same effect. Lowering the expected inflation rate, even into the negative range, there's no discontinuity where suddenly that makes businesses unprofitable or gives rise to recession. So, having a target of 0% or lower is different from hitting 0% or lower when people were expecting something higher. Here, they're expecting that, so it's not disruptive.
- 8.73** The third reason he gave—and this has been echoed by some other economists—is that when the Fed comes into a recession, we typically cut our interest rate target. If we had inflation at 0%, nominal interest rates would be pretty low to begin with, like 1%, because a rough estimate is that the nominal interest rate is 1% higher than the inflation rate, because the real interest rate is around 1%. That's the difference. Well, if nominal interest rates were 1%, the Fed couldn't cut interest rates much to fight the recession. Our hands would be tied, or as it's sometimes put, we would run out of ammunition for fighting the recession.
- 8.74** Of course, in the financial crisis, 2008 to 2010, the Fed lowered its interest rate target to about 0%, and then didn't think there was much else they could do. So they worried about running out of ammunition. What that complaint assumes is that there's nothing more they can do when their interest target has hit 0%. But that is an artifact of having an interest rate target as the instrument through which they manage monetary policy. When really, what their focus should be is the rate of growth of the monetary aggregates, the base and M2.
- 8.75** Of course, the Fed did engage in quantitative easing during the Great Recession. Quantitative easing, that is, expanding the Fed's balance sheet, is effective at putting more money into the economy, which is all the Fed can do to cure a recession when the recession is due to an inadequate supply of money. So a recession that's due to an unsatisfied excess demand for money can be remedied by providing more money; it doesn't require anything about interest rates. The Fed could pursue its monetary policy by focusing on the change in the monetary base and the change in money held by the public, M2. They don't need to pursue an interest rate target. So they've hamstrung

their own policy by insisting on formulating it in terms of "how low we want the interest rate target, and that's our signal as to whether we're expanding the monetary base enough."

- 8.76** I mentioned there's an alternative to inflation or price-level targeting, and that's nominal income targeting. The argument as to why this is better than inflation targeting is a little bit subtle, but let me run it past you this way. Let me name some economists who have suggested this: Scott Sumner, whom I mentioned earlier, George Selgin.
- 8.77** Friedrich Hayek, in his book "Prices and Production" in the 1930s, said that a central bank that's not constrained by the gold standard should aim to stabilize total nominal spending. In terms of the equation of exchange, that's $M \text{ times } V$, total money spending in the economy. And that means if you have 2% real economic growth and no change in velocity, you should have 2% falling prices. That would be okay with them. So stabilize nominal GDP.
- 8.78** What's the case for doing that rather than stabilizing the price index? Consider the following case. There's an oil price shock. OPEC raises the price of oil. This has happened before. How does that affect the U.S. economy? Well, we're an oil—at least we were, an oil-importing country—so this makes us poorer. This reduces our potential output because we import oil in order to produce other things. So, this is a loss of the natural rate of output. The economy's productive potential has been reduced. If monetary policy doesn't change, then the price level goes up to reflect real income going down. No change in $M \text{ times } V$.
- 8.79** Would it be better to try to cancel that rise in the price level by reducing the money supply? That's what you have to do to stabilize the price level in the face of this adverse supply shock. But that would hit the economy with a double whammy. Not only has there been this real shock that reduces the productive potential of the economy, now you have a negative monetary shock, a shrinkage in the quantity of money, that reduces spending even more and may make the recession even worse than it needs to be. You may drive the output of the economy below its new potential.
- 8.80** That is sometimes recognized by people who say they're in favor of an inflation target. "Yeah, we should suspend the target when there are real shocks." But then, you don't really have an inflation target or a price level target; you have a nominal income target.
- 8.81** What do we do about a Fed that hasn't even lived up to its own inflation target? Is there a remedy for poor performance by the central bank? There's a big debate going on recently about how much independence the Federal Reserve should have. Independence is useful if the Fed is pursuing a lower inflation policy than Congress or the President wants, and they're pressuring it. So, from the point of view of the public, it would be a good thing if they didn't give in to that pressure. So independence is good for that.

- 8.82** But independence is not good for remedying a central bank that is not well-managed, that chooses the wrong goal to pursue, like pays too much attention to output and employment, or that executes its policies in an incompetent way. To remedy that, you need to give the central bank instructions. You need a rule, not just independence. You need a commitment to an inflation target or a nominal income target. And by rule, I mean something that really has teeth: penalize them in some way when they don't reach their target.
- 8.83** But the Fed's current target is self-imposed. When they don't reach it, they say, "Sorry, we'll try to get back to it in due course." There's no discipline, there's no penalty. The Fed can always redefine the target so that it's not violating the target as currently defined. And I mentioned they redefined it from a symmetric target to an asymmetric target. But there's no penalty for failure to reach their goals, unlike the case of the New Zealand Central Bank, where the goal was imposed on the Central Bank by the legislature, and the Central Bank was being held accountable for it.
- 8.84** We need an instruction manual that the central bank is required to follow—that's a monetary rule. Lots of different rules have been proposed over the years: money growth rules, price level rules—that is, a price level or inflation target that really has teeth—a nominal GDP target that has teeth.
- 8.85** One of the most common rules that central banks around the world have is an exchange rate target. That's not the same as an inflation target or a nominal income target, but it makes sense in the context of a small economy that is highly dependent on external trade. They probably do the best for their economy by eliminating the risk of devaluation of their currency, eliminating exchange rate fluctuations, by a credible commitment to a fixed exchange rate.
- 8.86** So how do you make a credible commitment to a fixed exchange rate? You have a currency board that just redeems the local currency for dollars, or you outright use dollars—that's what Panama does. Or euros, if you're on the periphery of the Eurozone—that's what Bulgaria does. A lot of countries, though, have a soft peg to the dollar—this is the exchange rate until we decide to choose it. That's not so effective because that's subject to people not believing it and testing it by trying to redeem for dollars and forcing the central bank to float because they run out of dollars.
- 8.87** A popular prescription for Fed behavior is called the Taylor rule, named after John Taylor from Stanford University. The Taylor rule is formulated in terms of an interest rate target, and that's because that's how the Fed formulates its own policies. They say, "Here's our current interest rate target." Taylor observed the way the Fed behaved during a period in which performance was pretty good, sometimes called the Great Moderation: inflation was low, around 2%, and business cycles were not severe, recessions were pretty mild.

- 8.88** He said that we can summarize the way the Fed behaved during this period by saying they responded anytime inflation started to rise by raising their interest rate target, which means tightening money. They responded when real output fell below their target by looser monetary policy. So it was a feedback mechanism. Taylor said that just explaining Fed policy in terms of these variables pretty well describes what the Fed is doing. This was a good period, so maybe we should commit the Fed to responding this way to movements in inflation and movements in real output. So it went from being a description to being a prescription.
- 8.89** But it's not really a rule. The distinction between rules and guidelines was made very clear in the movie Ghostbusters. I don't know if you remember the scene where the woman who is possessed is floating above the bed, and one of the Ghostbusters is interviewing her, and she starts coming on to him, and he says, "I make it a rule never to get involved with possessed people." And then she starts seriously flirting with him, and he says, "Well, it's more of a guideline than a rule," which means my commitment to it is very weak. To make it a rule, there need to be real enforcement mechanisms.
- 8.90** After the period that Taylor studied, the Fed deviated from the Taylor rule, from the formula they seemed to implicitly be following, especially in the period running up to the financial crisis. During the period of the housing bubble, the Fed was keeping its interest rate target close to 0%, well, down at 1%, for a long time, even after the economy started to recover from the dot-com recession. That arguably provided cheap credit that fed the housing bubble. Only belatedly did the Fed start to raise its interest rate target as the economy started to overheat. But they raised it very slowly. It was another few years. It wasn't until the end of 2006 that they got back to the range of interest rate targets that Taylor would have recommended. In a later book, Taylor says — called "Getting Off Track" — that's why we had the housing bubble, because the Fed didn't follow the Taylor rule. It made credit too cheap.
- 8.91** What was the Fed's reason for that? They said, "We're in a new era. There's a global glut of savings. So we don't think that the normal interest rate or the real equilibrium interest rate is around 2% anymore. We think it's much lower. So we think our interest rate target should be much lower." It turns out they were wrong. Those interest rates couldn't be sustained. The economy started to inflate through excess money growth, and they had to start raising the target to tighten money.
- 8.92** So there we are with improvements in monetary policy from a fiat central bank. We can try to fasten a rule on it, but we need a rule with teeth. The inflation target seems to be the most popular proposal. Nominal income target, I think, is technically preferable, but I would settle for an inflation target that was not just a target, but a real commitment, a dismissal for failure to meet the target.
- 8.93** There is another way to try to secure better money, and that is to switch to a different monetary standard, something other than fiat money, where we rely on trying to control a federal agency to give us the quality of money we want. This alternative was

proposed back in the '70s, when inflation was in double digits, by the economist F. A. Hayek, whom I've mentioned earlier. Hayek won a Nobel Prize in 1974 for work he had done in the '30s. And he said, "Wow, this is great. Now people are paying attention to me. I stopped talking about money after Bretton Woods because nobody was listening to me, but now maybe people will listen to me."

- 8.94** Here's what he said, "Why do we have high inflation? It's like asking, 'Why do we have low-quality postal service?' Because we don't have competition, because the government has a monopoly, and they don't allow competition in the provision of money. At the very least, they should allow people to use money from neighboring countries or any country in the world, or gold, whatever existing money they want." Then, he broadened the proposal and said, "Why don't we let private enterprise in on the competition? Maybe they would come up with a more reliable money."
- 8.95** I've been inspired by that proposal of Hayek's. Of course, things are now different in that we do have privately-issued money-like assets in the form of cryptocurrencies. In particular, Bitcoin is a potential alternative monetary standard. Here I'm going to plug my latest book: in my 2023 book, "Better Money: Gold, Fiat, or Bitcoin," I look at the question of whether Bitcoin would be a reliable monetary standard. Not to give it away, but my answer is probably not as reliable as the gold standard was. We know the gold standard restrains inflation. We've got plenty of evidence of that. But to say more about Bitcoin, Peterson Academy is going to have to invite me back to give another series of lectures. I will stop here. Thanks very much.
- 8.96** Thank you. I was curious if there is a target deficit number. You said we're around 30 trillion in the United States. At what point does this become really concerning? Do you have any figure, in which things would get, I don't know, hyperinflation or just really bad?
- 8.97** It wouldn't take hyperinflation to erode our current debt; 5% to 7% inflation for a number of years would put a pretty big dent in it. That's basically what we saw in the period after the Second World War: not hyperinflation, but higher than we would like inflation, 4% up to double digits in the '70s. But when does the debt become really concerning? Well, I think it's really concerning now. It's hard to find historical examples of countries that had debt of more than 120% who brought it back down without inflation. So that's the concern.
- 8.98** I mean, in principle, if you didn't keep increasing it above 30 trillion, as the economy grows, that will become a smaller share of GDP. But you have to be committed to a balanced budget if you're going to keep it from growing, and we're nowhere near a credible commitment to a balanced budget.
- 8.99** There have been attempts. The last time we had a surplus was a result of an agreement between a Republican Congress led by Newt Gingrich and Bill Clinton being a moderate Democrat. As the economy recovered, as real output grew, they didn't ramp up

government spending to exhaust the growth in tax revenues, and we actually ended up with a surplus at the end of this boom period. But that hasn't been repeated.

8.100 In recent years, we've been running record deficits as a percent of GDP in peacetime in a growing economy, in a non-recession economy. So that's concerning because it doesn't look like there's going to be an opportunity to pay down the debt or even to stabilize the dollar amount of the debt because there's no discipline on the spending side. Both parties are happy to run big deficits.